

American Option Pricing Formulation and Numerical Analysis

Research project *supervised by Professor Steven Campbell*

by

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Author's Declaration

I declare that this thesis is the result of my own work and has not been submitted for any other degree or professional qualification. All sources used have been appropriately acknowledged.

Abstract

This report examines the pricing of American options using the Least Squares Monte Carlo (LSMC) method, with a particular focus on Laguerre polynomial basis functions. We benchmark LSMC against binomial tree models and analyze differences in computational efficiency and accuracy. Our numerical results demonstrate that Laguerre polynomials yield smoother and more stable early-exercise boundaries, while also reducing computational demands. A comprehensive sensitivity analysis reveals key parameter effects: higher volatility lowers exercise boundaries but increases the likelihood of early exercise; interest rates influence put-option incentives; and dividend yields have the most substantial impact on exercise behavior.

We find that fourth-degree Laguerre polynomials offer an effective balance between approximation accuracy and model complexity, with diminishing improvements beyond approximately 200 time steps and 50,000 simulation paths. Our convergence study and basis-function analysis provide practical guidelines for implementing LSMC in trading and risk management contexts. We conclude by identifying promising extensions, including multi-asset pricing and machine-learning-based regression frameworks.

Chapter 1

Introduction

This report is dedicated to studying the American option pricing problem. We begin by examining the fundamental mathematical notations and structures necessary to formulate and understand the problem. Next, we explore its theoretical foundations, including optimal stopping theory, free-boundary problems, and dynamic programming principles. We then discuss various numerical approaches for solving the problem, such as finite difference methods, binomial trees, and Monte Carlo simulations, highlighting their advantages and limitations.

Our research focuses particularly on the enhancement of the Least Squares Monte Carlo (LSM) method through the implementation of Laguerre polynomial basis functions. We demonstrate how this approach leads to more efficient computation and smoother representations of the early exercise boundary compared to traditional methods. Through detailed sensitivity analysis, we examine how volatility, interest rates, and dividend yields impact the early exercise decision process, offering practical insights into option valuation.

The report includes extensive numerical experiments that establish optimal parameters for the LSM method, including the ideal Laguerre polynomial degree and the necessary number of time steps and simulation paths for convergence. We provide comparative analyses of boundary estimation methods and investigate the early exercise premium across various market conditions. This work contributes to both the theoretical understanding of American option pricing and offers practical applications for financial modeling and trading strategy development.

Chapter 2

Literature Review

The pricing of American options—a class of derivatives that can be exercised at any point before expiration—has long been associated with the mathematical theory of optimal stopping. Foundational contributions to this theory were made by [Fakoev \(1970\)](#), [Bismut & Skalli \(1971\)](#), [Neveu \(1975\)](#), and [Shiryaev \(1978\)](#), who framed optimal stopping as the problem of choosing a stopping time to maximize the expected payoff from a stochastic process. These early results laid the groundwork for applying optimal stopping principles to financial derivatives such as American options.

The formal modeling of option prices was revolutionized by [Black & Scholes \(1973\)](#), who introduced a continuous-time framework based on stochastic calculus. [Merton \(1973\)](#) extended their work to include broader classes of stochastic processes. Together, these models form the backbone of modern option pricing theory. However, these models primarily address European-style options, which lack the early-exercise feature.

To accommodate early exercise, [Cox *et al.* \(1979\)](#) introduced the binomial tree model. This discrete-time framework is computationally tractable and well-suited to valuing American options, particularly due to its flexibility in handling varying conditions at each time step. Further theoretical insights were provided by [Bensoussan \(1984\)](#), who framed the American option pricing problem within a variational inequality formulation, emphasizing the existence and characterization of free boundaries.

[Hull & White \(1987\)](#) investigated option pricing under stochastic volatility conditions and demonstrated that the traditional Black-Scholes model tends to overestimate option prices, especially for long maturities. They also introduced numerical techniques for dealing with correlated volatility, significantly expanding the practical applicability of pricing models. [Carr \(1996\)](#) contributed a novel randomization technique for American puts, offering a probabilistic reinterpretation that enhances numerical approximation.

In a parallel development, [Davis & Karatzas \(1994\)](#) reframed the optimal stopping problem from a stochastic process viewpoint to a deterministic control framework, particularly in discrete time. Their approach demonstrated that stochastic problems could be decomposed into families of deterministic optimization problems, and their work elegantly connected with the prophet inequality of Krengel and Sucheston, broadening its implications for decision-making in finance. As financial models grew in complexity, especially with path-dependent or high-dimensional features, simulation-based methods gained prominence. [Longstaff & Schwartz \(2001\)](#) introduced the Least Squares Monte Carlo (LSM) method, which uses re-

gression over simulated paths to estimate continuation values. This approach provides a flexible alternative to finite-difference or tree-based methods, especially for American options with multiple sources of uncertainty. [Caffisch & Chaudhary \(2004\)](#) expanded on LSM by incorporating quasi-random sampling and martingale techniques, offering improved performance bounds and practical implementations in challenging scenarios like American Asian options.

An important complementary contribution was made by [Rogers \(2002\)](#), who introduced a dual Monte Carlo method for American option valuation. This method simulates both the payoff paths and a carefully chosen Lagrangian martingale, using their difference to construct an upper bound on the option price. Notably, this bound becomes sharp when the martingale is optimally chosen. Even simple martingale constructions showed surprising accuracy in numerical tests. Additionally, this approach naturally yields hedging strategies and risk estimates, enriching the practical insights available from simulation-based pricing.

For a more unified and rigorous mathematical treatment, [Peskir & Shiryaev \(2006\)](#) provided a comprehensive exposition of optimal stopping and free-boundary problems. Their work integrates stochastic analysis, variational inequalities, and boundary characterizations, offering a robust foundation for advanced financial applications.

Complementing these developments are a number of key texts that provide theoretical depth and pedagogical clarity. [Karatzas & Shreve \(1998\)](#) synthesizes stochastic finance theory in a rigorous probabilistic framework, while [Shreve \(2004\)](#) and [Shreve \(2004\)](#) offer an accessible two-part series on discrete and continuous models. [Privault \(2022\)](#) brings modern market examples to stochastic finance theory, and [Campbell \(2024\)](#) presents up-to-date lecture notes tailored for graduate-level instruction in financial stochastic methods.

Together, these contributions form a cohesive and evolving body of knowledge that underpins the valuation of American options and the broader application of optimal stopping in financial mathematics. The insights and tools developed across these works will inform the modeling, analysis, and numerical strategies employed in the remainder of this thesis.

Chapter 3

Preliminaries

Now we must establish the precise definitions of the key concepts: Snell envelope, risk-neutral measure, smooth pasting, dynamic programming, and partial differential equations. These definitions are crucial for our understanding of the problem formulation in Section 4.4.1, which is an essential component enabling us to construct our analysis and numerical calculation.

3.1 Probabilistic Concepts

Definition 1 (σ -algebra). *Let Ω be a nonempty set, and let $\mathcal{F}$ be a collection of subsets of Ω . Then $\mathcal{F}$ is called a σ -algebra (also known as a σ -field) if it satisfies the following properties:*

- (i) *The empty set is included: $\emptyset \in \mathcal{F}$.*
- (ii) *If a set A is in $\mathcal{F}$, then its complement A^c is also in $\mathcal{F}$.*
- (iii) *If a countable collection of sets $A_1, A_2, \dots \in \mathcal{F}$, then their union $\bigcup_{n=1}^{\infty} A_n$ also belongs to $\mathcal{F}$.*

Example 1. *Let $\Omega = \{1, 2\}$. Then the collection $\mathcal{F} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ is a σ -algebra over Ω .*

Definition 2 (Probability Measure). *Let Ω be a nonempty set, and let $\mathcal{F}$ be a σ -algebra of subsets of Ω . A probability measure $\mathbb{P}$ is a function that assigns to each set $A \in \mathcal{F}$ a value in $[0, 1]$, denoted by $\mathbb{P}(A)$, and satisfies the following conditions:*

- (i) **Normalization:** $\mathbb{P}(\Omega) = 1$.
- (ii) **Countable additivity:** *For any sequence of disjoint sets $A_1, A_2, \dots \in \mathcal{F}$,*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

Example 2. Let $\Omega = \{1, 2\}$ and $\mathcal{F} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Define $\mathbb{P}(\{1\}) = 0.4$, $\mathbb{P}(\{2\}) = 0.6$. Then $\mathbb{P}$ is a probability measure since $\mathbb{P}(\Omega) = 0.4 + 0.6 = 1$.

Definition 3 (Random Variable). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable is a real-valued function $X : \Omega \rightarrow \mathbb{R}$ such that for every Borel subset $B \subset \mathbb{R}$, the set

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$$

belongs to the σ -algebra $\mathcal{F}$.

Example 3. Let $\Omega = \{1, 2\}$, and define $X(1) = 5$, $X(2) = 10$. Then X is a random variable because for any Borel set $B \subset \mathbb{R}$, the preimage $X^{-1}(B)$ is a subset of Ω and hence in $\mathcal{F}$.

Definition 4 (Lebesgue Measure on $\mathbb{R}$). *Shreve (2004)* Let $\mathcal{B}(\mathbb{R})$ be the Borel σ -algebra on $\mathbb{R}$, i.e. the smallest σ -algebra containing all intervals of the form $[a, b]$. The Lebesgue measure $\mathcal{L} : \mathcal{B}(\mathbb{R}) \rightarrow [0, \infty]$ is the unique countably additive set function satisfying:

(i) (Length) For any real numbers $a \leq b$,

$$\mathcal{L}([a, b]) = b - a.$$

(ii) (Null empty set)

$$\mathcal{L}(\emptyset) = 0.$$

(iii) (Countable additivity) Whenever $\{B_n\}_{n \geq 1} \subset \mathcal{B}(\mathbb{R})$ are pairwise disjoint,

$$\mathcal{L}\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mathcal{L}(B_n).$$

In essence, $\mathcal{L}$ extends the usual notion of “length” on intervals to all Borel sets and underpins the modern theory of Lebesgue integration.

Example 4. The Lebesgue measure of the interval $[2, 5]$ is $\mathcal{L}([2, 5]) = 5 - 2 = 3$.

Definition 5 (Expectation). Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The expectation (or expected value) of X is defined as

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega),$$

where the right-hand side is a Lebesgue integral of X with respect to the probability measure $\mathbb{P}$.

This definition is valid if X is integrable, that is, if

$$\mathbb{E}[|X|] = \int_{\Omega} |X(\omega)| d\mathbb{P}(\omega) < \infty,$$

or if $X \geq 0$ almost surely. In the latter case, $\mathbb{E}[X]$ may be infinite.

Theorem 1 (Properties of Expectation). *Shreve (2004) Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The expectation operator $\mathbb{E}$ satisfies the following properties:*

(i) **Discrete formula:** *If X takes finitely many values $x_0, x_1, \dots, x_n$, then:*

$$\mathbb{E}[X] = \sum_{k=0}^n x_k \mathbb{P}(X = x_k).$$

In particular, if Ω is finite:

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega).$$

(ii) **Integrability:** *X is integrable if and only if*

$$\mathbb{E}[|X|] < \infty.$$

(iii) **Comparison:** *If $X \leq Y$ almost surely and X, Y are either integrable or nonnegative, then*

$$\mathbb{E}[X] \leq \mathbb{E}[Y].$$

Moreover, if $X = Y$ almost surely and either is integrable or nonnegative, then:

$$\mathbb{E}[X] = \mathbb{E}[Y].$$

(iv) **Linearity:** *If $\alpha, \beta \in \mathbb{R}$, and X, Y are integrable or nonnegative, then:*

$$\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y].$$

(v) **Jensen's Inequality:** *If φ is a convex, real-valued function on $\mathbb{R}$ and $\mathbb{E}[|X|] < \infty$, then:*

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

Definition 6 (Filtration). *Let Ω be a nonempty set and let $T > 0$ be fixed. Suppose that for each time $t \in [0, T]$, there is a σ -algebra $\mathcal{F}(t) \subseteq 2^\Omega$. If for all $0 \leq s \leq t \leq T$, we have*

$$\mathcal{F}(s) \subseteq \mathcal{F}(t),$$

then the family $\{\mathcal{F}(t)\}_{0 \leq t \leq T}$ is called a filtration.

Example 5. *Let $\Omega = \{H, T\}^3$ be the sample space of three coin tosses. Define $\mathcal{F}(0)$ as $\{\emptyset, \Omega\}$, $\mathcal{F}(1)$ as the σ -algebra generated by the outcome of the first toss, $\mathcal{F}(2)$ by the first two tosses, and so on. Then $\{\mathcal{F}(t)\}_{t=0}^3$ is a filtration.*

Interpretation. A filtration represents the accumulation of information over time. That is, at any time t , the σ -algebra $\mathcal{F}(t)$ contains all the events whose outcomes are known by time t . If $\omega \in \Omega$ is the true outcome, then at time t , we know whether or not ω lies in any event in $\mathcal{F}(t)$.

Definition 7 (Martingales, Submartingales, and Supermartingales). [Shreve \(2004\)](#) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\{\mathcal{F}_t\}_{0 \leq t \leq T}$ be a filtration. Let $\{M_t\}_{0 \leq t \leq T}$ be an adapted process satisfying $\mathbb{E}[|M_t|] < \infty$ for all t .

(i) M_t is a **martingale** if for all $0 \leq s \leq t \leq T$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s.$$

(ii) M_t is a **submartingale** if for all $0 \leq s \leq t \leq T$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] \geq M_s.$$

(iii) M_t is a **supermartingale** if for all $0 \leq s \leq t \leq T$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] \leq M_s.$$

Example 6. Let $M(t) = W(t)$ be a standard Brownian motion. Then $M(t)$ is a martingale with respect to its natural filtration $\mathcal{F}(t) = \sigma(W(s) : s \leq t)$ because $\mathbb{E}[W(t) \mid \mathcal{F}(s)] = W(s)$ for $s \leq t$.

Example 7. Let $M(t) = W(t)^2 - t$, where $W(t)$ is standard Brownian motion. By Itô's formula,

$$d(W(t)^2 - t) = 2W(t) dW(t),$$

which has zero drift. Therefore $M(t)$ is a **martingale**.

Example 8. A classical example of a supermartingale derived from Brownian motion is

$$M(t) = \frac{|W(t)|}{\sqrt{1+t}}.$$

Using Itô–Tanaka's formula,

$$d|W(t)| = \text{sgn}(W(t)) dW(t) + dL_t^0,$$

and therefore

$$dM(t) = \frac{\text{sgn}(W(t))}{\sqrt{1+t}} dW(t) - \frac{|W(t)|}{2(1+t)^{3/2}} dt + \frac{1}{\sqrt{1+t}} dL_t^0.$$

The drift term is nonpositive, so $M(t)$ is a **supermartingale**.

3.2 Stochastic Processes

A **stochastic process** is a collection of random variables $\{X_t\}_{t \in T}$ defined on a probability space, where t is usually time. It models systems evolving with randomness.

Definition 8 (Adapted Stochastic Process). *Shreve (2004)* Let Ω be a nonempty sample space equipped with a filtration $\{\mathcal{F}(t)\}_{0 \leq t \leq T}$. Let $\{X(t)\}_{0 \leq t \leq T}$ be a collection of random variables indexed by time $t \in [0, T]$. We say that $\{X(t)\}$ is an adapted stochastic process if, for each $t \in [0, T]$, the random variable $X(t)$ is $\mathcal{F}(t)$ -measurable.

Interpretation. An adapted process is one where the value of $X(t)$ is determined by the information available up to time t . That is, the process does not “look into the future.”

Definition 9 (Markov Process). *Shreve (2004)* Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\{\mathcal{F}(t)\}_{0 \leq t \leq T}$ be a filtration of sub- σ -algebras of $\mathcal{F}$. Consider an adapted stochastic process $\{X(t)\}_{0 \leq t \leq T}$.

We say that $X(t)$ is a Markov process if for all $0 \leq s \leq t \leq T$, and for every nonnegative, Borel-measurable function f , there exists a Borel-measurable function g such that:

$$\mathbb{E}[f(X(t)) \mid \mathcal{F}(s)] = g(X(s)).$$

Interpretation. The future behavior of a Markov process depends only on its present state $X(s)$, not on the full past history. This is known as the Markov property. The Markov property underlies the connection between stochastic process dynamics and their associated partial differential equation formulations in financial modeling.

Lemma 1 (Independence and Conditional Expectation). *Shreve (2004)* Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. Suppose that the random variables $X_1, \dots, X_K$ are measurable with respect to $\mathcal{G}$, and that $Y_1, \dots, Y_L$ are independent of $\mathcal{G}$.

Let f be a Borel-measurable function of both $x_1, \dots, x_K$ and $y_1, \dots, y_L$. Define the function

$$g(x_1, \dots, x_K) := \mathbb{E}[f(x_1, \dots, x_K, Y_1, \dots, Y_L)],$$

which averages out the randomness from the variables $Y_1, \dots, Y_L$.

Then the conditional expectation of $f(X_1, \dots, X_K, Y_1, \dots, Y_L)$ given $\mathcal{G}$ is:

$$\mathbb{E}[f(X_1, \dots, X_K, Y_1, \dots, Y_L) \mid \mathcal{G}] = g(X_1, \dots, X_K).$$

Definition 10 (Scaled Symmetric Random Walk). *Campbell (2024)* Fix a time horizon $T > 0$ and integer $N \geq 1$, and set $\Delta t = T/N$. A scaled symmetric random walk $\{X_{k\Delta t}^{(N)}\}_{k=0}^N$ is the discrete-time process defined by

$$X_0^{(N)} = 0 \quad \text{and} \quad X_{k\Delta t}^{(N)} = X_{(k-1)\Delta t}^{(N)} + \sqrt{\Delta t} Z_k, \quad k = 1, \dots, N,$$

where $Z_1, \dots, Z_N$ are independent Rademacher random variables satisfying $\Pr(Z_k = 1) = \Pr(Z_k = -1) = \frac{1}{2}$.

Lemma 2 (Mean and Variance of the Scaled Random Walk). *Campbell (2024)* With $X^{(N)}$ as above,

$$\mathbb{E}[X_T^{(N)}] = 0, \quad \text{Var}[X_T^{(N)}] = T.$$

Sketch of Proof. Since the Z_k are iid with $\mathbb{E}[Z_k] = 0$ and $\text{Var}(Z_k) = 1$, one finds

$$\mathbb{E}[X_T^{(N)}] = \sum_{k=1}^N \sqrt{\Delta t} \mathbb{E}[Z_k] = 0,$$

and by independence,

$$\text{Var}[X_T^{(N)}] = \sum_{k=1}^N (\sqrt{\Delta t})^2 \text{Var}(Z_k) = \sum_{k=1}^N \Delta t = T.$$

Definition 11 (Brownian Motion (Wiener Process)). [Campbell \(2024\)](#) A Brownian motion (or Wiener process) is a continuous-time stochastic process $\{W_t\}_{t \geq 0}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ satisfying:

- (i) $W_0 = 0$ almost surely.
- (ii) (**Independent, stationary increments**) For any $0 \leq s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t - s),$$

and if $[s, t]$ and $[u, v]$ are disjoint time intervals then $W_t - W_s$ is independent of $W_v - W_u$.

- (iii) (**Path continuity**) With probability one, the sample paths $t \mapsto W_t(\omega)$ are continuous functions of t .

Definition 12 (Filtration for Brownian Motion). [Shreve \(2004\)](#) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a Brownian motion $\{W(t)\}_{t \geq 0}$. A filtration for $\{W(t)\}$ is a family of sub- σ -algebras $\{\mathcal{F}(t)\}_{t \geq 0}$ of $\mathcal{F}$ satisfying:

- (i) **Nondecreasing information.** For all $0 \leq s \leq t$,

$$\mathcal{F}(s) \subseteq \mathcal{F}(t).$$

Thus, as time advances, no previously known event is “forgotten.”

- (ii) **Adaptedness.** For each $t \geq 0$, the value $W(t)$ is $\mathcal{F}(t)$ -measurable. In other words, all information up to time t suffices to determine $W(t)$.
- (iii) **Independent future increments.** For any $0 \leq s < t$, the increment $W(t) - W(s)$ is independent of $\mathcal{F}(s)$. That is, future Brownian increments are independent of the past.

Theorem 2 (Brownian Motion is a Martingale). [Shreve \(2004\)](#) Let $\{W(t)\}_{t \geq 0}$ be a Brownian motion with its natural filtration $\{\mathcal{F}(t)\}_{t \geq 0}$. Then for all $0 \leq s \leq t$,

$$\mathbb{E}[W(t) \mid \mathcal{F}(s)] = W(s).$$

Proof. Fix $0 \leq s \leq t$. Write

$$W(t) = (W(t) - W(s)) + W(s).$$

Since $W(t) - W(s)$ is independent of $\mathcal{F}(s)$ and has mean zero,

$$\mathbb{E}[W(t) - W(s) \mid \mathcal{F}(s)] = \mathbb{E}[W(t) - W(s)] = 0.$$

Moreover, $W(s)$ is $\mathcal{F}(s)$ -measurable, so

$$\mathbb{E}[W(s) \mid \mathcal{F}(s)] = W(s).$$

Combining these,

$$\mathbb{E}[W(t) \mid \mathcal{F}(s)] = \mathbb{E}[W(t) - W(s) \mid \mathcal{F}(s)] + \mathbb{E}[W(s) \mid \mathcal{F}(s)] = 0 + W(s) = W(s),$$

which establishes the martingale property. $\square$

Theorem 3 (Brownian Motion is Markov). *Shreve (2004) Let $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with its natural filtration $\{\mathcal{F}(t)\}_{t \geq 0}$. Then for every Borel-measurable function f and all $0 \leq s \leq t$,*

$$\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)] = g(W(s)),$$

where

$$g(x) = \mathbb{E}[f(x + (W(t) - W(s)))].$$

Hence $W(t)$ satisfies the Markov property.

Proof. To verify the Markov property, see definition 9, fix $0 \leq s \leq t$ and let f be any Borel-measurable function. We seek a function g for which

$$\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)] = g(W(s)).$$

Observe that

$$W(t) = W(s) + (W(t) - W(s)).$$

Since $W(s)$ is $\mathcal{F}(s)$ -measurable and the increment $W(t) - W(s)$ is independent of $\mathcal{F}(s)$, the Independence Lemma 1 implies

$$\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)] = \mathbb{E}[f(W(s) + (W(t) - W(s))) \mid \mathcal{F}(s)].$$

Treating $W(s)$ as a constant x inside the conditional expectation gives

$$\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)] = g(W(s)), \quad \text{where} \quad g(x) = \mathbb{E}[f(x + (W(t) - W(s)))].$$

Finally, since $W(t) - W(s) \sim N(0, t - s)$, we can write

$$g(x) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{\infty} f(x+y) \exp\left(-\frac{y^2}{2(t-s)}\right) dy.$$

Equivalently, defining the Gaussian transition density

$$p(\tau, x, y) = \frac{1}{\sqrt{2\pi\tau}} \exp\left(-\frac{(y-x)^2}{2\tau}\right), \quad \tau = t - s,$$

we obtain

$$\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)] = \int_{-\infty}^{\infty} f(y) p(t - s, W(s), y) dy.$$

This shows precisely that $\mathbb{E}[f(W(t)) \mid \mathcal{F}(s)]$ depends only on $W(s)$, confirming that $W(t)$ is a Markov process. $\square$

Definition 13 (First Passage Time). *Shreve (2004) Let $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion and fix a real level m . The first passage time to m is*

$$\tau_m = \inf\{t \geq 0 : W(t) = m\},$$

with the convention $\tau_m = \infty$ if $W(t)$ never hits m .

3.3 Mathematical Foundations of Derivative Pricing

When it comes to pricing formulation, we will utilize mathematical concepts such as stochastic processes, stopping times, and risk-neutral measure. These tools form the foundation of modern financial mathematics and allow us to derive fair values for various financial derivatives. The use of stopped processes helps in modeling contingent claims where timing matters, while the risk-neutral measure enables simplification of pricing by treating all assets as if they grow at the risk-free rate. The theorems and examples provided serve to establish a rigorous basis for option pricing models, including equivalence results between American and European options under certain conditions.

Definition 14 (Stopped Process). *Shreve (2004) If $\{Z(t)\}$ is any stochastic process and τ a stopping time, the stopped process at τ is*

$$Z(t \wedge \tau),$$

where $t \wedge \tau = \min\{t, \tau\}$. In particular, if $Z(t) = \exp(\sigma W(t) - \frac{1}{2}\sigma^2 t)$, then $Z(t \wedge \tau_m)$ remains a martingale under $\mathbb{P}$.

Definition 15 (Risk-neutral measure). *A risk-neutral measure is a probability measure under which the present value of all future cash flows is calculated by discounting at the risk-free rate, regardless of the actual risk or uncertainty of those cash flows.*

Example 9. *Consider a simple financial asset with a 1-year maturity. If the asset has an expected payoff of 110 with a current price of 100, then under the risk-neutral measure, the price is the discounted value of the expected payoff. Assuming the risk-free rate is 5% the price of the asset would be:*

$$\text{Price} = \frac{110}{(1 + 0.05)} = 104.76$$

Theorem 4 (Equivalence of American and European Calls for Non-Dividend-Paying Stocks). *For a non-dividend-paying stock with strike price K and maturity T , the value of an American call option equals that of the corresponding European call option:*

$$C^{\text{Am}}(S_t, t) = C^{\text{Eu}}(S_t, t) \quad \text{for all } 0 \leq t < T.$$

Proof. The key idea is to show that it is never optimal to exercise the American call early. Suppose there exists a time $t < T$ where $S(t) - K > 0$ for the first time and the holder considers exercising immediately. Doing so would yield the immediate payoff

$$S(t) - K.$$

Instead, consider waiting a small amount of time Δt . If the holder then decides to exercise at $t + \Delta t$, the expected *discounted* payoff in the risk-neutral world is

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\Delta t} (S(t + \Delta t) - K)^+],$$

where $(x)^+ = \max\{x, 0\}$ and r is the risk-free rate.

Because $(S(t + \Delta t) - K)^+ \geq S(t + \Delta t) - K$, it follows that

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\Delta t} (S(t + \Delta t) - K)^+] \geq e^{-r\Delta t} \mathbb{E}^{\mathbb{Q}}[S(t + \Delta t) - K] = e^{-r\Delta t} (\mathbb{E}^{\mathbb{Q}}[S(t + \Delta t)] - K).$$

Under the risk-neutral measure, the expected stock price $\mathbb{E}^{\mathbb{Q}}[S(t + \Delta t)]$ equals $S(t)e^{r\Delta t}$ in the continuous-compounding setting. Hence,

$$e^{-r\Delta t} (\mathbb{E}^{\mathbb{Q}}[S(t + \Delta t)] - K) = e^{-r\Delta t} (S(t)e^{r\Delta t} - K) = S(t) - K.$$

Thus, the expected discounted payoff from waiting is *at least* $S(t) - K$, which is the immediate exercise payoff. Therefore, exercising early provides no advantage when the underlying stock pays no dividends.

Since an American call is never exercised early in this scenario, its value must coincide with that of the European call:

$$C^{\text{Am}}(S_t, t) = C^{\text{Eu}}(S_t, t).$$

□

Source: <https://www.cs.rpi.edu/~magdon/courses/cf/notes/american.pdf>

3.4 Optimal Stopping

This section introduces the mathematical foundations of optimal stopping theory and stochastic control, essential for valuing American-style derivatives and solving dynamic decision-making problems in finance. Key concepts include stopping times, controlled stochastic processes, and the dynamic programming principle (DPP), which provides a recursive framework for evaluating optimal strategies. We also explore the risk-neutral valuation of options, the role of martingales, and the Snell envelope in determining optimal exercise policies. Concepts such as put-call parity, smooth pasting conditions, and delta hedging are discussed to illustrate practical applications in pricing and managing financial derivatives.

Definition 16 (Stopping Time). *Campbell (2024)*

A stopping time τ is a non-negative random variable taking values in $\mathbb{R}_+ \cup \{\infty\}$, defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, such that for every time $t \geq 0$, the event

$$\{\omega \in \Omega : \tau(\omega) \leq t\}$$

belongs to the sigma-algebra $\mathcal{F}_t$. This means that it is possible to determine whether τ has occurred by observing the market information up to time t .

Example 10 (Hitting Time of a Stock Level). Let $S = \{S_t\}_{t \geq 0}$ be a continuous stock price process. Then the first time the stock hits a fixed threshold $a > 0$, defined by

$$\tau_a = \inf\{t \geq 0 : S_t = a\},$$

is a stopping time.

Definition 17 (Dynamic Programming Principle for Optimal Stopping). *Campbell (2024)*
The Dynamic Programming Principle (DPP) for optimal stopping problems expresses the recursive structure of the value function $V(t, x)$ as:

$$V(t, x) = \mathbb{E}_{t,x} \left[\int_t^{\tau \wedge \Theta} e^{-r(s-t)} f(X_s) ds + e^{-r(\tau-t)} G(X_\tau) \mathbb{1}_{\tau < \Theta} + e^{-r(\Theta-t)} V(\Theta, X_\Theta) \mathbb{1}_{\tau \geq \Theta} \right],$$

for all checkpoint times Θ in the admissible set of stopping times $\mathcal{T}_{t,T}$. This principle reflects that the optimal value at (t, x) equals the best expected reward gained from stopping either at τ or continuing until Θ .

The components of the equation are defined as follows:

- $V(t, x)$: The value function at time t and state x , representing the maximum expected reward achievable from that point onward.
- τ : The stopping time, indicating when a decision-maker chooses to stop based on the process's evolution.
- Θ : An intermediate checkpoint time satisfying $t \leq \Theta \leq T$, used to split the horizon for recursion.
- $f(X_s)$: The running reward earned continuously as long as the process continues.
- $G(X_\tau)$: The terminal reward received upon stopping at time τ .
- $\mathbb{1}_{\tau < \Theta}$: An indicator function equal to 1 if stopping occurs before Θ .
- $\mathbb{1}_{\tau \geq \Theta}$: An indicator function equal to 1 if the process continues beyond Θ .

Definition 18 (Minimum of Stopping Times $\tau \wedge \Theta$). *Campbell (2024)* The notation $\tau \wedge \Theta$ refers to the minimum (i.e., the earlier occurrence) of two stopping times τ and Θ , defined as:

$$\tau \wedge \Theta = \begin{cases} \tau, & \text{if } \tau \leq \Theta, \\ \Theta, & \text{if } \Theta \leq \tau. \end{cases}$$

This quantity ensures that the process is evaluated at the earliest time when either the decision to stop is triggered or a predetermined checkpoint is reached.

Definition 19 (Controlled Process). *Campbell (2024)* In stochastic optimal control, the **controlled process** X_t^ν is governed by the stochastic differential equation (SDE)

$$dX_t^\nu = \mu(t, X_t^\nu, \nu_t)dt + \sigma(t, X_t^\nu, \nu_t)dW_t,$$

where:

- X_t^ν : The state process.
- ν_t : The control variable.
- $\mu(t, x, \nu)$: The drift term.
- $\sigma(t, x, \nu)$: The diffusion term.

Definition 20 (Admissible Controls). *Campbell (2024)* A control process ν_t is **admissible** if it belongs to the set

$$A := \left\{ \nu : \mathbb{E} \left[\int_0^T \nu_t^2 dt \right] < \infty, \quad \nu_t \text{ is predictable} \right\}.$$

Predictability here implies that ν_t is based on the information available up to time t ; for instance, continuous or left-continuous adapted processes are inherently predictable, while jump processes require additional considerations.

Definition 21 (Objective Function and Optimization Goal). *Campbell (2024)* The performance of a control ν is quantified by the **objective function**

$$J[\nu] = \mathbb{E} \left[\int_0^T f(t, X_t^\nu, \nu_t) dt + G(X_T^\nu) \right],$$

where:

- $f(t, X_t^\nu, \nu_t)$: The running reward or penalty.
- $G(X_T^\nu)$: The terminal reward or penalty.

The **optimization goal** is to determine the optimal control ν^* such that

$$\nu^* = \arg \max_{\nu \in A} J[\nu].$$

Definition 22 (Dynamic Programming Principle (DPP)). *Campbell (2024)* Define the value function associated with a control ν as

$$U^\nu(t, x) = \mathbb{E}_{t,x} \left[\int_t^T f(s, X_s^\nu, \nu_s) ds + G(X_T^\nu) \right],$$

and the corresponding **utility function** as

$$U(t, x) = \sup_{\nu \in \mathcal{A}_{t,T}} U^\nu(t, x),$$

where $\mathcal{A}_{t,T}$ denotes the set of admissible controls starting at time t . By considering a strategy that is arbitrary over $[t, t+h]$ and optimal thereafter, the DPP states that for any $h \in (0, T-t)$

$$U(t, x) = \sup_{\nu \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[\int_t^{t+h} f(s, X_s^\nu, \nu_s) ds + U(t+h, X_{t+h}^\nu) \right].$$

Definition 23 (Dynamic Programming Equation (DPE)). [Campbell \(2024\)](#) Assume the value function $U(t, x)$ is sufficiently smooth so that Itô's lemma applies. For a fixed control w over $[t, t+h]$ and initial state (t, x) , one has

$$U(t+h, X_{t+h}^w) = U(t, x) + \int_t^{t+h} \left(\partial_t U(s, X_s^w) + \mathcal{L}^w U(s, X_s^w) \right) ds + \int_t^{t+h} \sigma(s, X_s^w, w_s) \partial_x U(s, X_s^w) dW_s,$$

with the infinitesimal generator defined as

$$\mathcal{L}^w = \mu(t, x, w) \partial_x + \frac{1}{2} \sigma(t, x, w)^2 \partial_{xx}.$$

By taking expectations (where the stochastic integral vanishes) and applying the DPP, the **Dynamic Programming Equation** is derived as

$$\sup_w \left\{ f(t, x, w) + \mathcal{L}^w U(t, x) \right\} + \partial_t U(t, x) = 0, \quad U(T, x) = G(x).$$

where $G(x)$ is the terminal condition.

The Dynamic Programming Equation (DPE) provides a Partial Differential Equation (PDE) formulation for the value function $U(t, x)$ in optimal control problems. This formulation is pivotal for determining the optimal control strategy by linking the system's dynamics, running and terminal rewards, and the evolution of the utility function.

Definition 24 (European Call Option). **European Call Option** is a type of financial contract that gives the owner the right, but not the obligation, to buy a stock at a fixed price, called the **strike price** K , on a specific future date, known as the **expiration date**. The buyer of the option benefits if the stock price S at expiration is higher than the strike price, as they can buy the stock at K and sell it at S for a profit. Unlike American options, a European call option can only be exercised at the expiration date, not before.

Definition 25 (Put-Call Parity). [Shreve \(2004\)](#) Let $C(t)$ and $P(t)$ denote the prices at time t of a European call and put option, respectively, with the same strike price K , maturity T , and underlying asset price $S(t)$. Let $r \geq 0$ be the continuously compounded risk-free interest rate. Then the put-call parity relation is given by:

$$C(t) - P(t) = S(t) - Ke^{-r(T-t)}.$$

Definition 26 (American Call Option). ***American Call Option** is a type of financial contract that gives the owner the right, but not the obligation, to buy a stock at a fixed price, called the **strike price** K , at any time up to and including a specific future date, known as the **expiration date**. The buyer of the option benefits if the stock price S at the time of exercise is higher than the strike price, as they can buy the stock at K and sell it at S for a profit. Unlike European options, an American call option can be exercised at any point before or on the expiration date, providing greater flexibility to the holder.*

Definition 27 (American Put Option). ***American Put Option** is a type of financial contract that gives the owner the right, but not the obligation, to sell a stock at a fixed price, called the **strike price** K , at any time up to and including a specific future date, known as the **expiration date**. The buyer of the option benefits if the stock price S at the time of exercise is lower than the strike price, as they can buy the stock at S and sell it at K for a profit. Unlike European options, an American put option can be exercised at any point before or on the expiration date, providing greater flexibility to the holder.*

Definition 28 (Discrete-time Martingale). *A stochastic process $(X_n)_{n \geq 0}$ adapted to a filtration $(\mathcal{F}_n)_{n \geq 0}$ is called a **martingale** if it satisfies the following conditions:*

1. (**Adaptedness**) X_n is $\mathcal{F}_n$ -measurable for all $n \geq 0$.
2. (**Integrability**) $\mathbb{E}[|X_n|] < \infty$ for all $n \geq 0$.
3. (**Martingale property**) $\mathbb{E}[X_{n+1} | \mathcal{F}_n] = X_n$ for all $n \geq 0$.

Definition 29 (Optimal Stopping Problem). *Given a stochastic process $\{X_t\}_{t \geq 0}$ defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, the optimal stopping problem is to find a stopping time τ^* that maximizes the expected reward:*

$$\tau^* = \arg \max_{\tau \in \mathcal{T}} \mathbb{E}[g(X_\tau)],$$

where $\mathcal{T}$ is the set of all stopping times with respect to $\{\mathcal{F}_t\}$, and g is a given reward function.

Example 11 (Coin Tossing Game). *Suppose you observe a sequence of i.i.d. coin tosses $X_1, X_2, \dots, X_n$ where $X_i = 1$ for heads and $X_i = 0$ for tails. At each time $t = 1, \dots, n$, you may choose to stop and receive a reward equal to the value of X_t . The goal is to choose a stopping time τ that maximizes the expected reward $\mathbb{E}[X_\tau]$.*

Since the coin is fair, $\mathbb{E}[X_t] = 0.5$ for all t , and thus any stopping time $\tau \in \{1, \dots, n\}$ yields the same expected reward. Therefore, all stopping rules are optimal in this case.

Definition 30 (Risk-Neutral Pricing). *Risk-neutral pricing is a method of valuing financial derivatives by assuming that all investors are indifferent to risk. Under the risk-neutral measure $\mathbb{Q}$, the price of a derivative is given by the discounted expected payoff:*

$$\text{Price at time } t = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} \cdot \text{Payoff} \mid \mathcal{F}_t \right],$$

where r is the risk-free interest rate and T is the maturity date.

Example 12 (European Call Option). *The risk-neutral price of a European call option with strike K and maturity T on a stock S_t is:*

$$C(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} \left[(S_T - K)^+ \mid \mathcal{F}_t \right].$$

Definition 31 (Options Hedging). *Options hedging is the practice of reducing the risk of an options position by taking offsetting positions in the underlying asset or other derivatives. A common method is delta hedging, where one adjusts the position in the underlying asset to neutralize small price movements.*

Example 13 (Delta Hedging). *For a European call option priced by the Black-Scholes model, the delta $\Delta = \frac{\partial C}{\partial S}$ tells how many units of the stock to hold to hedge one short option position. The portfolio is rebalanced continuously to maintain a delta-neutral position.*

Definition 32 (Snell Envelope). *Lamberton (2009) The Snell envelope of a stochastic process $\{X_t\}_{t \geq 0}$ is the smallest supermartingale $\{Y_t\}$ such that $Y_t \geq X_t$ for all t . It characterizes the value process in optimal stopping problems:*

$$Y_t = \sup_{\tau \geq t} \mathbb{E}[X_\tau \mid \mathcal{F}_t],$$

where the supremum is over all stopping times $\tau \geq t$.

Example 14 (American Option Valuation). *The Snell envelope of the discounted payoff process $e^{-rt}(S_t - K)^+$ gives the value of an American call option. The option should be exercised when the payoff equals the Snell envelope, which determines the optimal stopping time.*

Definition 33 (Smooth Pasting Condition). *In optimal stopping problems, the smooth pasting condition requires that the value function $V(x)$ and its first derivative match the payoff function $g(x)$ at the optimal stopping boundary x^* :*

$$V(x^*) = g(x^*), \quad V'(x^*) = g'(x^*).$$

Example 15. *Consider the perpetual American call option with payoff $g(x) = (x - K)^+$. Let $V(x)$ be the value function. The smooth pasting condition at the optimal exercise boundary x^* requires:*

$$V(x^*) = x^* - K, \quad V'(x^*) = 1.$$

These conditions help determine x^ and solve for $V(x)$.*

Example 16. *Consider a fair coin-tossing game where a player starts with an initial fortune $X_0 = 0$. At each time step n , they win 1 dollar with probability 0.5 and lose 1 dollar with probability 0.5. Define the stochastic process $(X_n)_{n \geq 0}$ by*

$$X_n = X_{n-1} + Y_n, \quad \text{where } Y_n \text{ is } \pm 1 \text{ with equal probability.}$$

The process (X_n) is a martingale because:

1. *It is adapted to the natural filtration $(\mathcal{F}_n)$, which consists of all information up to time n .*

2. It has finite expectation since $\mathbb{E}[|X_n|] < \infty$.

3. The martingale property holds:

$$\mathbb{E}[X_{n+1} | \mathcal{F}_n] = \mathbb{E}[X_n + Y_{n+1} | \mathcal{F}_n] = X_n + \mathbb{E}[Y_{n+1} | \mathcal{F}_n] = X_n.$$

Thus, (X_n) is a martingale.

3.5 Black-Scholes Model

Under the Black-Scholes model:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

The price of a derivative with payoff $F(S_T)$ is given by:

$$P_t = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [F(S_T) | \mathcal{F}_t].$$

For call and put options:

$$\begin{aligned} P_t^{\text{Call}} &= S_t \Phi(d_t^+) - K e^{-r(T-t)} \Phi(d_t^-), \\ P_t^{\text{Put}} &= K e^{-r(T-t)} \Phi(-d_t^-) - S_t \Phi(-d_t^+), \end{aligned}$$

where:

$$d_t^{\pm} = \frac{\ln(S_t/K) + (r \pm \frac{\sigma^2}{2})(T-t)}{\sigma \sqrt{T-t}}.$$

Proof. The Black-Scholes Partial Differential Equation (PDE) for a derivative price $P(t, S_t)$ is:

$$\frac{\partial P}{\partial t} + rS_t \frac{\partial P}{\partial S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} - rP = 0,$$

with terminal condition $P(T, S_T) = F(S_T)$, where F is the payoff function.

Let $\tau = T - t$ and $x = \ln(S_t)$. Define a transformed function $u(\tau, x)$ related to P through

$$P(t, S_t) = e^{\alpha x + \beta \tau} u(\tau, x),$$

where constants α, β are chosen to eliminate first-order terms.

Substituting into the Black-Scholes PDE reduces it to the standard heat equation:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

Using known methods (e.g., Fourier transform or convolution with the heat kernel), solve for $u(\tau, x)$, then back-substitute to obtain $P(t, S_t)$.

Final solution: For a European call or put option with strike K , the closed-form solution is:

$$P_t^{\text{Call}} = S_t \Phi(d_t^+) - K e^{-r(T-t)} \Phi(d_t^-),$$

$$P_t^{\text{Put}} = Ke^{-r(T-t)}\Phi(-d_t^-) - S_t\Phi(-d_t^+),$$

where

$$d_t^\pm = \frac{\ln(S_t/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}},$$

and $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. □

Example 17 (Black-Scholes Model Calculation). *Given a stock price $S = 100$, strike price $K = 105$, time to expiration $T = 1$ year, risk-free rate $r = 5\%$, and volatility $\sigma = 20\%$, we compute the option price using the Black-Scholes formula.*

First, calculate the parameters:

$$\begin{aligned} d_1 &= \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\ln(100/105) + (0.05 + 0.02)}{0.2} \\ &= \frac{-0.0488 + 0.07}{0.2} = \frac{0.0212}{0.2} = 0.106, \\ d_2 &= d_1 - \sigma\sqrt{T} = 0.106 - 0.2 = -0.094. \end{aligned}$$

Using standard normal values:

$$N(d_1) \approx 0.5423, \quad N(d_2) \approx 0.4625.$$

Apply the Black-Scholes formula:

$$\begin{aligned} C &= SN(d_1) - Ke^{-rT}N(d_2) \\ &= 100 \cdot 0.5423 - 105 \cdot e^{-0.05} \cdot 0.4625 \\ &= 54.23 - 105 \cdot 0.9512 \cdot 0.4625 \\ &= 54.23 - 105 \cdot 0.4396 = 54.23 - 46.16 = 8.07. \end{aligned}$$

Therefore, the price of the European call option is approximately \$8.07.

In practice, the volatility σ used in the Black-Scholes model often differs from the implied volatility observed in the market, leading to discrepancies between theoretical and actual option prices.

Chapter 4

American Options Pricing

4.1 Introduction to Options

Options are financial derivatives that give the holder the right, but not the obligation, to buy (call option) or sell (put option) an underlying asset at a predetermined price (the strike price) before or at a specific date (the expiration or maturity date). European options can only be exercised at maturity, while American options allow for early exercise at any time before expiration.

4.1.1 Option Payoffs

The fundamental value proposition of options lies in their terminal payoff structure. Figure 4.1 illustrates the payoff functions for both call and put options at maturity.

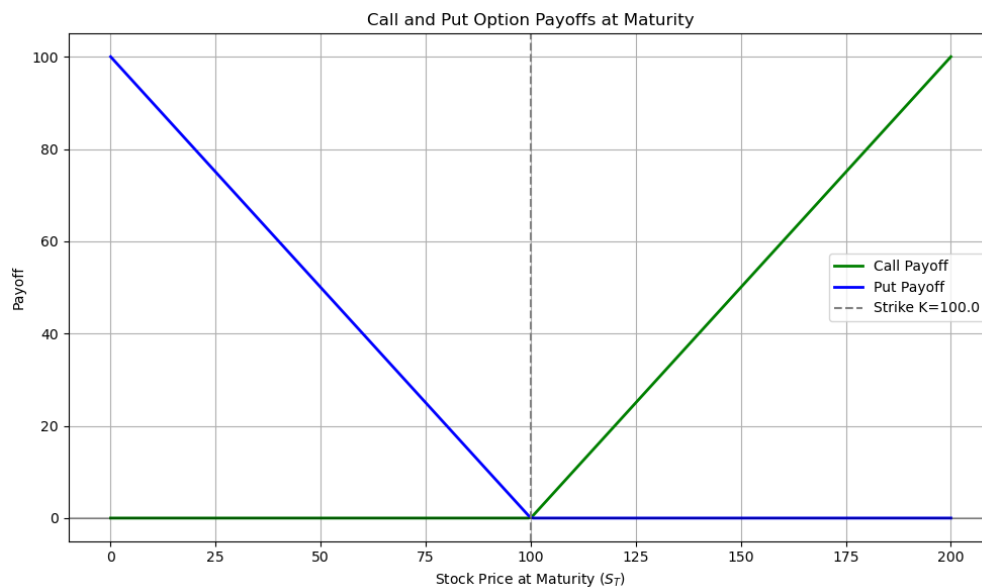


Figure 4.1: Call and Put Option Payoffs at Maturity with Strike Price $K = 100$. The call option (green) provides a positive payoff when the stock price exceeds the strike price, while the put option (blue) generates value when the stock price falls below the strike.

Mathematically, for a strike price K , the payoff functions at maturity for call and put options are:

$$\text{Call Payoff} = \max(S_T - K, 0) \quad (4.1)$$

$$\text{Put Payoff} = \max(K - S_T, 0) \quad (4.2)$$

where S_T represents the price of the underlying asset at maturity T .

For example, with a strike price $K = 100$, a call option would yield a payoff of \$20 if the underlying asset price at maturity is $S_T = 120$, while it would be worthless if $S_T = 80$. Conversely, a put option with the same strike would yield \$20 if $S_T = 80$ but would be worthless if $S_T = 120$.

4.2 Option Valuation Fundamentals

4.2.1 Intrinsic and Time Value

The pricing of options involves two key components: intrinsic value and time value. The intrinsic value represents the payoff if the option were exercised immediately, while the time value accounts for the potential of the option to increase in value before expiration.

Figure 4.2 illustrates the relationship between the intrinsic value and the theoretical price of a European put option as calculated using the Black-Scholes formula with zero interest rate.

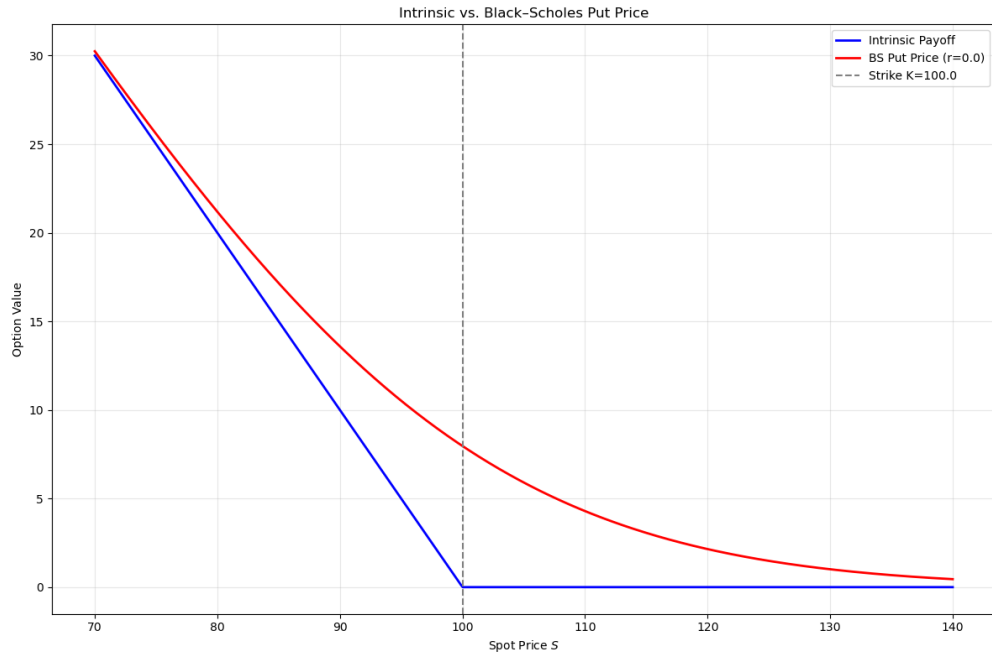


Figure 4.2: Comparison of Intrinsic Payoff and Black-Scholes Put Price with Strike $K = 100$ and $r = 0$. The difference between the Black-Scholes price (red) and the intrinsic value (blue) represents the time value of the option.

For a European put option, the Black-Scholes price exceeds the intrinsic value $\max(K - S, 0)$ at all stock prices except at deep in-the-money and deep out-of-the-money regions. This difference represents the time value of the option, which reflects the probability that the option will become more valuable before expiration due to the underlying asset's price movements.

4.2.2 European vs. American Options

The fundamental distinction between European and American options lies in their exercise flexibility. While European options can only be exercised at maturity, American options permit exercise at any point during their lifetime. This seemingly simple difference creates significant complexity in the valuation process.

For European options, the Black-Scholes-Merton model provides a closed-form analytical solution, calculated as the discounted expected payoff under the risk-neutral measure:

$$V_0^E = e^{-rT} \mathbb{E}^{\mathbb{Q}}[h(S_T)] \quad (4.3)$$

where $h(S_T)$ represents the terminal payoff function:

$$h(S_T) = \begin{cases} \max(S_T - K, 0) & \text{for call options} \\ \max(K - S_T, 0) & \text{for put options} \end{cases} \quad (4.4)$$

and $\mathbb{Q}$ denotes the risk-neutral probability measure.

American options, however, require solving an optimal stopping problem, where the holder must determine not only whether to exercise, but when. This is mathematically expressed as:

$$V_0^A = \sup_{\tau \in \mathcal{T}_{0,T}} \mathbb{E}^{\mathbb{Q}}[e^{-r\tau} h(S_\tau)] \quad (4.5)$$

where $\mathcal{T}_{0,T}$ represents the set of all stopping times with values in $[0, T]$.

This formulation leads to the fundamental constraint for American options:

$$V_t \geq h(S_t) \quad \text{for all } t \in [0, T] \quad (4.6)$$

For put options specifically, this becomes:

$$V_t \geq \max(K - S_t, 0) \quad \text{for all } t \in [0, T] \quad (4.7)$$

This constraint, absent in European option pricing, creates a free boundary problem that generally precludes closed-form solutions and necessitates numerical approximation methods.

4.3 The Early Exercise Premium

The early exercise premium constitutes the additional economic value inherent in American options relative to their European counterparts. This premium quantifies the financial benefit derived from the optionality of exercising prior to maturity, representing one of the fundamental valuation challenges in derivatives pricing theory.

4.3.1 Effect of Interest Rates on Option Pricing

Interest rates exert a significant influence on early exercise decisions, particularly for put options where the temporal value of money directly impacts optimal exercise strategies. Figures 4.3a and 4.3b illustrate the relationship between intrinsic value and theoretical European put prices under the Black-Scholes framework across varying interest rate regimes.

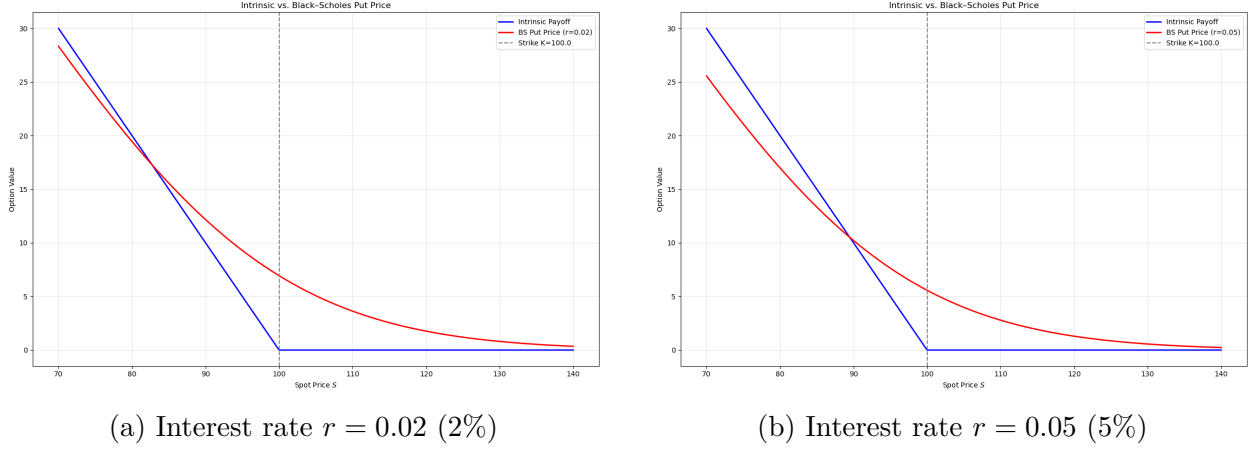


Figure 4.3: Comparative analysis of intrinsic payoff and Black-Scholes put price with strike $K = 100$ under different interest rate environments. The increasing divergence between curves at higher interest rates demonstrates the economic incentive for early exercise.

The economic rationale underlying these observations stems from the opportunity cost of capital: as interest rates rise, the present value of the future strike price payment diminishes, increasing the relative value of receiving the strike early. Consequently, for an American put, higher interest rates strengthen the incentive to exercise early and reinvest the proceeds at the prevailing interest rate, rather than deferring exercise until maturity.

4.3.2 The Free Boundary Problem

The valuation of American options can be formally expressed as a free boundary problem within the framework of stochastic calculus and partial differential equations. Under the risk-neutral measure in the Black-Scholes framework, the option price $V(S, t)$ must satisfy the Black-Scholes partial differential equation within the continuation region:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q)S \frac{\partial V}{\partial S} - rV = 0 \quad (4.8)$$

subject to the boundary constraints:

$$V(S, t) \geq h(S) \quad \forall S, t \quad (4.9)$$

$$V(S, T) = h(S) \quad \forall S \quad (4.10)$$

where $h(S) = \max(K - S, 0)$ for put options, σ represents the volatility parameter, r denotes the risk-free rate, and q signifies the continuous dividend yield.

The distinguishing characteristic of this problem is the free boundary $S^*(t)$ —the early exercise boundary—which delineates the continuation region from the exercise region. For put options, this boundary satisfies the conditions:

$$V(S, t) > \max(K - S, 0) \quad \text{for } S > S^*(t) \quad (\text{Continuation Region}) \quad (4.11)$$

$$V(S, t) = \max(K - S, 0) \quad \text{for } S \leq S^*(t) \quad (\text{Exercise Region}) \quad (4.12)$$

Furthermore, to ensure mathematical consistency at the boundary, the smooth pasting condition must be satisfied:

$$\frac{\partial V}{\partial S}(S^*(t), t) = \frac{\partial h}{\partial S}(S^*(t)) \quad (4.13)$$

This free boundary is endogenous to the problem and must be determined contemporaneously with the option price, which generally precludes closed-form analytical solutions except under specific constraints. Determining this boundary is mathematically equivalent to solving the optimal stopping problem for the American option holder.

The intrinsic complexity of this free boundary problem necessitates sophisticated numerical approaches for accurate pricing. The subsequent chapters present three principal methodologies for valuing American options: the Binomial Tree method (Section 4.5), Monte Carlo methods (Section 4.6), and the Least Squares Monte Carlo method (Section 4.7). The empirical quantification of the early exercise premium using these methods, particularly with Laguerre polynomial basis functions, will be examined in detail in Chapter 11, where we analyze the premium's magnitude, stability, and statistical properties across various parameter configurations.

4.4 Perpetual American Options: A Special Case

Perpetual American options, which have no expiration date ($T \rightarrow \infty$), represent a special case that provides valuable theoretical insights into the free boundary problem. Unlike their finite-maturity counterparts, perpetual American options admit closed-form solutions, as demonstrated by [Karatzas & Shreve \(1998\)](#) and [Peskir & Shiryaev \(2006\)](#).

4.4.1 Problem Formulation

For a perpetual American put option with strike price K , the pricing problem can be formulated as an infinite-horizon optimal stopping problem:

$$V(t, x) = \sup_{\tau \in \mathcal{T}_{t, \infty}} \mathbb{E}_{t, x}^{\mathbb{Q}} \left[e^{-r(\tau - t)} (K - S_{\tau})^+ \right] \quad (4.14)$$

where under the risk-neutral measure $\mathbb{Q}$, the stock price follows a geometric Brownian motion:

$$dS_u = rS_u du + \sigma S_u dW_u^{\mathbb{Q}}, \quad S_t = x \quad (4.15)$$

A key observation, as noted by [Shiryaev \(1978\)](#), is that this problem possesses time-homogeneity. Since the time horizon is infinite and the coefficients in the stochastic differential equation are time-independent, the value function depends only on the current stock price, not on time. This time-invariance property simplifies the problem considerably, reducing it to:

$$\hat{V}(x) = \sup_{\tau \in \mathcal{T}_{0,\infty}} \mathbb{E}_{0,x}^{\mathbb{Q}} \left[e^{-r\tau} (K - S_\tau)^+ \right] \quad (4.16)$$

4.4.2 Analytical Solution Approach

For the perpetual American put option, the free boundary problem simplifies significantly compared to the finite-maturity case. By applying Itô's formula and the strong Markov property, one can derive that the value function $\hat{V}(x)$ must satisfy the following ordinary differential equation in the continuation region ([Campbell, 2024](#)):

$$\mathcal{L}\hat{V}(x) - r\hat{V}(x) = 0 \quad (4.17)$$

where $\mathcal{L}$ is the infinitesimal generator of the diffusion process:

$$\mathcal{L} = rx \frac{d}{dx} + \frac{1}{2} \sigma^2 x^2 \frac{d^2}{dx^2} \quad (4.18)$$

This yields the second-order ODE:

$$r\hat{V}(x) - rx \frac{d\hat{V}}{dx}(x) - \frac{\sigma^2}{2} x^2 \frac{d^2\hat{V}}{dx^2}(x) = 0 \quad (4.19)$$

The structure of the continuation and stopping regions for the perpetual American put follows a threshold policy. Through convexity arguments and the monotonicity properties of the value function, it can be proven that there exists a unique critical stock price b^* such that:

$$\mathcal{C} = \{x : x > b^*\} \quad (\text{Continuation Region}) \quad (4.20)$$

$$\mathcal{S} = \{x : x \leq b^*\} \quad (\text{Stopping Region}) \quad (4.21)$$

To solve this homogeneous second-order ODE, we employ a power function ansatz:

$$\hat{V}(x) = Ax^\gamma, \quad x > b^* \quad (4.22)$$

Substituting this ansatz into the ODE yields:

$$\left(r - r\gamma - \frac{\sigma^2}{2} \gamma(\gamma - 1) \right) Ax^\gamma = 0 \quad (4.23)$$

For non-trivial solutions, the characteristic equation must be satisfied:

$$\frac{\sigma^2}{2} \gamma^2 + \left(r - \frac{\sigma^2}{2} \right) \gamma - r = 0 \quad (4.24)$$

This quadratic equation has roots:

$$\gamma_1 = 1 \quad (4.25)$$

$$\gamma_2 = -\frac{2r}{\sigma^2} \quad (4.26)$$

The general solution is therefore:

$$\hat{V}(x) = A_1 x^{\gamma_1} + A_2 x^{\gamma_2} = A_1 x + A_2 x^{-\frac{2r}{\sigma^2}} \quad (4.27)$$

Since $\hat{V}(x)$ must remain bounded as $x \rightarrow \infty$, we must set $A_1 = 0$, leaving us with:

$$\hat{V}(x) = A_2 x^{-\frac{2r}{\sigma^2}}, \quad x > b^* \quad (4.28)$$

For notational simplicity, we denote A_2 as A .

4.4.3 Determining the Free Boundary

To completely characterize the solution, we must determine both the constant A and the critical threshold b^* . Following the theory of optimal stopping, these parameters are obtained by imposing the value-matching and smooth-pasting conditions at the boundary, as formalized by [van Moerbeke \(1976\)](#) and [Peskir & Shiryaev \(2006\)](#):

$$\hat{V}(b^*) = K - b^* \quad (\text{Value Matching}) \quad (4.29)$$

$$\left. \frac{d\hat{V}}{dx} \right|_{x=b^*} = -1 \quad (\text{Smooth Pasting}) \quad (4.30)$$

The value-matching condition ensures continuity of the value function across the boundary, while the smooth-pasting condition (also known as high-contact or smooth-fit) ensures differentiability at the boundary.

Substituting our ansatz into these conditions yields the system:

$$A(b^*)^{-\frac{2r}{\sigma^2}} = K - b^* \quad (4.31)$$

$$-\frac{2r}{\sigma^2} A(b^*)^{-\frac{2r}{\sigma^2}-1} = -1 \quad (4.32)$$

From the second equation, we obtain:

$$A = \frac{\sigma^2}{2r} (b^*)^{\frac{2r}{\sigma^2}+1} \quad (4.33)$$

Substituting this into the first equation and solving for b^* :

$$\frac{\sigma^2}{2r} (b^*)^{\frac{2r}{\sigma^2}+1} \cdot (b^*)^{-\frac{2r}{\sigma^2}} = K - b^* \quad (4.34)$$

Simplifying:

$$\frac{\sigma^2}{2r}b^* = K - b^* \quad (4.35)$$

Which yields:

$$b^* = \frac{2Kr}{\sigma^2 + 2r} \quad (4.36)$$

Having determined b^* , we can now express the constant A as:

$$A = (K - b^*)(b^*)^{\frac{2r}{\sigma^2}} = \frac{\sigma^2}{2r} \cdot K \cdot \left(\frac{2r}{\sigma^2 + 2r} \right)^{1 + \frac{2r}{\sigma^2}} \quad (4.37)$$

4.4.4 Closed-Form Solution

Thus, the closed-form solution for the perpetual American put option value is:

$$\hat{V}(x) = \begin{cases} Ax^{-\frac{2r}{\sigma^2}}, & x > b^* \\ K - x, & x \leq b^* \end{cases} \quad (4.38)$$

where $b^* = \frac{2Kr}{\sigma^2 + 2r}$ and $A = \frac{\sigma^2}{2r}(b^*)^{1 + \frac{2r}{\sigma^2}}$.

This solution has several important implications, as noted by [Karatzas \(1988\)](#) and [Davis & Karatzas \(1994\)](#):

1. The optimal stopping time is given by the first passage time of the stock price to the boundary b^* :

$$\tau^* = \inf\{t \geq 0 : S_t \leq b^*\} \quad (4.39)$$

2. The critical threshold b^* is strictly less than the strike price K , implying that it is never optimal to exercise when the option is at-the-money or out-of-the-money. Specifically, $b^* < K$ since $\frac{2r}{\sigma^2 + 2r} < 1$ for all $r, \sigma > 0$.

3. The comparative statics of the threshold b^* reveal that:

$$\frac{\partial b^*}{\partial r} = \frac{2K\sigma^2}{(\sigma^2 + 2r)^2} > 0 \quad (4.40)$$

$$\frac{\partial b^*}{\partial \sigma} = -\frac{4Kr\sigma}{(\sigma^2 + 2r)^2} < 0 \quad (4.41)$$

Thus, b^* increases with interest rate r and decreases with volatility σ , reflecting the economic incentives for early exercise: higher interest rates increase the opportunity cost of waiting, while higher volatility increases the potential upside from continuation.

4.4.5 Theoretical Significance

The perpetual American option case demonstrates that the free boundary problem becomes tractable when time-dependence is eliminated. As observed by [Karatzas \(1988\)](#), this special case provides insight into the limiting behavior of finite-maturity American options as time to maturity increases.

Lamberton (2009) notes that this closed-form solution serves as an important benchmark for validating numerical methods, as any approach for pricing American options should correctly converge to the perpetual solution as the time to maturity increases. Additionally, Bismut & Skalli (1971) highlight the connection between this solution and the broader theory of optimal stopping for Markov processes.

The perpetual case also illustrates the fundamental role of the smooth-pasting condition in optimal stopping problems. This principle, sometimes referred to as the "high-contact" condition, was rigorously justified by Peskir & Shiryaev (2006) as a necessary optimality condition derived from the variational inequalities associated with the free boundary problem.

While perpetual options have limited practical application, their analytical tractability offers fundamental insights into the structure of optimal exercise policies and the valuation of American-style derivatives. The explicit characterization of the exercise boundary as a constant threshold demonstrates how the complexity of the time-dependent free boundary problem reduces to a simpler form in the perpetual case.

The contrast with the time-dependent case underscores the necessity of numerical methods for finite-maturity American options, which we explore in the following sections.

4.5 The Binomial Tree Method

The binomial tree method, pioneered by Cox *et al.* (1979), constitutes a discrete-time numerical framework for option valuation, with particular efficacy for American-style derivatives. This approach merits consideration due to its conceptual transparency, computational tractability, and capacity to accommodate the early exercise optionality inherent in American contracts.

4.5.1 Theoretical Foundation

The fundamental premise of the binomial model involves representing the stochastic evolution of the underlying asset price through a recombining lattice structure. At each discrete time node, the asset price undergoes either an upward or downward movement with corresponding risk-neutral probabilities. Within the canonical Cox-Ross-Rubinstein (CRR) formulation, the multiplicative factors governing these price transitions, denoted by u and d respectively, are parameterized as follows:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} \quad (4.42)$$

where $\sigma \in \mathbb{R}^+$ represents the annualized volatility of the underlying asset, and $\Delta t \in \mathbb{R}^+$ denotes the temporal discretization interval.

The risk-neutral probabilities associated with the respective upward and downward movements are derived through the arbitrage-free pricing principle:

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \quad 1 - p = \frac{u - e^{(r-q)\Delta t}}{u - d} \quad (4.43)$$

where $r \in \mathbb{R}$ is the risk-free interest rate, and $q \in \mathbb{R}^+$ represents the continuous dividend yield of the underlying asset.

Figure 4.4 below illustrates a 3-step binomial tree for an American put option, highlighting the node structure, asset prices, and the transition dynamics:

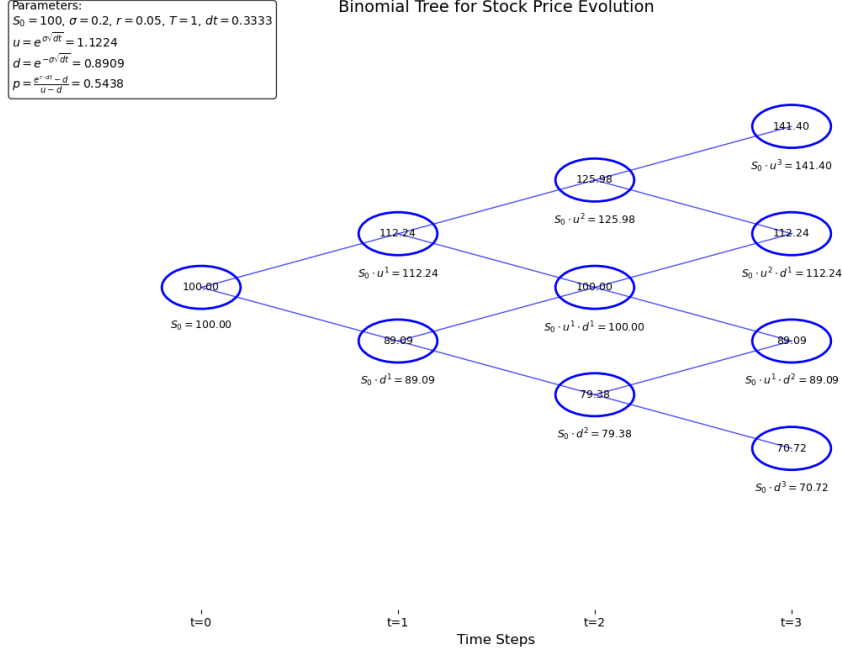


Figure 4.4: 3-Step Binomial Tree for American Option

4.5.2 Recursive Valuation Procedure

Let us examine, without loss of generality, the pricing of an American put option with strike price $K \in \mathbb{R}^+$. The valuation follows the principle of dynamic programming, which enables us to decompose this complex decision problem into a sequence of simpler subproblems through backward recursion.

The terminal condition at maturity (time T) for each node (N, j) in the lattice is characterized by the payoff function:

$$V_{N,j} = \max(K - S_{N,j}, 0) \quad (4.44)$$

where $S_{N,j}$ represents the asset price at the terminal node (N, j) , computed as $S_{N,j} = S_0 u^j d^{N-j}$.

At each preceding node (i, j) where $i \in \{0, 1, \dots, N-1\}$ and $j \in \{0, 1, \dots, i\}$, the option value is determined through backward induction by evaluating the optimal decision between immediate exercise and continuation. This optimal stopping problem can be formally expressed via the Bellman equation:

$$V_{i,j} = \max \left\{ \max(K - S_{i,j}, 0), e^{-r\Delta t} (pV_{i+1,j+1} + (1-p)V_{i+1,j}) \right\} \quad (4.45)$$

This equation encapsulates the fundamental trade-off in American option valuation: at each decision point, the option holder must determine whether immediate exercise generates greater value than continued holding.

For analytical clarity, we decompose the valuation at each non-terminal node (i, j) into two components.

The Immediate Exercise Value (Intrinsic Value) represents the payoff derived from exercising the option at the current node, expressed as:

$$EV_{i,j} = \max(K - S_{i,j}, 0) \quad (4.46)$$

The Continuation Value (Holding Value) is the risk-neutral expected discounted value of maintaining the option position, computed as the probability-weighted average of subsequent node values:

$$CV_{i,j} = e^{-r\Delta t} \mathbb{E}^{\mathbb{Q}} [V_{i+1,\cdot} \mid S_{i,j}] = e^{-r\Delta t} [pV_{i+1,j+1} + (1-p)V_{i+1,j}] \quad (4.47)$$

where $\mathbb{E}^{\mathbb{Q}}[\cdot]$ denotes the expectation under the risk-neutral measure $\mathbb{Q}$.

The optimal value at node (i, j) is therefore:

$$V_{i,j} = \max\{EV_{i,j}, CV_{i,j}\} \quad (4.48)$$

This formulation induces a partition of the state space into the continuation region $\mathcal{C}$ and the stopping (exercise) region $\mathcal{S}$:

$$\mathcal{C}_i = \{j : CV_{i,j} > EV_{i,j}\} \quad (4.49)$$

$$\mathcal{S}_i = \{j : CV_{i,j} \leq EV_{i,j}\} \quad (4.50)$$

At each time step i , the collection of indices j for which exercise is optimal forms a discrete approximation of the optimal exercise boundary.

Figure 4.5 illustrates the option value components at different time points for an American put option with strike $K = 100$. The plots visualize the relationship between the option value, continuation value, and intrinsic value across different underlying stock prices, and they clearly demonstrate the optimal exercise boundary.

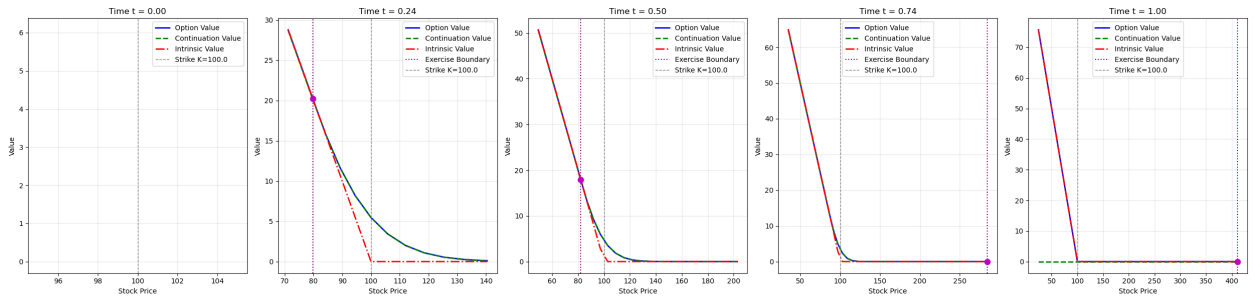


Figure 4.5: Option value components for an American put option with strike $K = 100$ at different time points. The blue solid line represents the option value, the green dashed line shows the continuation value, and the red dash-dot line indicates the intrinsic value. The vertical dotted line marks the exercise boundary, below which early exercise is optimal.

The recursive optimization procedure propagates backward through the lattice structure, culminating in the determination of the option's present value $V_{0,0}$ at the initial node. As $N \rightarrow \infty$, this discrete approximation converges to the continuous-time solution of the free boundary problem associated with American option pricing, as established by [Cox *et al.* \(1979\)](#).

It is worth noting that the binomial tree not only yields the option price but also characterizes the optimal exercise strategy. For each node (i, j) , the optimal action (exercise or continue) can be determined by comparing $EV_{i,j}$ and $CV_{i,j}$, providing a complete state-contingent exercise policy.

4.5.3 Binomial Algorithm

Algorithm 1 Binomial Tree Algorithm for Pricing American Options

Require: Number of steps N , strike price K , risk-free rate r , volatility σ , time to maturity T , initial stock price S_0 , option type (**call** or **put**)

```

1:  $\Delta t \leftarrow T/N$ 
2:  $u \leftarrow e^{\sigma\sqrt{\Delta t}}$ ,  $d \leftarrow 1/u$ 
3:  $p \leftarrow \frac{e^{r\Delta t} - d}{u - d}$  {Risk-neutral probability}
4: Initialize stock prices at maturity:  $S[i] \leftarrow S_0 \cdot u^i \cdot d^{N-i}$  for  $i = 0 \dots N$ 
5: for  $i = 0$  to  $N$  do
6:   if option is call then
7:      $V[i] \leftarrow \max(S[i] - K, 0)$ 
8:   else
9:      $V[i] \leftarrow \max(K - S[i], 0)$ 
10:  end if
11: end for
12: for  $n = N - 1$  down to  $0$  do
13:   for  $i = 0$  to  $n$  do
14:      $S[i] \leftarrow S[i]/u$  {Move one step back in stock price}
15:     Continuation  $\leftarrow e^{-r\Delta t} \cdot (p \cdot V[i + 1] + (1 - p) \cdot V[i])$ 
16:     if option is call then
17:       Exercise  $\leftarrow \max(S[i] - K, 0)$ 
18:     else
19:       Exercise  $\leftarrow \max(K - S[i], 0)$ 
20:     end if
21:      $V[i] \leftarrow \max(\text{Exercise}, \text{Continuation})$ 
22:   end for
23: end for
24: return  $V[0]$  {Option value at initial time}

```

4.6 Monte Carlo Methods

Monte Carlo methods constitute a robust class of numerical algorithms extensively employed in mathematical finance, particularly valuable for derivative pricing where closed-form solutions are either unavailable or computationally intractable. At their core, these methods involve simulating numerous paths of underlying stochastic processes to approximate the expected payoffs of financial instruments.

4.6.1 Asset Price Simulation

The foundation of Monte Carlo methods is the simulation of asset price paths following a stochastic process. For standard option pricing, the underlying asset is typically modeled using geometric Brownian motion (GBM):

$$dS_t = (r - d)S_t dt + \sigma S_t dW_t \quad (4.51)$$

where r is the risk-free rate, d is the dividend yield, σ is the volatility, and W_t is a standard Brownian motion. In the risk-neutral pricing framework, the drift term is set to $r - d$ to ensure arbitrage-free pricing.

For numerical implementation, this continuous-time stochastic differential equation is typically discretized using an Euler-Maruyama scheme or exact sampling. The resulting discrete-time approximation for the asset price at time t_{i+1} given the price at time t_i is:

$$S_{t_{i+1}} = S_{t_i} \exp \left(\left(r - d - \frac{\sigma^2}{2} \right) \Delta t + \sigma \sqrt{\Delta t} Z_i \right) \quad (4.52)$$

where $Z_i \sim \mathcal{N}(0, 1)$ are independent standard normal random variables and $\Delta t = \frac{T}{N}$ with N being the number of time steps.

Figure 4.6 illustrates a sample of simulated GBM paths using the following parameters: initial stock price $S_0 = 100$, risk-free rate $r = 0.03$, volatility $\sigma = 0.2$, time to maturity $T = 1$ year, dividend yield $d = 0$, and strike price $K = 100$. The simulation uses 10,000 paths with 120 time steps, though only 20 representative paths are displayed for clarity.

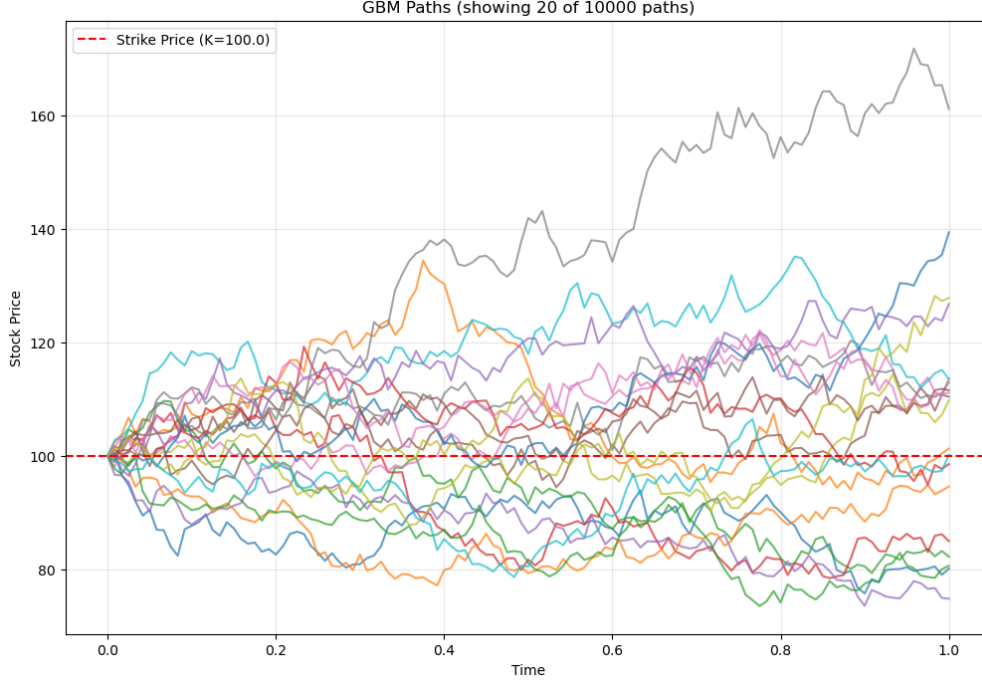


Figure 4.6: Simulated GBM Paths for Stock Price (showing 20 of 10,000 paths) with Strike Price $K = 100$, $S_0 = 100$, $r = 0.03$, $\sigma = 0.2$, $T = 1$ year, and $d = 0$

The red dashed line at $K = 100$ represents the strike price, which is particularly relevant for American options where the decision to exercise early depends on whether the option is in-the-money (i.e., whether $S_t < K$ for puts or $S_t > K$ for calls) at each possible exercise time.

A key theoretical aspect of the GBM model is that the terminal distribution of stock prices follows a log-normal distribution. Figure 4.7 shows the histogram of terminal stock prices at maturity $T = 1$ for all 10,000 simulated paths using the same parameters. The distribution confirms the theoretical expectation, with the log-normal probability density function overlaid in red. The terminal stock price S_T follows:

$$S_T = S_0 \exp \left(\left(r - d - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z \right) \quad (4.53)$$

where $Z \sim \mathcal{N}(0, 1)$. This implies that $\ln(S_T/S_0)$ is normally distributed with mean $(r - d - \sigma^2/2)T$ and variance $\sigma^2 T$.

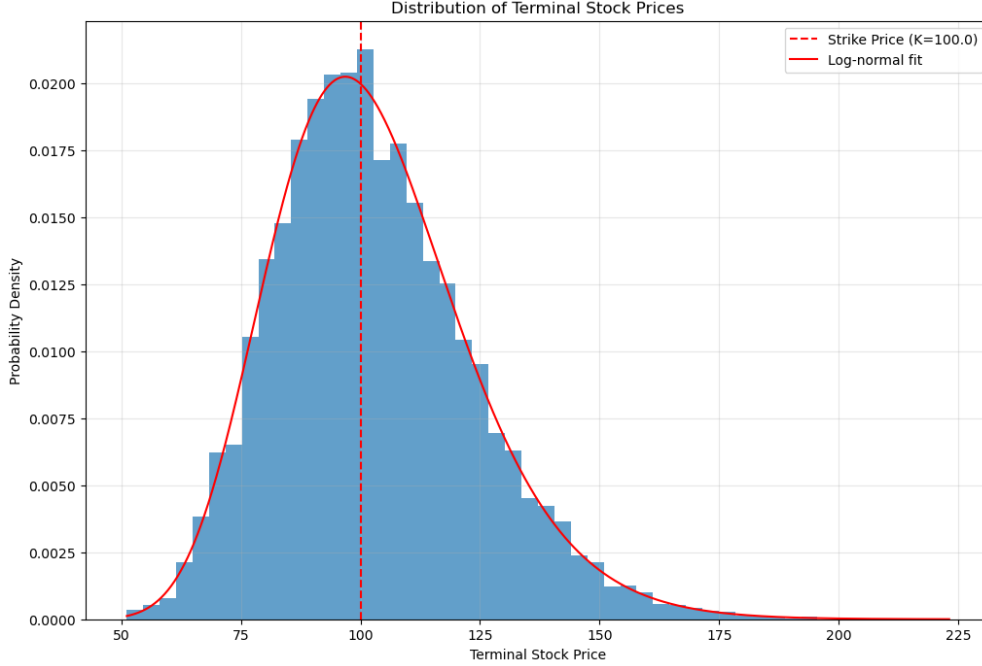


Figure 4.7: Distribution of Terminal Stock Prices at Maturity ($T = 1$) for 10,000 Simulated Paths with Log-Normal Fit. Parameters: $S_0 = 100$, $r = 0.03$, $\sigma = 0.2$, $T = 1$ year, and $d = 0$

For European options, only this terminal distribution matters for pricing, as the option can only be exercised at maturity. The option price is determined by the expected discounted payoff with respect to this distribution. However, for American options, the entire path becomes relevant since early exercise decisions must be made based on the asset price at each potential exercise date.

These simulated paths form the basis for Monte Carlo option pricing. For European options, the Monte Carlo method's implementation is relatively direct. One simulates paths $\{S_t^{(i)}\}_{t=0}^T$ of the underlying asset from the initial time $t = 0$ to the maturity T . The option price is estimated as the discounted expected payoff:

$$V_0 \approx e^{-rT} \frac{1}{n} \sum_{i=1}^n h(S_T^{(i)}), \quad (4.54)$$

where $h(S_T)$ is the payoff function at maturity. According to [Boyle, Broadie & Glasserman \(1997\)](#), as the number of simulations n increases, this estimator converges to the true value with statistical error declining proportionally to $1/\sqrt{n}$.

4.6.2 American Option Pricing

Monte Carlo methods for American options involve additional complexities due to the option's early exercise feature, represented mathematically as an optimal stopping problem. This necessitates determining an optimal exercise strategy or boundary.

Early approaches by [Tilley \(1993\)](#) introduced bundling algorithms, grouping similar paths to estimate continuation values. Subsequent developments by [Broadie & Glasserman \(1997\)](#) presented the stochastic mesh method, efficiently addressing high-dimensional American option problems. Their method constructs a mesh of paths and leverages backward induction to approximate optimal exercise decisions, significantly enhancing computational efficiency.

An important advancement was made by [Rogers \(2002\)](#), who reformulated the American option pricing problem into a dual optimization problem. This yields an upper bound for the option's price, complementary to lower bounds typically derived from suboptimal exercise policies:

$$V_0 = \inf_{M \in \mathcal{M}} \mathbb{E} \left[\max_{0 \leq t \leq T} (h(S_t) - M_t) \right], \quad (4.55)$$

where $\mathcal{M}$ represents the space of martingales with $M_0 = 0$.

4.6.3 Theoretical Framework

The Monte Carlo methodology for American options fundamentally relies on solving an optimal stopping problem, formalized as follows [Karatzas \(1988\)](#); [Peskir & Shiryaev \(2006\)](#):

$$V_0 = \sup_{\tau \in \mathcal{T}} \mathbb{E} \left[e^{-r\tau} h(S_\tau) \right], \quad (4.56)$$

where τ is an admissible stopping time in the set $\mathcal{T}$. Dynamic programming provides the recursive characterization of the solution, facilitating computational implementation:

$$V_t(S_t) = \max \{ h(S_t), C_t(S_t) \}, \quad (4.57)$$

where the continuation value $C_t(S_t)$ is defined as:

$$C_t(S_t) = \mathbb{E}_t \left[e^{-r\Delta t} V_{t+\Delta t}(S_{t+\Delta t}) \right]. \quad (4.58)$$

Monte Carlo implementation of this recursion involves backward induction on simulated paths, necessitating efficient estimation of the continuation value at each timestep. This formulation establishes a theoretical bridge to the subsequent LSM method discussed in Section ??, which offers a particularly efficient approach to this estimation problem.

4.6.4 Monte Carlo Algorithm

Algorithm 2 Standard Monte Carlo Algorithm for American Option Pricing

Require: Number of paths M , time steps N , strike K , risk-free rate r , volatility σ , initial asset price S_0 , time to maturity T , payoff function $h(x) = (K - x)^+$ for puts

- 1: $\Delta t \leftarrow T/N$ {Time step size}
- 2: Initialize $S[m][0] \leftarrow S_0$ for $m = 1 \dots M$ {Initialize all paths}
- 3: Initialize indicator $\text{Active}[m] \leftarrow \text{true}$ for $m = 1 \dots M$ {Track if option is still active}
- 4: Initialize $\text{Value}[m] \leftarrow 0$ for $m = 1 \dots M$ {Store option payoff for each path}
- 5: Initialize $\text{ExerciseTime}[m] \leftarrow -1$ for $m = 1 \dots M$ {Track exercise time}
- {Forward simulation of asset price paths}
- 6: **for** $n = 0$ to $N - 1$ **do**
- 7: **for** $m = 1$ to M **do**
- 8: Generate standard normal random variable $Z \sim \mathcal{N}(0, 1)$
- 9: $S[m][n + 1] \leftarrow S[m][n] \cdot \exp\left((r - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t} \cdot Z\right)$
- 10: **end for**
- 11: **end for**
- {Backward induction for early exercise decisions}
- 12: Calculate terminal payoffs: $\text{Value}[m] \leftarrow h(S[m][N])$ for $m = 1 \dots M$
- 13: Set exercise times for paths with positive terminal payoff: $\text{ExerciseTime}[m] \leftarrow N$ if $\text{Value}[m] > 0$
- 14: **for** $n = N - 1$ down to 0 **do**
- 15: $\text{ITM} \leftarrow \{m \mid h(S[m][n]) > 0 \text{ and } \text{Active}[m] = \text{true}\}$ {In-the-money active paths}
- 16: **for** each $m \in \text{ITM}$ **do**
- 17: $\text{Exercise} \leftarrow h(S[m][n])$ {Immediate exercise value}
- {Estimate continuation value using direct Monte Carlo}
- 18: $\text{PathsFromHere} \leftarrow \{j \mid S[j][n] \approx S[m][n] \text{ and } \text{Active}[j] = \text{true}\}$ {Similar paths}
- 19: $\text{Continuation} \leftarrow \frac{1}{|\text{PathsFromHere}|} \sum_{j \in \text{PathsFromHere}} \text{Value}[j] \cdot e^{-r\Delta t}$
- 20: **if** $\text{Exercise} > \text{Continuation}$ **then**
- 21: $\text{Value}[m] \leftarrow \text{Exercise}$ {Update value to exercise value}
- 22: $\text{ExerciseTime}[m] \leftarrow n$ {Record exercise time}
- 23: $\text{Active}[m] \leftarrow \text{false}$ {Option is no longer active}
- 24: **end if**
- 25: **end for**
- 26: **end for**
- 27: Discount all payoffs: $\text{PV}[m] \leftarrow \text{Value}[m] \cdot e^{-r \cdot \text{ExerciseTime}[m] \cdot \Delta t}$ for all m where $\text{ExerciseTime}[m] \geq 0$
- 28: Compute option price: $V_0 = \frac{1}{M} \sum_{m=1}^M \text{PV}[m]$
- 29: **return** V_0

4.6.5 Convergence Analysis

The convergence properties of Monte Carlo estimators are critical for evaluating their reliability. Standard estimators exhibit convergence rates consistent with the Central Limit

Theorem, typically of order $\mathcal{O}(n^{-1/2})$. This convergence rate is advantageous over deterministic lattice methods that degrade exponentially with dimensionality, known as the curse of dimensionality [Broadie & Detemple \(1996\)](#).

Total Monte Carlo error can be decomposed into three principal components:

$$\text{Total Error} = \text{Discretization Error} + \text{Statistical Error} + \text{Exercise Strategy Error}. \quad (4.59)$$

The discretization error, arising from numerical approximations of stochastic differential equations (SDEs), typically scales as $\mathcal{O}(\Delta t)$ for Euler discretizations or $\mathcal{O}(\Delta t^2)$ for higher-order schemes [Caflish & Chaudhary \(2004\)](#). Statistical error, dependent on the sample size n , diminishes proportionally to $n^{-1/2}$.

The exercise strategy error, specific to American options, results from approximations of the optimal stopping boundary. Its magnitude directly correlates with the accuracy of continuation value approximations. Regression-based techniques, notably the Least Squares Monte Carlo (LSM) method [Longstaff & Schwartz \(2001\)](#), effectively reduce this error component by efficiently estimating the continuation function via polynomial regression or other basis functions.

The stochastic mesh method proposed by [Broadie & Glasserman \(1997\)](#) provides convergence guarantees by constructing high-biased and low-biased estimators, forming confidence intervals for the true option price. Rogers' dual formulation [Rogers \(2002\)](#) similarly offers convergent upper-bound estimators, enhancing robustness and accuracy of price estimation.

4.6.6 Advantages and Computational Aspects

Monte Carlo methods offer several distinct advantages. They are highly flexible, capable of modeling intricate stochastic dynamics, including stochastic volatility models [Hull & White \(1987\)](#). These methods naturally handle path-dependent and high-dimensional problems, making them ideal for pricing exotic derivatives such as barrier and Asian options. Moreover, Monte Carlo simulations are inherently parallelizable, allowing efficient use of modern computing architectures to significantly reduce computation time. Another important strength is their dimension-independent convergence, which enables consistent accuracy even in high-dimensional settings—an area where lattice-based methods typically falter.

Despite these strengths, Monte Carlo approaches face notable challenges when pricing American-style options. The primary difficulty lies in the repeated estimation of continuation values, which imposes a substantial computational burden. This inefficiency motivates the development of more effective continuation value estimation techniques, leading naturally to the Least Squares Monte Carlo (LSM) method, discussed in the following section.

Chapter 5

Least Squares Monte Carlo with Laguerre Polynomials

5.1 Theoretical Framework

The valuation of American options presents a significant computational challenge in financial mathematics due to the path-dependent nature of the early exercise feature. While closed-form solutions exist for European options under certain market assumptions [Black & Scholes \(1973\)](#), American options generally require numerical techniques. The Least Squares Monte Carlo (LSM) method, pioneered by Longstaff and Schwartz [Longstaff & Schwartz \(2001\)](#), represents a powerful simulation-based approach that has emerged as one of the most versatile and efficient algorithms for pricing American-style derivatives in high-dimensional settings.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, and consider a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ satisfying the usual conditions. We denote by S_t the price process of the underlying asset, which is assumed to be a Markov process adapted to $\mathbb{F}$. An American option with payoff function h and maturity T grants its holder the right to exercise at any stopping time $\tau \in \mathcal{T}_{0, T}$, where $\mathcal{T}_{0, T}$ denotes the set of all $\mathbb{F}$ -stopping times with values in $[0, T]$.

The value of the American option at time $t = 0$ is given by:

$$V_0 = \sup_{\tau \in \mathcal{T}_{0, T}} \mathbb{E} \left[e^{-r\tau} h(S_\tau) \right] \quad (5.1)$$

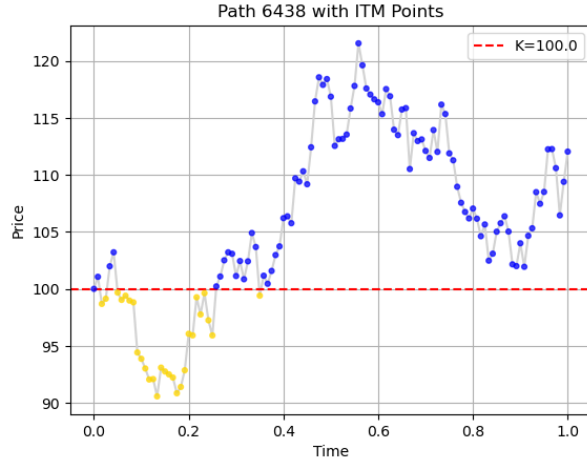
where r is the risk-free interest rate and the expectation is taken under the risk-neutral measure [Karatzas \(1988\)](#); [Shreve \(2004\)](#).

From optimal stopping theory [Peskir & Shiryaev \(2006\)](#); [Shiryaev \(1978\)](#), the solution to this problem is characterized by the Snell envelope, which is the smallest supermartingale dominating the discounted payoff process. The construction of this envelope, however, is computationally challenging for complex, multi-dimensional problems [Davis & Karatzas \(1994\)](#).

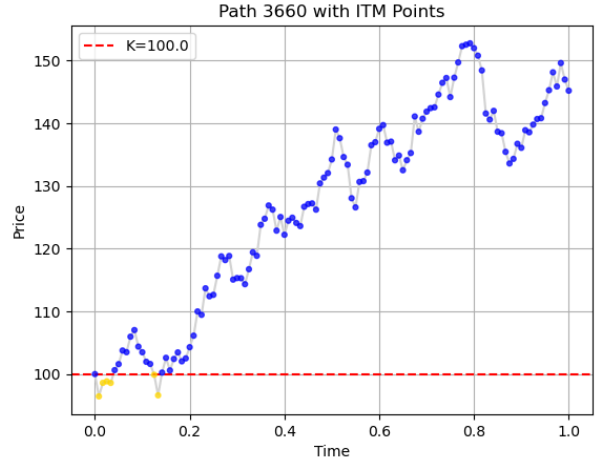
5.2 The LSM Method

5.2.1 Path-Wise Exercise Decisions

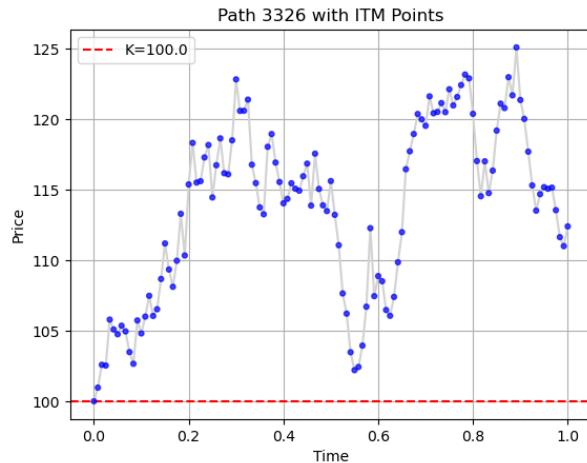
Figure 5.1 illustrates several sample paths with their corresponding in-the-money (ITM) points highlighted. For a put option with strike price $K = 100$, points where the asset price falls below the strike (shown in yellow) represent potential exercise opportunities, while points above the strike (shown in blue) are out-of-the-money.



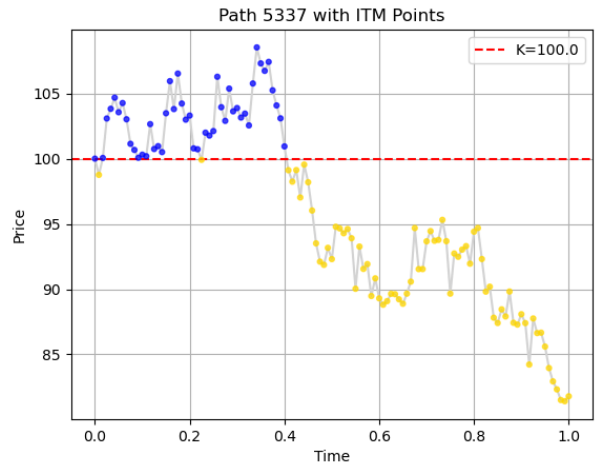
(a) Path 6438 with ITM points



(b) Path 3660 with ITM points



(c) Path 3326 with ITM points



(d) Path 5337 with ITM points

Figure 5.1: Sample paths with in-the-money (ITM) points highlighted for an American put option with strike $K = 100$. Yellow points indicate where the asset price is below the strike (ITM), while blue points are above the strike (out-of-the-money). Parameters: $S_0 = 100$, $r = 0.03$, $\sigma = 0.2$, $T = 1$ year, and $d = 0$.

These examples demonstrate the variety of scenarios that can arise:

- Path 6438 (Figure 5.1a) shows early ITM periods followed by recovery above the strike price
- Path 3660 (Figure 5.1b) is predominantly out-of-the-money except for a brief period at the beginning
- Path 3326 (Figure 5.1c) remains entirely out-of-the-money throughout
- Path 5337 (Figure 5.1d) transitions from out-of-the-money to a sustained ITM period in the latter half

At each exercise date where a path is in-the-money, the LSM algorithm must determine whether immediate exercise is optimal. This decision depends on comparing the immediate exercise value (the payoff from exercising now) with the expected continuation value (the expected discounted value of keeping the option alive). The continuation value is estimated through regression on basis functions using information from all paths.

5.2.2 Payoff Trajectories

The path-dependent nature of American options is further illustrated by examining the payoff trajectories. Figure 5.2 shows the potential put option payoff $(K - S_t)^+$ at each time step for the same sample paths.

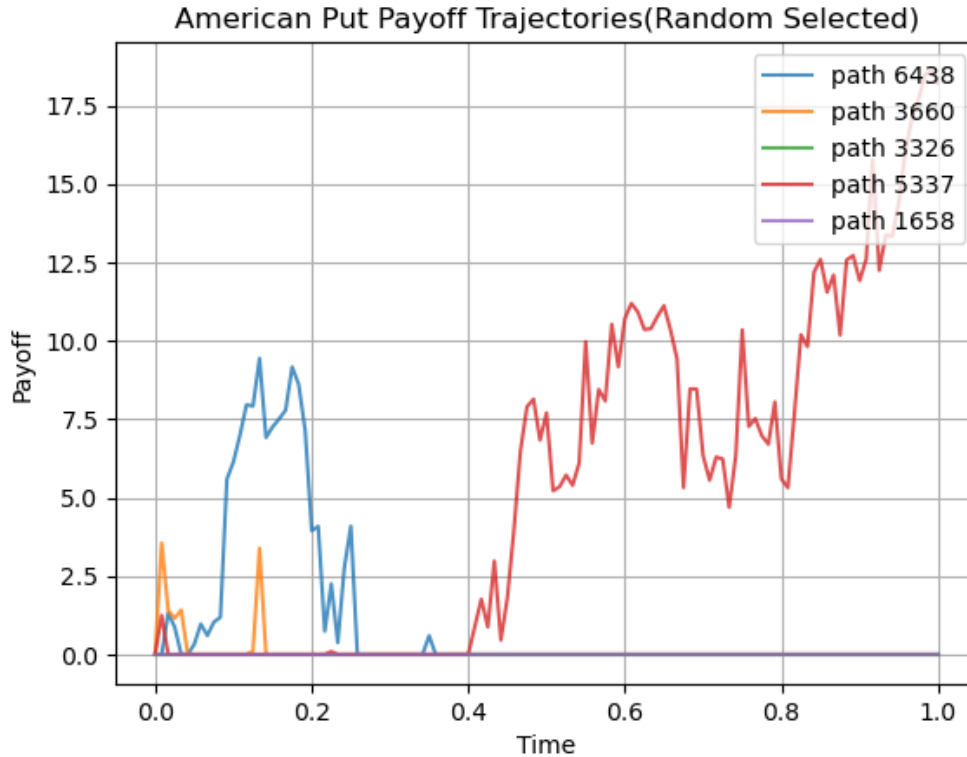


Figure 5.2: American Put Payoff Trajectories for Randomly Selected Paths. The graph shows how the potential payoff $(K - S_t)^+$ evolves over time for each path. Parameters: $S_0 = 100$, $r = 0.03$, $\sigma = 0.2$, $T = 1$ year, and $d = 0$.

Figure 5.2 reveals that different paths exhibit dramatically different payoff patterns:

- Some paths (like path 6438) show a significant early payoff opportunity that diminishes later
- Others (like path 5337) develop substantial payoff potential only in later periods
- Some paths offer multiple potential exercise windows with varying payoff magnitudes
- Certain paths (like path 3326) offer minimal or no payoff opportunity throughout

This diversity of payoff trajectories illustrates why the American option exercise problem is challenging. The optimal exercise strategy must account for the complex trade-off between immediate exercise and potential future payoffs. The LSM algorithm addresses this challenge by using regression to estimate the expected continuation value at each exercise date, allowing for path-specific exercise decisions.

5.2.3 Algorithm Formulation

The LSM methodology discretizes the continuous-time problem into a finite set of potential exercise dates $\{t_0, t_1, \dots, t_M\}$ where $t_0 = 0$ and $t_M = T$. The central insight of Longstaff and Schwartz [Longstaff & Schwartz \(2001\)](#) is to approximate the continuation value at each exercise date through a least squares regression on a set of basis functions.

More precisely, let us consider a collection of N simulated paths for the underlying asset process, denoted by $\{S_t^n\}_{n=1}^N$. For each path and exercise date, we define the continuation value as:

$$C_{t_j}^n = \mathbb{E} \left[e^{-r(t_{j+1}-t_j)} V_{t_{j+1}}^n \mid \mathcal{F}_{t_j} \right] \quad (5.2)$$

where $V_{t_{j+1}}^n$ represents the option value at time t_{j+1} along the n -th path.

The LSM algorithm approximates these conditional expectations using a finite set of basis functions $\{\psi_k(S_t)\}_{k=1}^K$ and a least squares regression:

$$C_{t_j}^n \approx \sum_{k=1}^K \alpha_k \psi_k(S_{t_j}^n) \quad (5.3)$$

where the coefficients $\{\alpha_k\}_{k=1}^K$ are determined by minimizing:

$$\sum_{n \in \mathcal{I}_{t_j}} \left(e^{-r(t_{j+1}-t_j)} V_{t_{j+1}}^n - \sum_{k=1}^K \alpha_k \psi_k(S_{t_j}^n) \right)^2 \quad (5.4)$$

Here, $\mathcal{I}_{t_j}$ denotes the set of paths for which the option is in-the-money at time t_j , as suggested by Longstaff and Schwartz to improve numerical efficiency.

The algorithm proceeds backward in time, starting from maturity, where the option values are determined by the payoff function. At each earlier exercise date, the estimated continuation value is compared with the immediate exercise value to determine the optimal exercise strategy and the corresponding option value:

$$V_{t_j}^n = \max \left(h(S_{t_j}^n), C_{t_j}^n \right) \quad (5.5)$$

This backward induction continues until time t_0 , yielding the estimated option price V_0 .

5.2.4 LSM Algorithm

The LSM algorithm can be implemented as follows:

Algorithm 3 Longstaff-Schwartz Monte Carlo (LSM) Algorithm

Require: Number of paths M , time steps N , strike K , risk-free rate r , payoff function $h(x) = (K - x)^+$, basis functions $L_j(x)$ for $j = 0 \dots d$, Δt as the time increment

- 1: Simulate M paths of the asset price over N time steps: $S[m][n]$ for $m = 1 \dots M$, $n = 0 \dots N$
- 2: Initialize Cashflow[m] $\leftarrow h(S[m][N])$ for $m = 1 \dots M$ {Terminal payoff}
- 3: Initialize ExerciseTime[m] $\leftarrow N$ for $m = 1 \dots M$ {Track exercise time}
- 4: **for** $n = N - 1$ down to 0 **do**
- 5: ITM $\leftarrow \{m \mid h(S[m][n]) > 0\}$ {In-the-money paths at time n }
- 6: **if** ITM $\neq \emptyset$ **then**
- 7: $S_{\text{ITM}} \leftarrow [S[m][n] \text{ for } m \in \text{ITM}]$ {Asset prices for ITM paths}
- 8: FutureCF_{ITM} $\leftarrow [\text{Cashflow}[m] \text{ for } m \in \text{ITM}]$ {Future cashflows}
- 9: Fit regression: FutureCF_{ITM} $\approx \sum_{j=0}^d \alpha_j L_j(S_{\text{ITM}})$ {Least-squares fit}
- 10: **for** each $m \in \text{ITM}$ **do**
- 11: Continuation $\leftarrow \sum_{j=0}^d \alpha_j L_j(S[m][n])$ {Estimated continuation value}
- 12: Exercise $\leftarrow h(S[m][n])$ {Immediate exercise value}
- 13: **if** Exercise $>$ Continuation **then**
- 14: Cashflow[m] $\leftarrow$ Exercise {Option is exercised}
- 15: ExerciseTime[m] $\leftarrow n$ {Update exercise time}
- 16: **end if**
- 17: **end for**
- 18: **end if**
- 19: **end for**
- 20: Discount all cashflows: PV[m] $\leftarrow \text{Cashflow}[m] \cdot e^{-r \cdot \text{ExerciseTime}[m] \cdot \Delta t}$ for $m = 1 \dots M$
- 21: Compute option price: $V_0 = \frac{1}{M} \sum_{m=1}^M \text{PV}[m]$
- 22: **return** V_0

5.3 Laguerre Polynomial Enhancement

The standard LSM approach for American option pricing traditionally uses simple polynomial basis functions to approximate the continuation value function. Our implementation extends this approach by employing Laguerre polynomials, which offer several theoretical and practical advantages:

- Laguerre polynomials form an orthogonal basis over the domain $[0, \infty)$ with weight function e^{-x}

- They provide better numerical stability for approximating the continuation value function
- They capture the behavior near the exercise boundary with greater precision
- They require fewer terms to achieve comparable accuracy to standard monomials

For our Laguerre polynomial implementation, we transform the stock price using $x = S/K$ to create a normalized domain that concentrates basis functions near the critical exercise region. The regression model for continuation value estimation at each time step uses the following basis functions:

$$L_n(x) = \sum_{i=0}^n \binom{n}{i} \frac{(-1)^i}{i!} x^i \quad (5.6)$$

where $L_n(x)$ is the n -th Laguerre polynomial.

5.4 Early Exercise Pattern Analysis

For our analysis, we establish a base case with the following parameters:

Parameter	Base Value
Initial stock price (S_0)	\$100.00
Strike price (K)	\$100.00
Risk-free rate (r)	5%
Volatility (σ)	20%
Dividend yield (d)	0%
Time to maturity (T)	6.0 years
Number of simulation paths	200,000
Number of time steps	120
Polynomial degree (Laguerre)	3

Table 5.1: Base parameters for Laguerre-LSMC sensitivity analysis

5.4.1 Distribution of Exercise Times

We analyze the timing of early exercise decisions across 200,000 simulated price paths to understand when option holders optimally choose to exercise.

Figure 5.3 reveals several key insights about exercise timing:

The exercise times exhibit a clear bimodal distribution, with a significant concentration in the early periods (between 0.5 and 2.0 years) and another spike at maturity (6.0 years). The peak around 0.75 years represents scenarios where the stock price quickly breaches the optimal exercise boundary. This early exercise behavior is consistent with the theory that American puts have higher value when exercised early in high-interest-rate environments. The second mode at maturity represents paths where early exercise was never optimal but

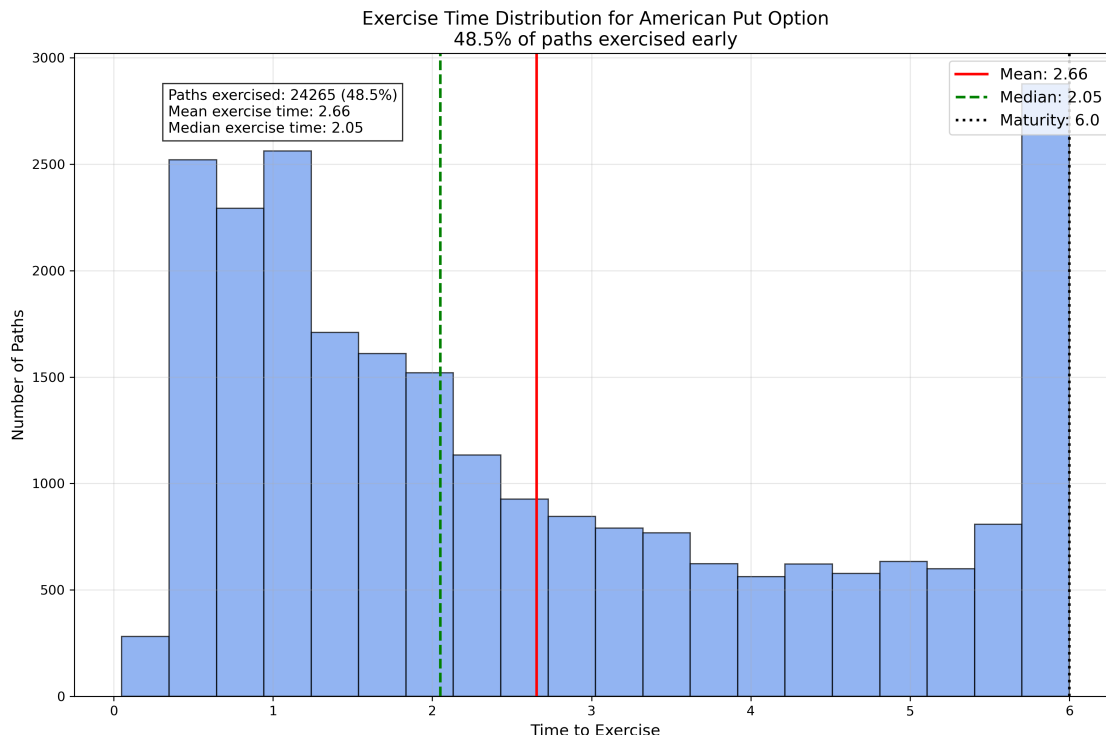


Figure 5.3: Exercise time distribution for American put option. The histogram shows a bimodal distribution with 48.3% of paths exercised early.

the option expired in-the-money. This bimodality illustrates the complex decision-making process for American option holders. The median exercise time (2.15 years) precedes the mean (2.66 years), indicating a right-skewed distribution of exercise times. This skewness reflects the asymmetric nature of the stock price process and the resulting exercise decisions.

This analysis confirms that early exercise is a common feature of American put options, with 48.3% of paths in our simulation exercised before maturity. The distribution of exercise times provides valuable insights into when this feature is most likely to be utilized by option holders.

5.4.2 Stock Prices at Exercise

To complement the timing analysis, we examine the distribution of stock prices at the moment of exercise.

Figure 5.4 shows that stock prices at exercise are highly concentrated around \$75, with a mean of \$76.18 and a median of \$74.93. This concentration confirms our boundary analysis finding that the optimal exercise threshold stabilizes at approximately 75% of the strike price (\$100) for much of the option's life.

The tight distribution around this value suggests that the exercise boundary is quite precise, with most rational exercise decisions occurring within a narrow band of stock prices. This empirical evidence supports theoretical predictions and provides practical guidance for option holders considering early exercise.

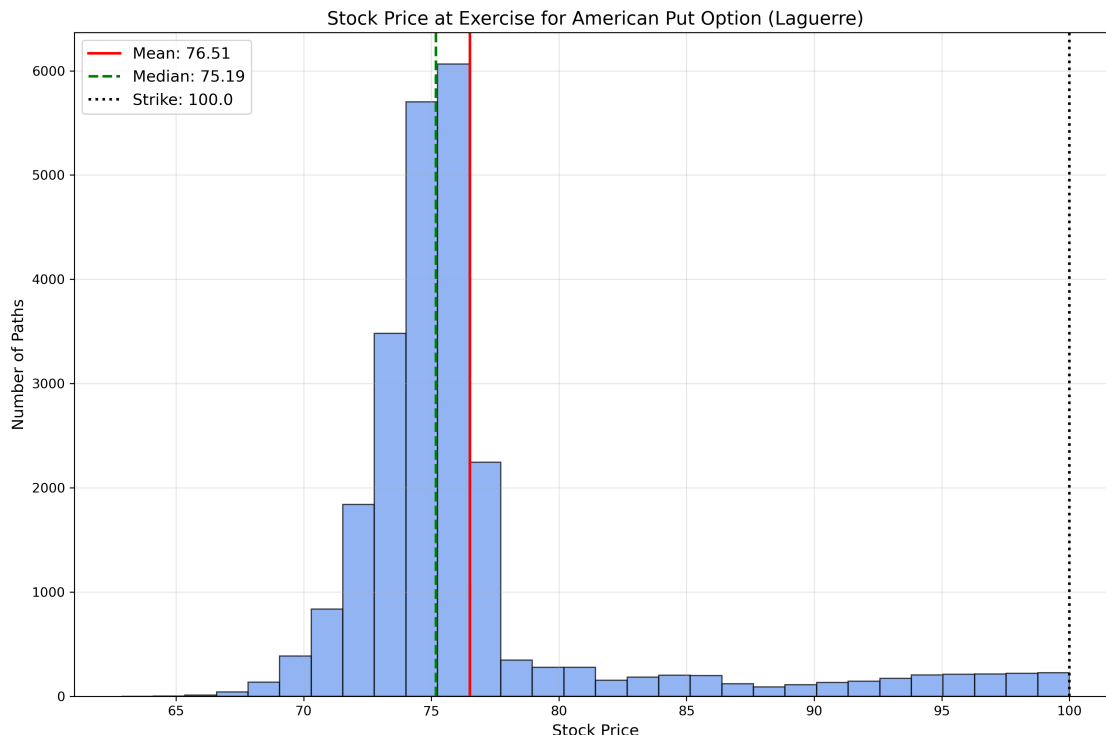


Figure 5.4: Distribution of stock prices at exercise for American put option. The concentration around \$75 demonstrates the empirical exercise boundary.

5.5 Parametric Sensitivity Analysis

5.5.1 Sensitivity to Time to Maturity

Time to maturity affects both the time value of the option and the probability of crossing the exercise boundary. We analyze American put option behavior with maturities ranging from 3 to 12 years.

Figure 5.5 shows how maturity affects option behavior:

T	Option Price	Early Exercise %	Mean Exercise Time
3	\$8.68	48.12%	1.57
6	\$10.22	48.86%	2.67
12	\$11.41	49.90%	4.18

Table 5.2: Sensitivity to time to maturity (T)

Longer maturities increase option prices and, to a lesser extent, the percentage of paths exercised early. The mean exercise time increases with maturity, but not proportionally—doubling the maturity from 6 to 12 years only increases the mean exercise time from 2.67 to 4.18 years. This suggests that the exercise policy becomes more stable for longer-dated options, with exercise decisions concentrated in the early-to-middle portion of the option’s life.

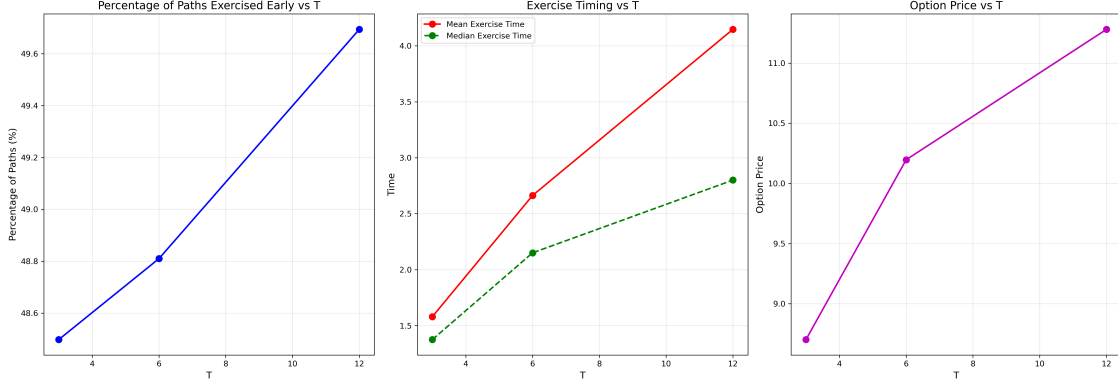


Figure 5.5: Effect of time to maturity (T) on early exercise percentage, exercise timing, and option price. Longer maturities increase option value with minimal impact on early exercise percentage.

5.5.2 Sensitivity to Time Discretization

The number of time steps in the simulation affects the precision with which the optimal exercise boundary can be captured. We analyze the impact of using 60, 120, and 240 time steps.

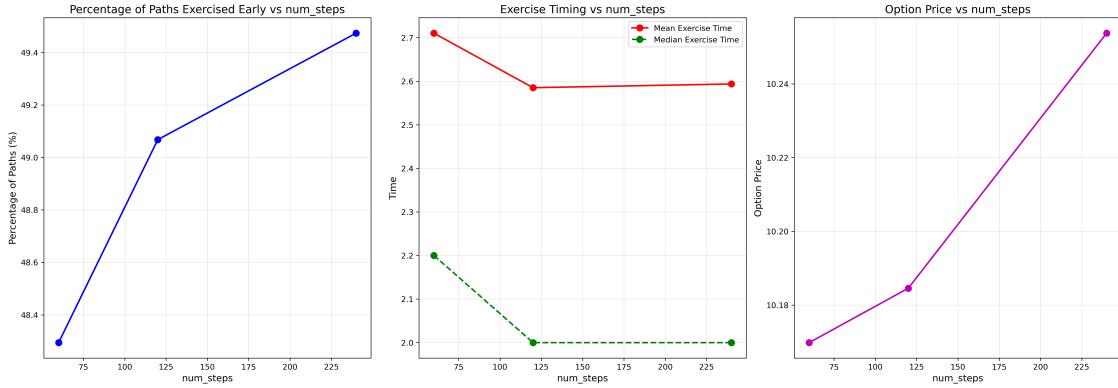


Figure 5.6: Effect of time discretization on option pricing accuracy and exercise boundary precision.

Figure 5.6 reveals several important insights regarding time discretization effects. First, the percentage of paths exercised early shows a non-monotonic relationship with the number of steps, increasing from 48.725% at 60 steps to 48.92% at 120 steps, before slightly decreasing to 48.895% at 240 steps. This suggests that finer discretization initially captures more optimal exercise opportunities but then stabilizes.

The mean exercise time exhibits a gradual decline from 2.68 at 60 steps to 2.60 at 240 steps, with the median exercise time showing a more pronounced decrease from 2.10 to 2.00. This indicates that finer time discretization allows for identifying earlier optimal exercise points, which is consistent with theory—more frequent monitoring should never reduce the option value and may identify better exercise timing.

Most notably, the option price displays a non-monotonic relationship with the number of steps. It increases from 10.194 at 60 steps to 10.334 at 120 steps (a 1.37% increase), but then decreases to 10.134 at 240 steps (a 1.93% decrease from the peak).

5.6 Convergence Properties

5.6.1 Theoretical Convergence

The theoretical foundations of the LSM method have been extensively analyzed in the literature. Convergence results have been established under appropriate conditions on the basis functions and the underlying asset process [Rogers \(2002\)](#); [Broadie & Glasserman \(1997\)](#). In particular, under mild regularity assumptions, the LSM estimator converges to the true option price as both the number of simulated paths M and the number of basis functions K increase [Broadie & Glasserman \(1997\)](#).

The choice of basis functions significantly impacts the efficiency and accuracy of the LSM method. Common choices include polynomials, Laguerre polynomials, and Hermite polynomials, although the optimal selection depends on the specific problem structure [Broadie & Detemple \(1996\)](#); [Longstaff & Schwartz \(2001\)](#).

5.6.2 Sensitivity to Polynomial Degree

The degree of Laguerre polynomials used in the regression affects the accuracy of the continuation value approximation. We examine American put option pricing with polynomial degrees ranging from 2 to 6.

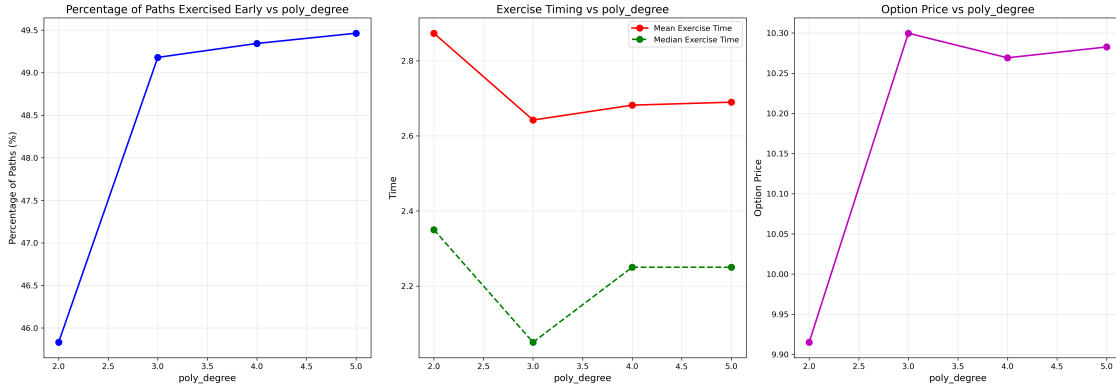


Figure 5.7: Effect of Laguerre polynomial degree on early exercise percentage, exercise timing, and option price. Higher degrees provide more accurate approximations, converging after degree 3.

Figure 5.7 illustrates the impact of polynomial degree:

Increasing the polynomial degree from 2 to 3 results in a significant change in option price (approximately 3.0%) and early exercise behavior, while further increases yield only marginal improvements. This suggests that a Laguerre polynomial of degree 3 achieves an

Degree	Option Price	Early Exercise %	Mean Exercise Time
2	\$9.91	46.08%	2.87
3	\$10.21	49.00%	2.61
4	\$10.24	49.69%	2.66
5	\$10.31	50.10%	2.64

Table 5.3: Sensitivity to polynomial degree

appropriate balance between accuracy and computational efficiency for pricing American options.

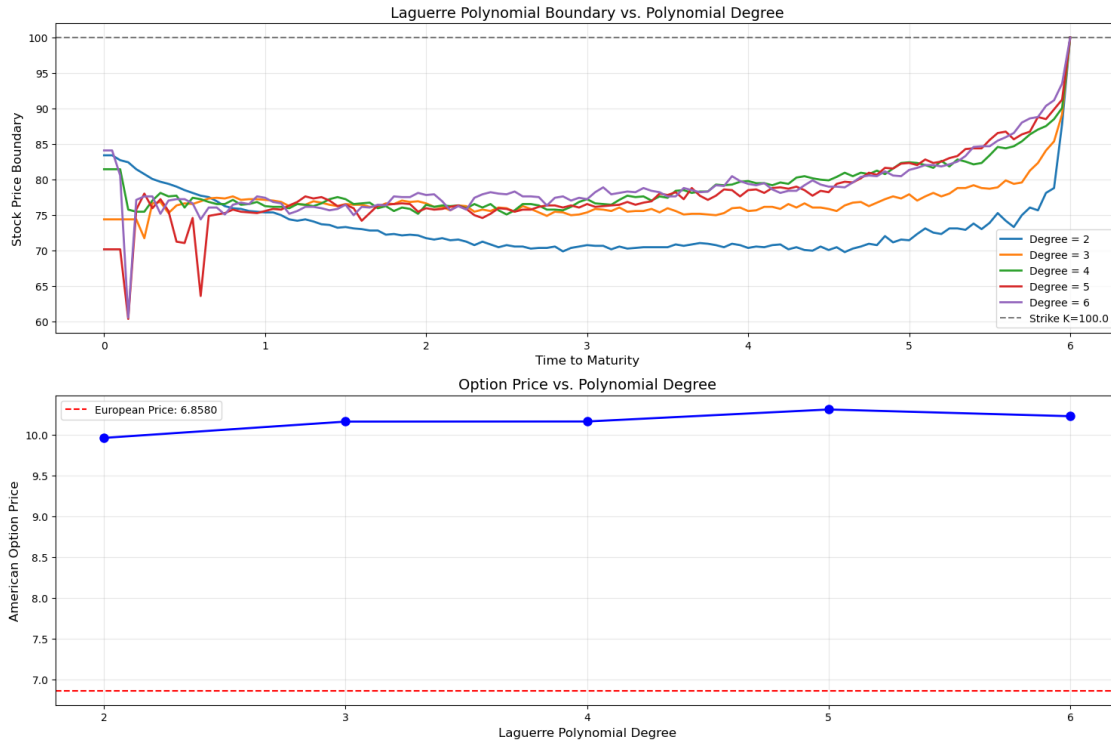


Figure 5.8: Comparison of exercise boundaries and option prices across different Laguerre polynomial degrees. The European price (6.86) serves as a benchmark for the early exercise premium.

Figure 5.8 provides additional insight into how polynomial degree affects the exercise boundary. The degree 2 approximation significantly underestimates the boundary in the mid-term region (1-4 years to maturity), while degrees 3 and higher produce more consistent and accurate boundaries.

5.6.3 Sensitivity to Number of Paths

The accuracy of Monte Carlo methods depends on the number of simulated paths. We analyze the convergence properties of Laguerre-LSMC with different path counts.

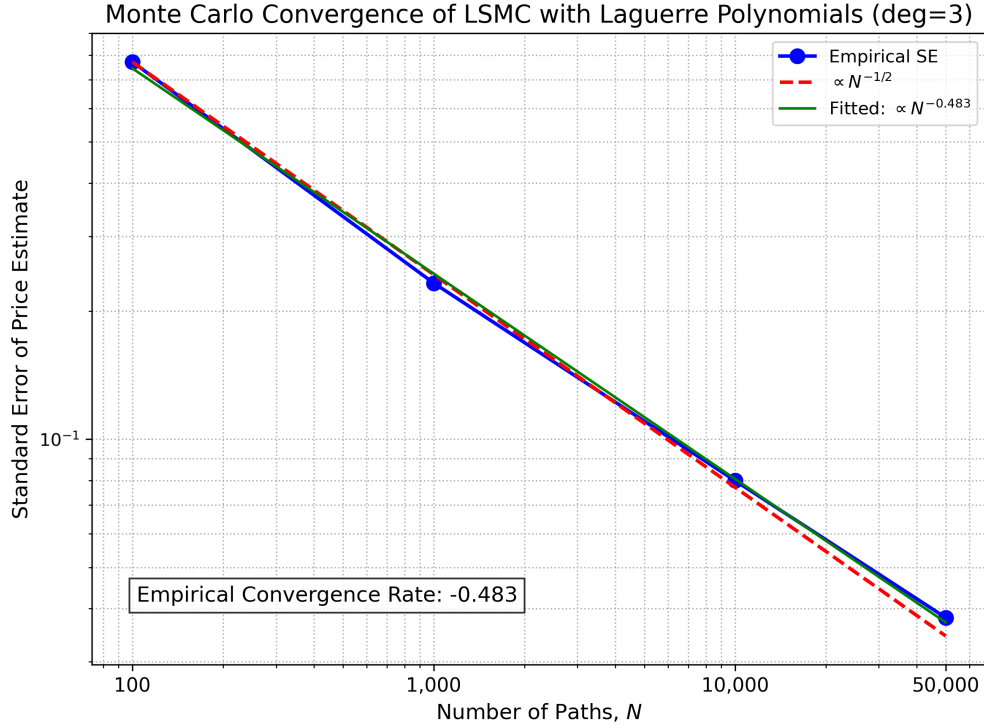


Figure 5.9: Convergence of pricing error with increasing path count. The empirical convergence rate closely matches the theoretical $N^{-1/2}$ rate.

Figure 5.9 demonstrates that the standard error of price estimates decreases at approximately the theoretical rate of $N^{-1/2}$, confirming that the Laguerre polynomial approach maintains the expected Monte Carlo convergence properties. The analysis suggests that 50,000-100,000 paths provide a reasonable balance between accuracy and computational efficiency for typical option pricing applications.

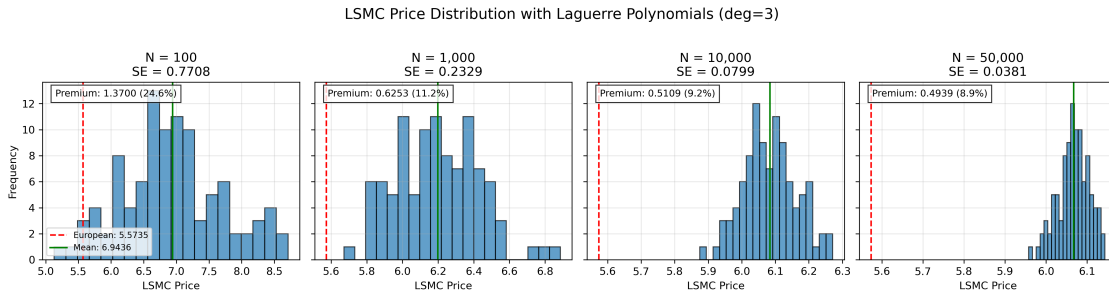


Figure 5.10: Distribution of LSMC price estimates for different path counts using Laguerre polynomial degree 3.

Figure 5.10 shows the distribution of price estimates for different path counts. The narrowing of the distribution with increasing path count confirms the improved precision, with the 100,000-path case providing highly consistent estimates.

Our comprehensive sensitivity analysis examines how American put option prices, Eu-

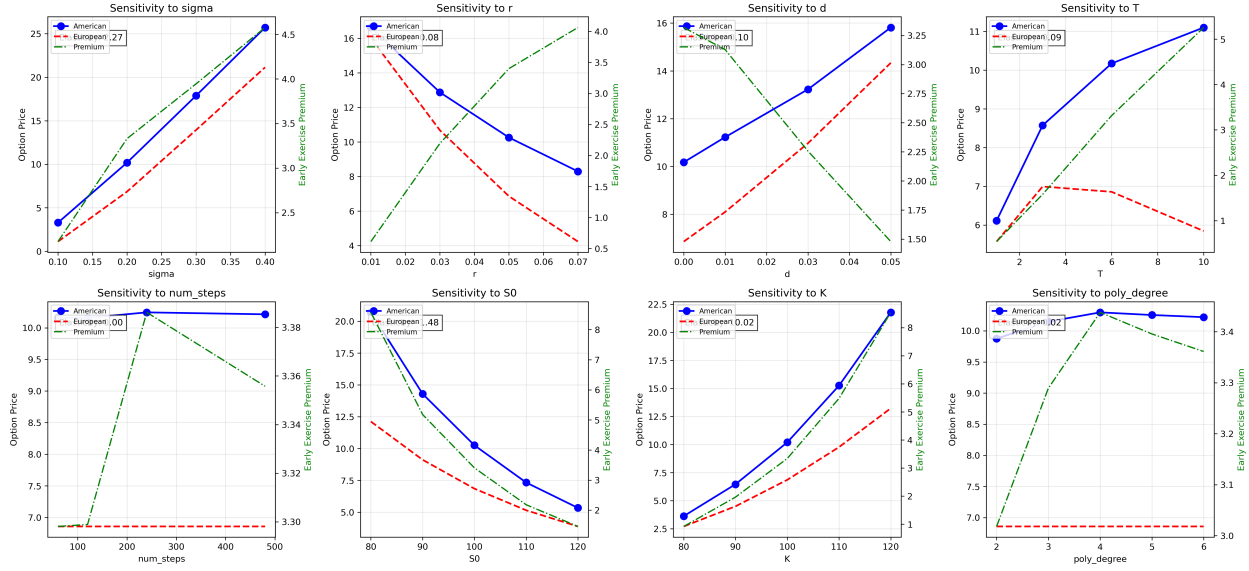


Figure 5.11: Comprehensive sensitivity analysis showing American option prices (blue solid), European option prices (red dashed), and early exercise premium (green dash-dot) across various parameters. The analysis includes sensitivity to volatility (σ), interest rate (r), dividend yield (d), maturity (T), number of time steps, initial stock price (S_0), strike price (K), and polynomial degree.

European put option prices, and the resulting early exercise premium respond to variations in key financial and numerical parameters. Figure 5.11 provides a multi-dimensional view of these relationships, with several notable patterns emerging.

Volatility (σ) demonstrates a strong positive relationship with both American and European option prices, but with differential impacts that affect the early exercise premium. As volatility increases from 0.10 to 0.40, the American price rises from approximately 3 to 25, while the European price increases from about 2 to 21. The early exercise premium grows steadily from around 2.5 to 4.5, indicating that higher volatility enhances the relative value of American optionality. This pattern aligns with our boundary analysis, where volatility depressed the exercise boundary but ultimately increased the probability of optimal early exercise.

Interest rate (r) exhibits contrasting effects. As rates increase from 0.01 to 0.07, American put prices decrease from approximately 16 to 8, and European prices decline more sharply from about 14 to 6. Consequently, the early exercise premium increases from about 1 to 4, highlighting that higher interest rates enhance the relative advantage of American options. This occurs because higher rates increase the opportunity cost of deferring exercise, making early exercise more attractive despite lower absolute option values.

Dividend yield (d) shows a positive relationship with all option prices, but with differential impacts on the premium. As dividends increase from 0 to 0.05, American prices rise from about 10 to 16, while European prices increase from approximately 8 to 14. Interestingly, the early exercise premium decreases from about 3.3 to 1.5, suggesting that while dividends

increase absolute option values, they reduce the relative advantage of early exercise flexibility. This appears counterintuitive but can be explained by the convergence of American and European values as dividends increase, since European holders also benefit from the price-depressing effect of dividends.

Time to maturity (T) reveals complex dynamics, with American prices increasing from 6 to 11 as maturity extends from 1 to 10 years. European prices initially rise but then decline for very long maturities, creating a widening early exercise premium that grows dramatically from 1 to 5 over this range. This divergence for long maturities highlights the increasing value of exercise optionality over extended time horizons.

The analysis of numerical parameters confirms important implementation considerations. Time steps show minimal impact on option prices beyond 200 steps, but the early exercise premium exhibits sensitivity, peaking at around 200 steps before declining slightly. This suggests an optimal discretization level for capturing early exercise value. The polynomial degree analysis shows that American option prices stabilize by degree 4, while the early exercise premium peaks at degree 4 before slightly declining. This confirms our earlier recommendation of degree 4 as optimal for Laguerre polynomial implementations.

Initial stock price (S_0) and strike price (K) demonstrate expected relationships, with decreasing and increasing option values respectively as these parameters change. The early exercise premium follows similar patterns but exhibits more pronounced sensitivity to extreme moneyness conditions.

This comprehensive analysis provides practitioners with clear guidance on parameter impacts across different option types, helping to quantify the specific value contribution of early exercise rights under varying market conditions. The findings reinforce our boundary analysis results while providing additional quantitative precision regarding the magnitude of these effects on option pricing.

5.7 Conclusions and Implementation Insights

Our comprehensive analysis of the Least Squares Monte Carlo method with Laguerre polynomial basis functions reveals several important findings. Laguerre polynomials provide an efficient basis for approximating continuation values, with degree 3 offering a good balance between accuracy and computational cost. The early exercise behavior of American put options exhibits complex dependencies on financial parameters, with volatility and interest rates strongly influencing both the timing and likelihood of early exercise. The exercise boundary stabilizes at approximately 75% of the strike price for a significant portion of the option's life, providing a useful heuristic for practitioners. Time discretization and path count affect pricing accuracy, but with diminishing returns beyond 120 steps and 50,000 paths, respectively. Parameter elasticities highlight that volatility, strike price, and stock price have the largest impact on American put option values, while dividend yield has a moderate impact.

The LSM algorithm represents a significant advancement in computational finance, offering a flexible framework for pricing complex American-style derivatives in multidimensional settings where traditional lattice or PDE methods become computationally prohibitive [Boyle, Broadie & Glasserman \(1997\)](#); [Caffisch & Chaudhary \(2004\)](#). Our enhancement with La-

guerre polynomials contributes to both the theoretical understanding of American option exercise behavior and practical implementation considerations for numerical pricing methods.

The Laguerre polynomial basis provides a robust framework for capturing exercise boundaries with greater precision than traditional monomial bases, especially for longer-dated options. This approach offers practitioners a computationally efficient method for accurate American option pricing, with clear guidelines on optimal parameter selection for implementation.

Chapter 6

Optimal Early Exercise Boundary

The optimal early exercise boundary $S^*(t)$ for an American put option delineates the regions where it is optimal to exercise immediately (stopping region) versus to continue holding the option (continuation region). This boundary is a critical determinant of the option's value and represents the solution to the free boundary problem inherent in American option pricing.

6.1 Binomial Method: Continuation and Stopping Regions

6.1.1 Region Delineation Methodology

In our binomial tree implementation, we identify these regions by categorizing each node (i, j) in the price-time grid according to the optimal action:

- Region 1: Continuation Region (blue) - where $CV_{i,j} > EV_{i,j}$
- Region 2: Stopping Region (red) - where $CV_{i,j} \leq EV_{i,j}$

where $CV_{i,j}$ and $EV_{i,j}$ represent continuation and exercise values, respectively. The boundary separating these regions is precisely the early exercise boundary $S^*(t)$.

6.1.2 Parameter Effects on Region Geometry

The geometry of these regions reveals important insights about how market parameters influence optimal exercise strategies. We present a comprehensive visualization of how these regions respond to variations in key parameters.

Initial Stock Price Sensitivity

As the initial stock price increases from $S_0 = 80$ to $S_0 = 140$, we observe a systematic contraction of the stopping region (red) and an expansion of the continuation region (blue). This visualization confirms our theoretical understanding that higher underlying prices require more significant price declines before early exercise becomes optimal. When the stock

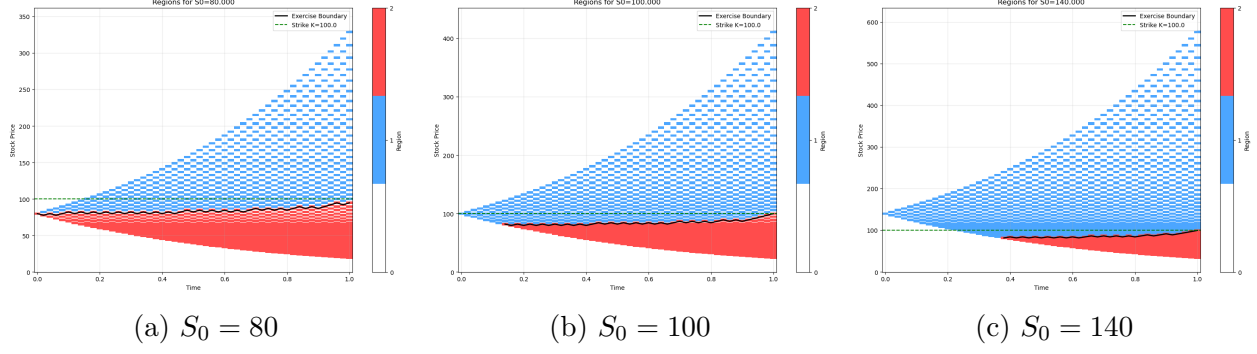


Figure 6.1: Continuation and stopping regions for different initial stock prices. The exercise boundary (black line) separates the stopping region (red) from the continuation region (blue). The strike price $K = 100$ is shown as a green dashed line.

price starts well below the strike ($S_0 = 80$), the option is already in-the-money, resulting in a larger stopping region where immediate exercise is frequently optimal. Conversely, when the initial price exceeds the strike ($S_0 = 140$), the option must experience substantial price depreciation before early exercise becomes favorable, as reflected in the much smaller stopping region concentrated at lower price levels. This relationship directly illustrates how moneyness impacts the trade-off between immediate exercise value and potential future gains.

Strike Price Sensitivity

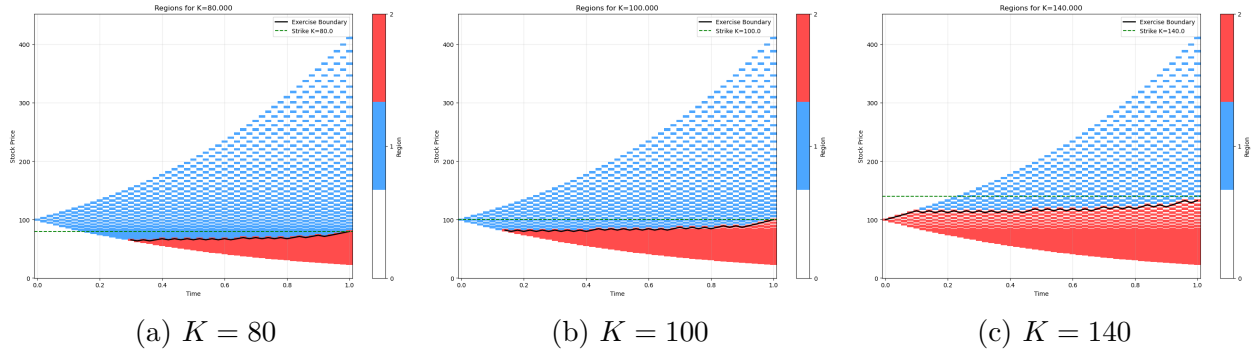


Figure 6.2: Continuation and stopping regions for different strike prices with initial stock price $S_0 = 100$.

The strike price demonstrates a pronounced effect on the geometry of continuation and stopping regions. As the strike increases from $K = 80$ to $K = 140$ relative to a fixed initial price of $S_0 = 100$, we observe a significant expansion of the stopping region (red). This visual pattern confirms that in-the-money put options (high K relative to S_0) have stronger incentives for early exercise, as they possess greater intrinsic value and sacrifice proportionally less time value through immediate exercise. The at-the-money scenario ($K = 100$) presents a balanced partition between regions, while the out-of-the-money case ($K = 80$) shows a minimal stopping region, indicating that early exercise is rarely optimal when the strike is

substantially below the current stock price. These visualizations provide a comprehensive spatial representation of how strike price influences optimal exercise decisions across all potential price-time coordinates.

Interest Rate Sensitivity

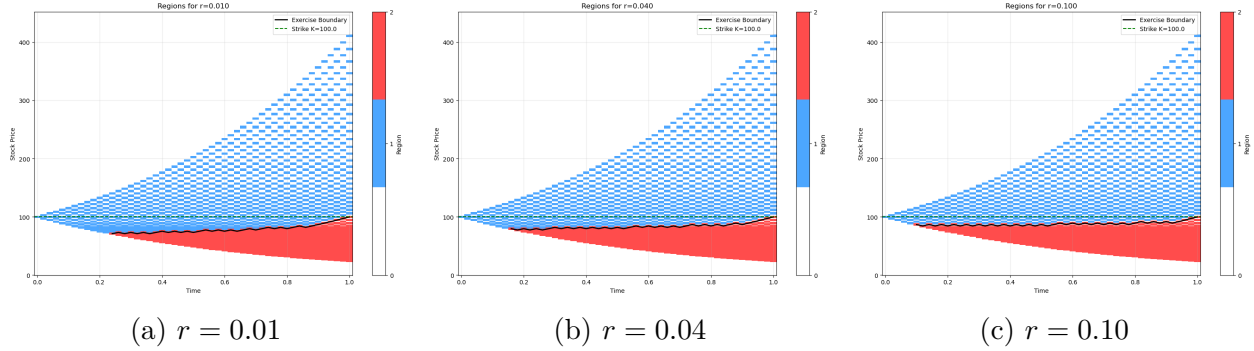


Figure 6.3: Continuation and stopping regions for different interest rates.

Interest rates exhibit a substantial impact on the early exercise decision landscape for American put options. As the risk-free rate increases from $r = 0.01$ to $r = 0.10$, we observe a systematic expansion of the stopping region (red) at the expense of the continuation region (blue). This relationship arises because higher interest rates increase the opportunity cost of delaying exercise—when rates are elevated, the present value of receiving the strike price earlier becomes more attractive. At low interest rates ($r = 0.01$), the opportunity cost of deferring exercise is minimal, resulting in a smaller stopping region and greater incentive to hold the option. Conversely, at high interest rates ($r = 0.10$), the stopping region expands significantly, indicating that immediate exercise becomes optimal across a much wider range of stock price levels. This visualization provides clear evidence of the discounting mechanism's role in optimal exercise decisions.

Volatility Sensitivity

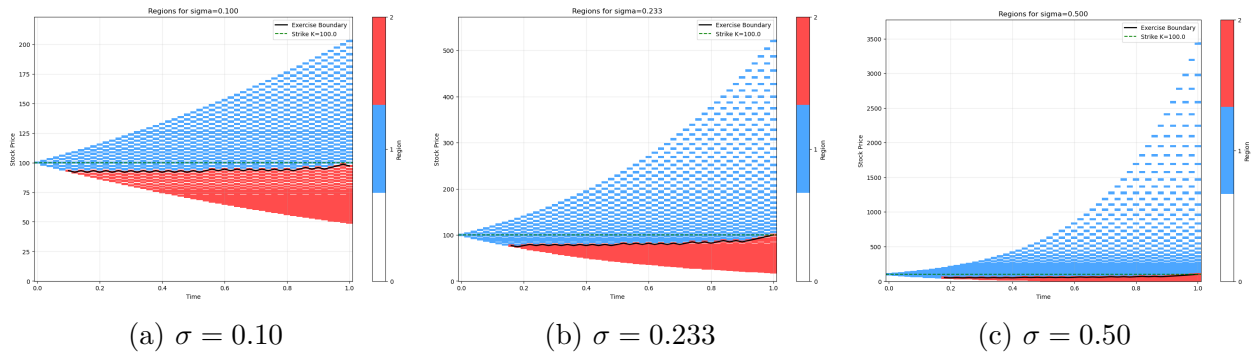


Figure 6.4: Continuation and stopping regions for different volatility values.

The effect of volatility on the exercise regions produces one of the most dramatic visual transformations in our analysis. As volatility increases from $\sigma = 0.10$ to $\sigma = 0.50$, two significant changes occur simultaneously: the overall price grid expands vertically, reflecting the wider range of possible future prices, while the stopping region (red) becomes proportionally smaller relative to the continuation region (blue). At low volatility ($\sigma = 0.10$), price movement is constrained, and the stopping region occupies a significant portion of the solution space, indicating frequent optimality of early exercise. In stark contrast, at high volatility ($\sigma = 0.50$), the stopping region shrinks to a narrow band at the bottom of the expanded price grid. This visualization powerfully demonstrates how increased volatility enhances the option's time value, making holders reluctant to sacrifice this value through early exercise except when the option is deeply in-the-money. The effect aligns with fundamental option pricing theory, where volatility is a primary driver of option value.

Time to Maturity Sensitivity

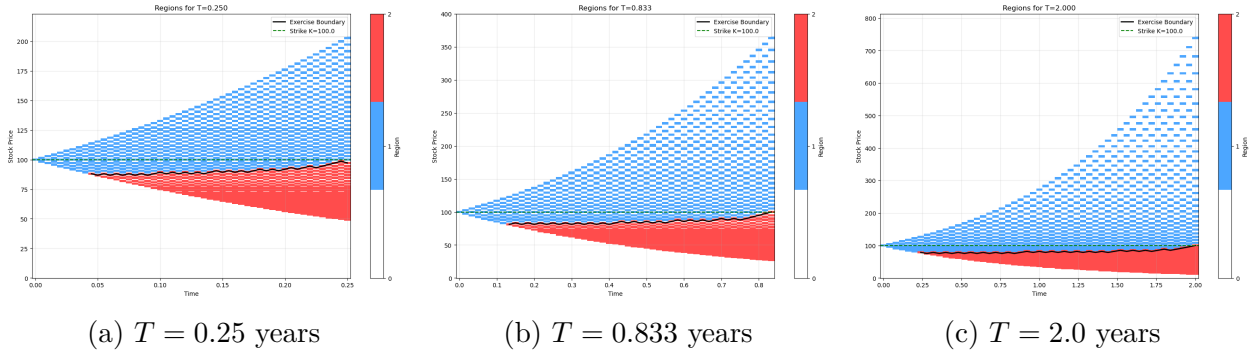


Figure 6.5: Continuation and stopping regions for different maturity periods.

Time to maturity fundamentally reshapes both the scale and structure of the exercise regions. As maturity extends from $T = 0.25$ to $T = 2.0$ years, the price grid expands horizontally (reflecting the longer time horizon) and vertically (accommodating the wider range of possible future prices). Short-term options ($T = 0.25$) exhibit compressed time scales with distinct, sharply delineated boundaries between regions. Medium-term options ($T = 0.833$) display more gradual boundary evolution, while long-term options ($T = 2.0$) reveal complex boundary dynamics that evolve more slowly during early and middle phases of the option's lifetime before accelerating toward the strike price near expiration. This visualization illustrates how the temporal dimension influences the opportunity cost of early exercise, with longer-dated options typically requiring more substantial price movements before early exercise becomes optimal, reflecting their greater embedded time value.

6.2 Comparative Analysis of Boundary Estimation Methods

6.2.1 Empirical vs. Laguerre Polynomial Approaches

In our analysis, we employ two complementary boundary estimation techniques:

- **Empirical Boundary:** Directly extracted from the simulated paths by identifying the critical stock prices where exercise actions change from continuation to stopping.
- **Laguerre Polynomial Boundary:** Obtained by applying regression with orthogonal Laguerre polynomials to smooth the exercise boundary function.

The empirical boundary approach provides an unbiased estimate but may exhibit statistical noise due to finite sampling. The Laguerre polynomial approach fits a functional form to these empirical observations, expressed as:

$$S_{\text{Laguerre}}^*(t_i) = \sum_{j=0}^M b_j L_j(t_i) \quad (6.1)$$

where $L_j(t_i)$ denotes the Laguerre polynomials and b_j are the regression coefficients. This approach inherently smooths the boundary, potentially filtering out noise while preserving the underlying structure.

6.3 Numerical Parameter Sensitivity

The accuracy and stability of the boundary approximation depend on several numerical parameters in our implementation. We analyze the impact of key numerical parameters on boundary estimation accuracy:

6.3.1 Time Steps Sensitivity

Increased time discretization enhances boundary smoothness. The empirical boundary shows greater sensitivity to the number of time steps compared to the Laguerre boundary, which maintains smoother profiles even with coarser discretization. This highlights the Laguerre method's advantage in mitigating discretization artifacts through its functional approximation approach. The empirical convergence rate as Δt decreases appears consistent with an $\mathcal{O}(\Delta t)$ pattern.

With fewer time steps (e.g., $N = 50$), the boundary exhibits a more pronounced zigzag pattern due to the coarse discretization of both time and prices. As the number of steps increases (e.g., $N = 200$), this pattern diminishes, and the boundary approaches a smoother curve.

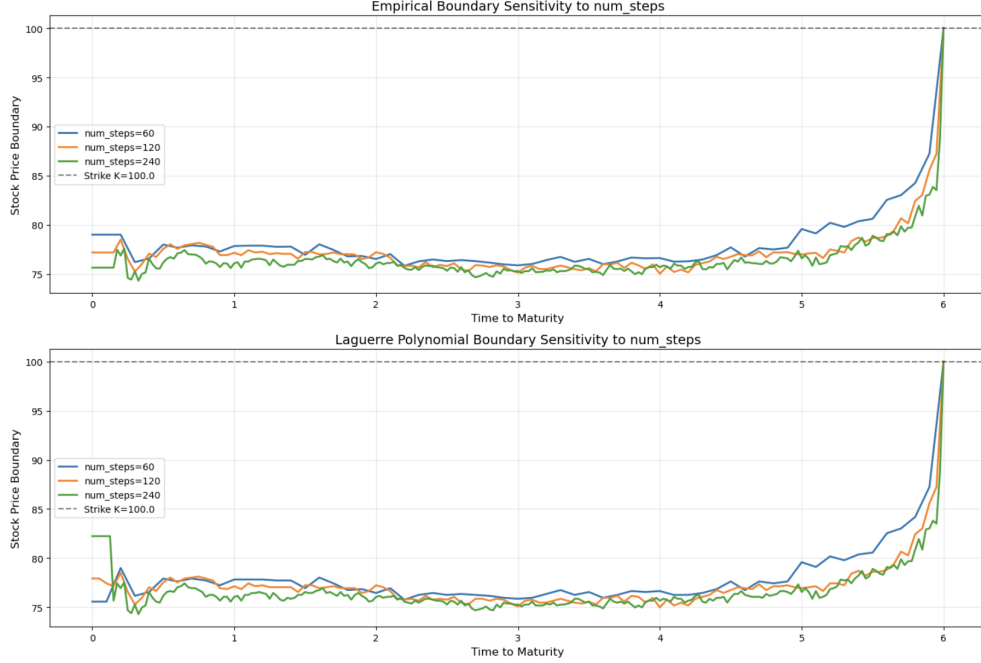


Figure 6.6: Effect of time steps on the exercise boundary. Time discretization affects smoothness, with 120 steps providing a balanced accuracy-efficiency trade-off.

6.3.2 Path Count Sensitivity

The boundary's statistical stability improves with increasing path count. Both methods converge as path count increases, but the Laguerre approach demonstrates superior stability at lower path counts. This suggests its effectiveness in capturing the underlying boundary structure even with limited sampling. The convergence follows the expected Monte Carlo rate of approximately $\mathcal{O}(1/\sqrt{N})$, with minimal improvements observed beyond 100,000 paths.

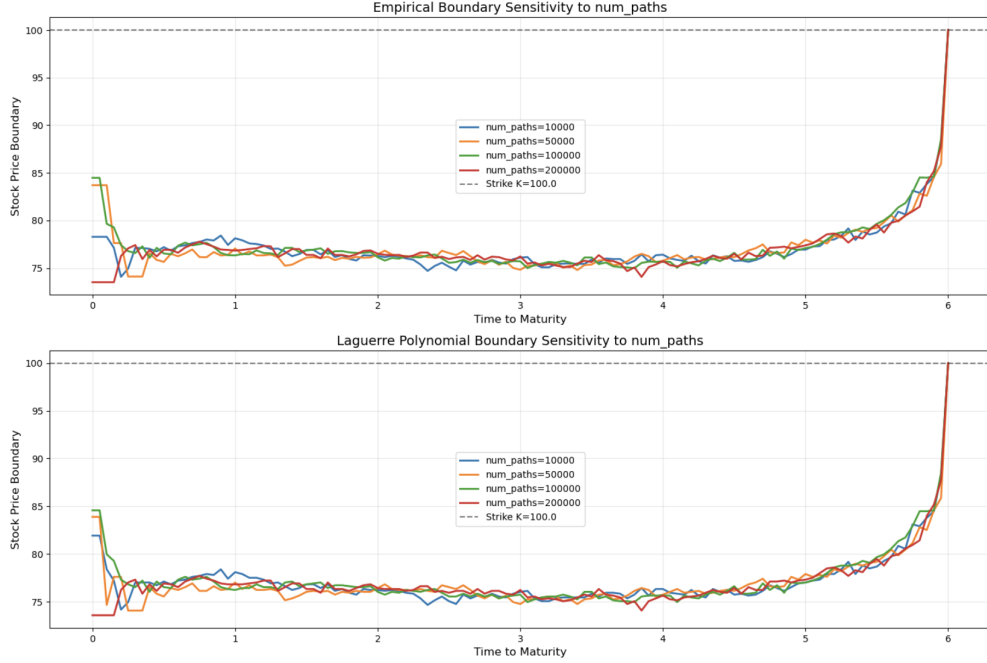


Figure 6.7: Path count sensitivity on boundary estimation. Path count exhibits a Monte Carlo-typical $O(1/\sqrt{N})$ convergence pattern with diminishing returns beyond 100,000 paths.

6.3.3 Polynomial Degree Sensitivity

Higher polynomial degrees capture more complex boundary behaviors. Degrees 4-5 appear to offer optimal balance, while degree 6 introduces oscillatory behavior, particularly near the beginning of the time horizon, suggesting potential overfitting in regions with sparse sampling. This underscores the importance of appropriate basis function selection when employing the Laguerre method.

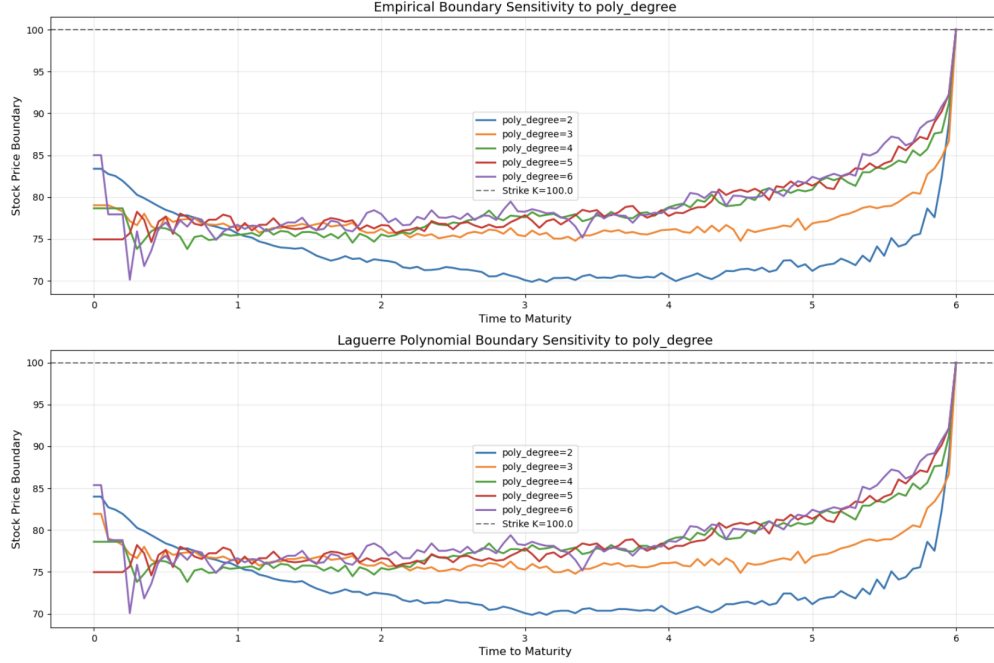


Figure 6.8: Polynomial degree sensitivity on the exercise boundary. Higher degrees (4-6) capture more complex boundary behavior but may introduce oscillations with degree 6, suggesting potential overfitting.

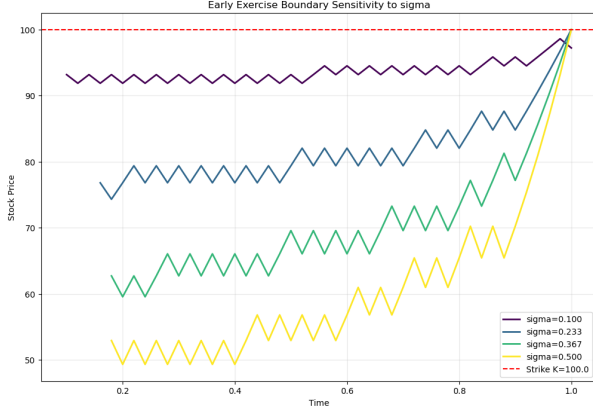
6.4 Market Parameter Sensitivity

6.4.1 Methodology

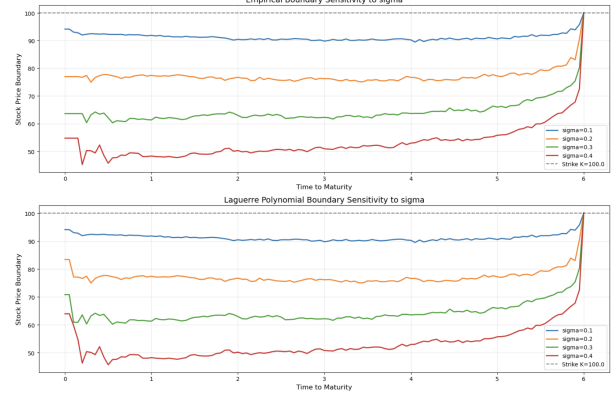
Our sensitivity analysis examines how the optimal exercise boundary changes when varying one parameter while holding all others constant. The base case parameters, unless otherwise varied, are:

- Initial stock price: $S_0 = 100$
- Strike price: $K = 100$
- Risk-free rate: $r = 0.05$
- Volatility: $\sigma = 0.2$
- Time to maturity: $T = 1.0$ year
- No dividends

For each parameter set, we extract the early exercise boundary at each time step and analyze its behavior.



(a) Binomial tree method



(b) Empirical and Laguerre polynomial methods

Figure 6.9: Early exercise boundary sensitivity to volatility σ using different estimation methods. Higher volatility substantially lowers the boundary, reducing the region where early exercise is optimal.

6.4.2 Volatility Sensitivity

Volatility emerges as one of the most influential parameters in determining the early exercise boundary position, with consistent effects captured across all estimation methodologies. As volatility increases from low ($\sigma = 0.1$) to high ($\sigma = 0.5$) levels, the exercise boundary is systematically depressed across all time points, with the effect most pronounced in the middle of the option's lifetime. This behavior aligns with fundamental option pricing theory, as increased volatility enhances the option's time value through greater upside potential, making early exercise less attractive. The binomial tree method (Fig. 6.9a) provides a complete visualization with clear separation between boundaries at different volatility levels, while the empirical and Laguerre methods (Fig. 6.9b) offer complementary perspectives with the Laguerre approach providing smoother transitions. For high volatility values, both methodologies confirm that the boundary remains substantially below the strike price until very close to maturity, indicating that early exercise is rarely optimal for highly volatile underlyings except in the immediate pre-expiration period.

6.4.3 Dividend Yield Sensitivity

Dividend yield demonstrates a strong positive correlation with the early exercise boundary level, an effect captured consistently across estimation methodologies. As the dividend yield increases from $d = 0.00$ to $d = 0.05$, the boundary elevates substantially across all time points, with particularly pronounced effects in the mid-to-late portions of the option's lifetime. This relationship is economically intuitive: higher dividends reduce the expected future asset value, thereby diminishing the opportunity cost of early exercise and making immediate exercise optimal at higher price levels. The empirical boundary exhibits more localized fluctuations compared to its Laguerre counterpart, particularly at higher dividend rates where sample paths may encounter complex exercise patterns. For substantial divi-

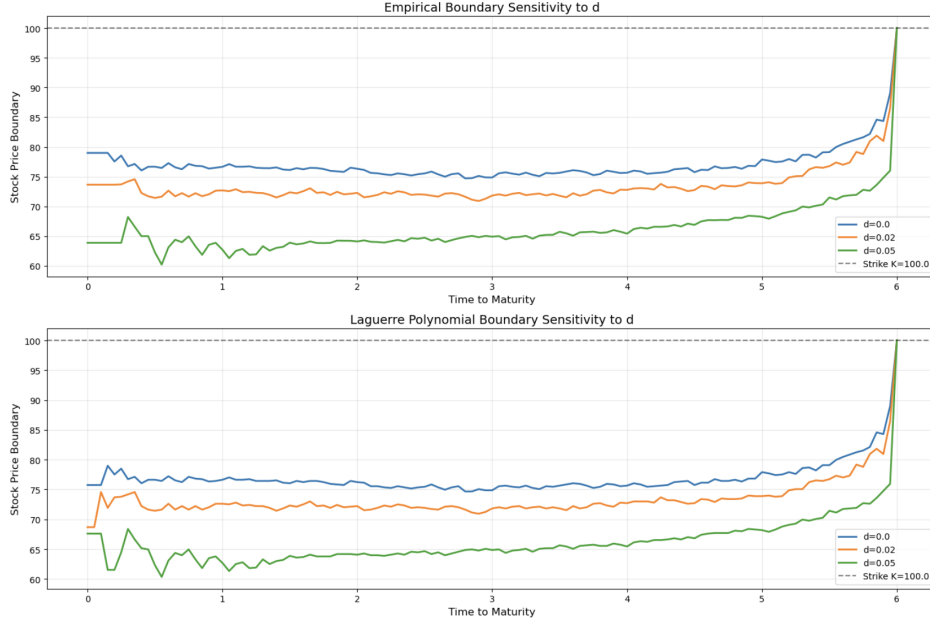
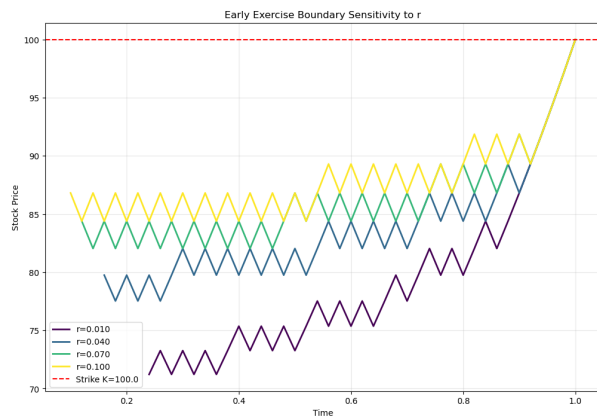


Figure 6.10: Early exercise boundary sensitivity to dividend yield d using empirical and Laguerre polynomial methods. Higher dividend yields lead to higher boundaries, expanding the region where early exercise is optimal.

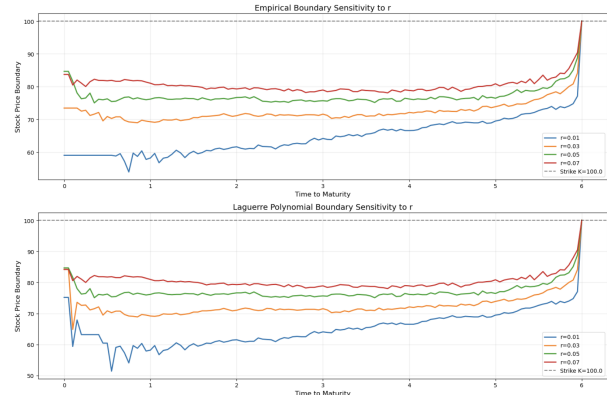
dividend yields ($d = 0.05$), the boundary approaches the strike price even well before maturity, indicating that dividends can fundamentally alter the early exercise calculus by significantly eroding the advantage of continued option holding. This strong sensitivity suggests that dividend announcements and policy changes represent critical monitoring points for holders of American put options.

6.4.4 Interest Rate Sensitivity

Interest rates display a clear inverse relationship with the early exercise boundary position across all estimation methods. As rates increase from $r = 0.01$ to $r = 0.10$, the boundary systematically shifts downward, expanding the region where early exercise is optimal. This effect stems from the increased present value of the strike price with higher interest rates, making immediate exercise more attractive relative to continued holding. The binomial method (Fig. 6.11a) captures this relationship with pronounced discrete steps, while the Laguerre approach (Fig. 6.11b) produces smoother transitions that filter out the noise evident in the empirical boundary. Interestingly, both methodologies demonstrate that the boundary's sensitivity to interest rates is strongest in the middle of the option's lifetime and diminishes near maturity, where all boundaries converge toward the strike price regardless of rate environment. This temporal pattern provides valuable insights for option holders, suggesting that interest rate changes have their most significant impact on exercise decisions during the middle portion of the option's life cycle.



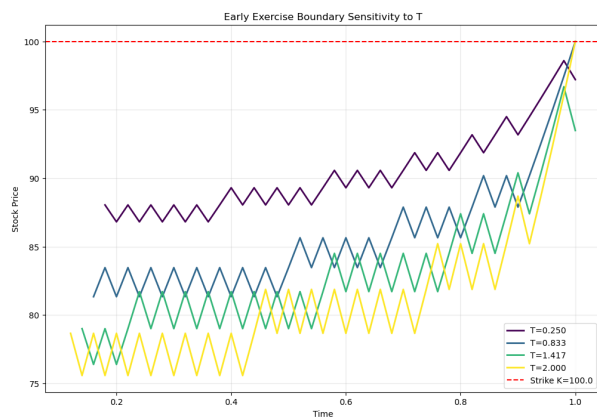
(a) Binomial tree method



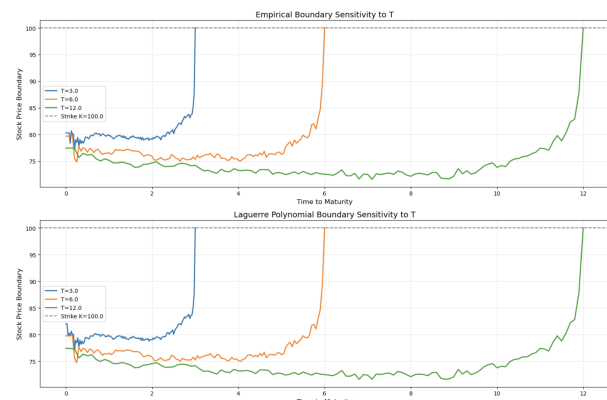
(b) Empirical and Laguerre polynomial methods

Figure 6.11: Early exercise boundary sensitivity to risk-free rate r using different estimation methods. Higher interest rates lead to lower boundaries, expanding the region where early exercise is optimal, due to the increased opportunity cost of deferring exercise.

6.4.5 Time to Maturity Sensitivity



(a) Binomial tree method



(b) Empirical and Laguerre polynomial methods

Figure 6.12: Early exercise boundary sensitivity to time to maturity T using different estimation methods. The boundary exhibits non-monotonic behavior with respect to maturity, highlighting the complex interplay between time value and exercise decisions.

The boundary's shape and position vary markedly with different maturity horizons, revealing complex temporal dynamics in optimal exercise patterns. Shorter maturities ($T = 0.25$) exhibit steeper boundary curves near expiration compared to longer maturities ($T = 2.0$), reflecting the compressed timeframe in which time value decays. The binomial representation (Fig. 6.12a) provides a comprehensive visualization of how the boundary evolves across different time horizons, with clear separation between maturity curves. The empirical and Laguerre approaches (Fig. 6.12b) offer complementary perspectives, with the

Laguerre method providing particularly consistent results across different time horizons. All methodologies confirm that shorter-term options show more rapid convergence to the strike price as expiration approaches, while longer-term options maintain lower boundaries until closer to expiration. This reflects the time-dependent influence of factors like interest rates and dividends on the optimal exercise decision, with longer horizons amplifying the effects of these continuous-time processes.

6.4.6 Initial Price Sensitivity

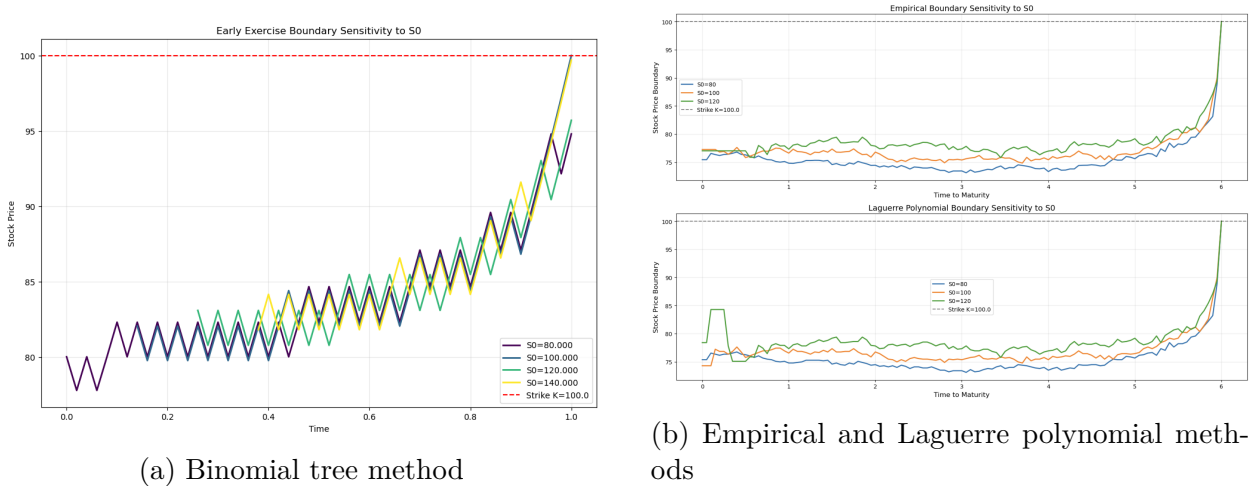
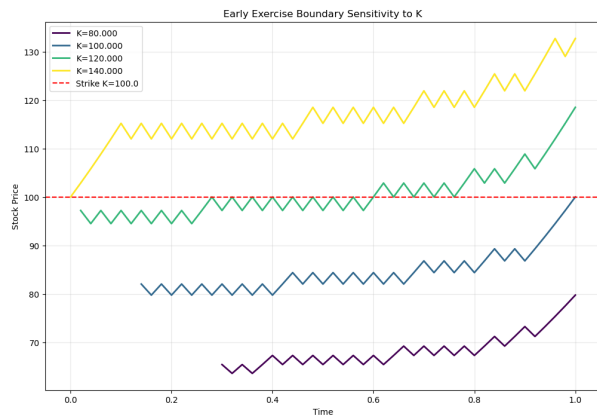


Figure 6.13: Early exercise boundary sensitivity to initial stock price S_0 using different estimation methods. Higher initial stock prices lead to higher early exercise boundaries, indicating that options with higher underlying prices require deeper price declines before early exercise becomes optimal.

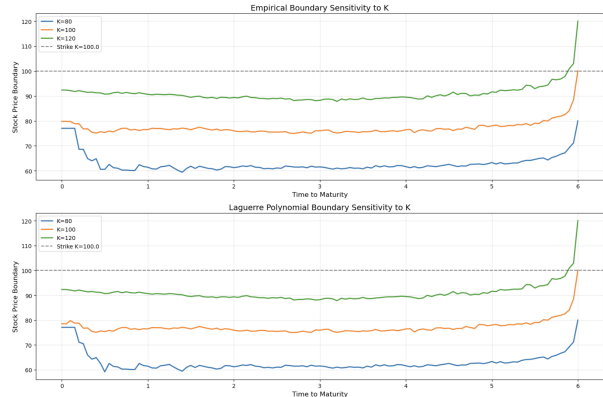
As the initial stock price increases from $S_0 = 80$ to $S_0 = 140$, the early exercise boundary shifts upward across all time points. This upward shift reflects the decreased likelihood of early exercise when the option starts further out-of-the-money, as the put option has less intrinsic value and thus a higher relative proportion of time value that would be sacrificed through early exercise. The relationship between initial price and boundary position is particularly informative for understanding how moneyness affects optimal exercise strategy. Options that begin deep in-the-money (low S_0 relative to strike) have boundaries that remain closer to the strike price, indicating that early exercise becomes optimal after relatively smaller price movements. Conversely, options that begin out-of-the-money (high S_0 relative to strike) require much more significant price declines before reaching their exercise boundaries. This visualization provides valuable strategic insight for option holders, illustrating how the optimal exercise threshold scales with the option's initial moneyness.

6.4.7 Strike Price Sensitivity

Higher strike prices consistently elevate the early exercise boundary across all time points, with an approximately linear scaling relationship between strike level and boundary position.



(a) Binomial tree method



(b) Empirical and Laguerre polynomial methods

Figure 6.14: Early exercise boundary sensitivity to strike price K using different estimation methods. The boundaries scale approximately linearly with the strike price, demonstrating that the relative distance between the boundary and strike remains consistent across different strike values.

This pattern emerges because higher strikes increase the intrinsic value of the put option, making early exercise more attractive at higher underlying price levels. The boundaries demonstrate remarkable scaling consistency, with the relative distance between the boundary and its corresponding strike remaining largely invariant across different absolute price levels. This suggests that the optimal exercise decision depends primarily on the proportional relationship between current price and strike, rather than on absolute price levels. From a practical perspective, this scaling property offers valuable simplification for option holders, as it indicates that optimal exercise strategies can be expressed in terms of price ratios rather than absolute thresholds, potentially allowing for more generalized decision rules across different strike values in the same asset class.

6.5 Early Exercise Analysis

Our analysis of early exercise behavior reveals clear relationships between model parameters and the likelihood of optimal early exercise. As shown in Figure 6.15, we examine how the percentage of simulated paths that result in early exercise varies across key financial and numerical parameters.

Volatility (σ) demonstrates one of the strongest positive relationships with early exercise frequency, with the percentage of paths exercised early increasing from approximately 38% at $\sigma = 0.10$ to over 63% at $\sigma = 0.40$. This seemingly counterintuitive result—given that higher volatility depresses the exercise boundary—can be explained by the increased likelihood of paths crossing a lower boundary at some point during their evolution when volatility is higher, despite the boundary itself being lower.

Interest rate (r) exhibits a clear negative correlation with early exercise percentage, decreasing from approximately 57% at $r = 0.01$ to 46% at $r = 0.07$. This negative relationship

Early Exercise Analysis Using Laguerre Polynomials

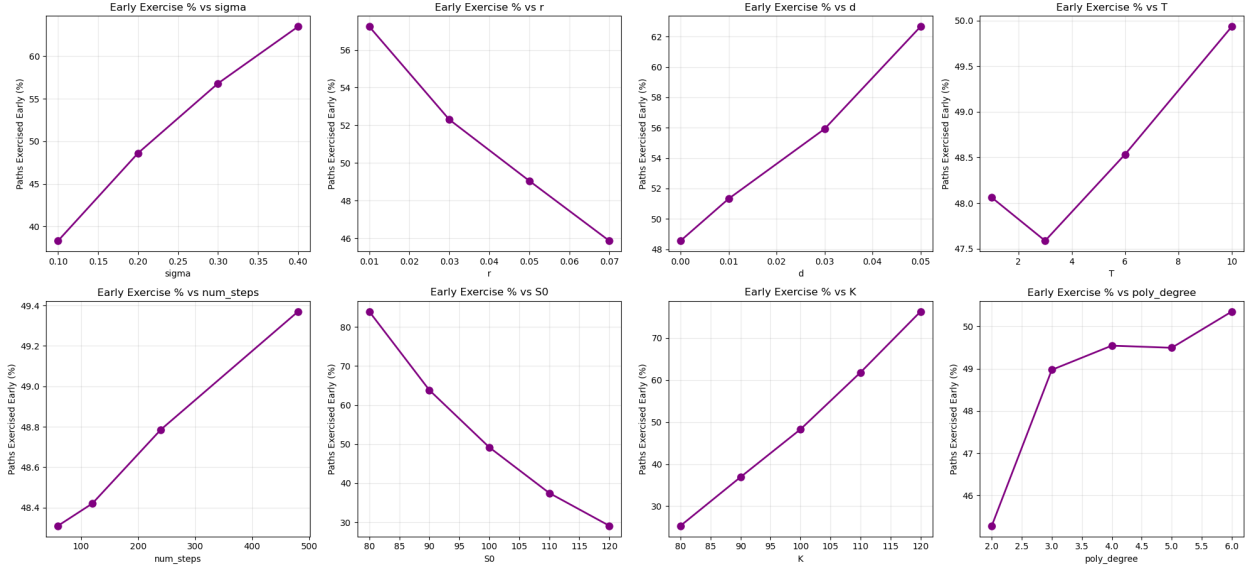


Figure 6.15: Sensitivity of early exercise percentage to various parameters. The figure shows the relationship between the percentage of paths exercised early and key parameters: volatility (σ), interest rate (r), dividend yield (d), maturity (T), number of time steps, initial stock price (S_0), strike price (K), and polynomial degree.

aligns with our boundary analysis, as higher interest rates lower the early exercise boundary, expanding the stopping region and thus increasing the probability that any given path will enter this region during the option's lifetime.

Dividend yield (d) shows a strong positive effect on early exercise likelihood, with the percentage increasing from 49% with no dividends to 62% with a 5% dividend yield. This result is consistent with the elevated exercise boundary observed at higher dividend rates, as dividends reduce the benefit of holding the option and increase the attractiveness of early exercise.

Time to maturity (T) reveals a non-monotonic relationship, with the early exercise percentage initially decreasing from $T = 1$ to $T = 3$ before increasing substantially for longer maturities. This U-shaped pattern suggests competing effects: longer maturities allow more time for paths to potentially cross the exercise boundary, while the boundary itself is lower for longer-dated options during most of their lifetime.

The numerical parameters also demonstrate interesting relationships. Increasing the number of time steps from 100 to 500 results in a modest but consistent increase in early exercise percentage, suggesting that finer time discretization captures more crossing events. The initial stock price (S_0) shows a strong negative relationship with early exercise percentage, decreasing from over 80% at $S_0 = 80$ to less than 30% at $S_0 = 120$. This reflects the moneyness effect, where options starting further in-the-money are much more likely to be exercised early.

Strike price (K) exhibits the opposite pattern, with early exercise percentage increasing from 26% at $K = 80$ to 75% at $K = 120$, again reflecting the moneyness effect from the

opposite direction. Finally, the polynomial degree used in the Laguerre approximation shows diminishing returns beyond degree 3, with early exercise percentages stabilizing between 49-50% for degrees 3 through 5, before slightly increasing for degree 6.

These relationships provide valuable insights for practitioners, highlighting which parameters have the most significant impact on early exercise behavior and offering guidance for managing positions with American-style options.

6.6 Conclusions and Implementation Insights

Our comparative analysis of boundary estimation methods reveals significant insights about both numerical approaches and the economic behavior of American options. While the binomial method offers computational simplicity with natural boundary extraction for moderate-sized problems, its discrete nature creates oscillatory boundary patterns. The LSM approach with Laguerre polynomials provides superior smoothness, especially where path sampling is sparse, and handles high-dimensional problems more effectively.

From an economic perspective, both methods confirm that increased volatility significantly depresses the early exercise boundary due to enhanced option time value under heightened uncertainty. For high volatility values ($\sigma > 0.3$), early exercise is rarely optimal except near expiration, suggesting portfolios with American puts on high-volatility assets should anticipate minimal early exercise until near maturity.

For put options, higher interest rates lower the exercise boundary by decreasing the present value of the future strike price receipt, thereby reducing the opportunity cost of early exercise. Dividend yields demonstrate the most dramatic impact, with higher yields substantially elevating the boundary because they reduce the expected future asset value. This suggests practitioners should maintain heightened vigilance around dividend announcements for portfolios containing American put options.

The Laguerre polynomial approach maintains stability with fewer simulated paths (50,000-100,000) compared to the empirical method and shows greater resilience to time step variations. Both methodologies confirm that all boundaries converge to the strike price at maturity, though this convergence manifests differently between methods.

In summary, while both boundary estimation approaches reach similar economic conclusions, they differ significantly in their numerical properties. The binomial method excels in computational simplicity, while the LSM approach with Laguerre polynomials offers superior boundary representation for complex problems. The optimal approach depends on the specific dimensionality, required accuracy, and computational constraints of the practical application.

Chapter 7

Monte Carlo Convergence Analysis

7.1 Price Distribution Analysis

The Monte Carlo simulation results provide valuable insights into the distribution of American put option prices using the Least Squares Monte Carlo (LSMC) method with Laguerre polynomials as basis functions. Our analysis spans multiple path counts from 100 to 50,000 to assess convergence properties and pricing stability.

7.1.1 Distribution Evolution with Increasing Path Count

For Laguerre polynomial degree 2, the price distribution demonstrates clear convergence behavior as the number of paths increases (Figure 7.1). With $N = 100$ paths, the distribution exhibits significant variability with a standard error (SE) of 0.8423, resulting in a wide and potentially unreliable price range. The estimated early exercise premium at this stage is 1.0572 (19.0% of the European price), which appears overestimated compared to results with higher path counts.

As we increase to $N = 1,000$ paths, the standard error decreases substantially to 0.2216, with the distribution becoming more centered. The early exercise premium decreases to 0.5039 (9.0%), suggesting that low path counts tend to overestimate the American option value. This pattern continues with $N = 10,000$ paths (SE = 0.0803, premium = 0.4230 or 7.6%) and $N = 50,000$ paths (SE = 0.0391, premium = 0.4110 or 7.4%), where both the distribution and premium estimates stabilize.

The most reliable estimate from our highest path count ($N = 50,000$) indicates an American put price of approximately 5.9845, compared to the Black-Scholes European put price of 5.5735, yielding an early exercise premium of approximately 7.4%.

Similar convergence patterns are observed for other polynomial degrees, with all converging toward a premium of approximately 7-10% at high path counts, though with slightly different specific values.

7.1.2 Standard Error Reduction and Confidence Intervals

The standard error of the price estimate decreases systematically with increasing path count, following an approximate power law relationship. For polynomial degree 2, the standard error

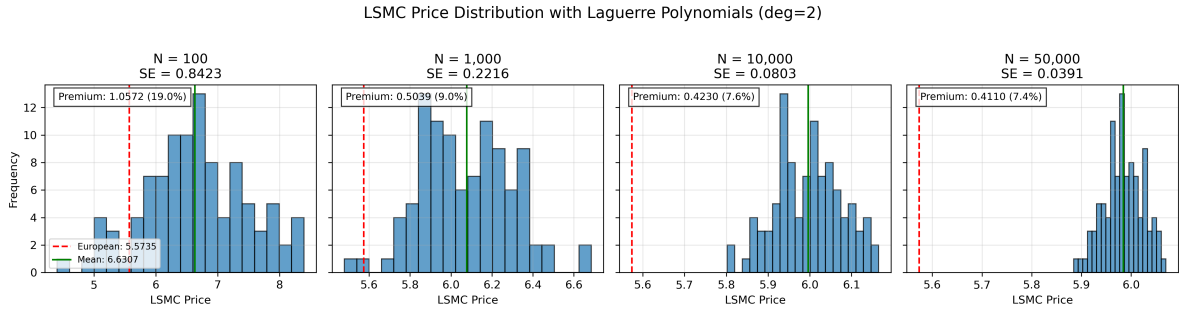


Figure 7.1: LSMC Price Distribution with Laguerre Polynomials (deg=2) at various path counts

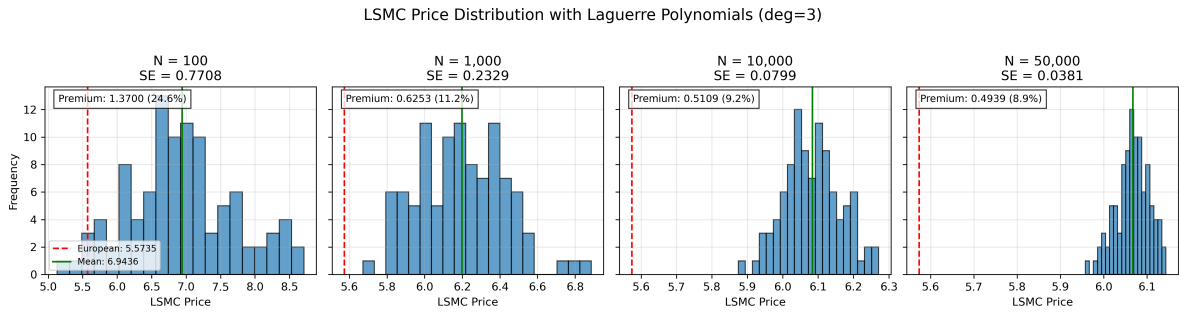


Figure 7.2: LSMC Price Distribution with Laguerre Polynomials (deg=3) at various path counts

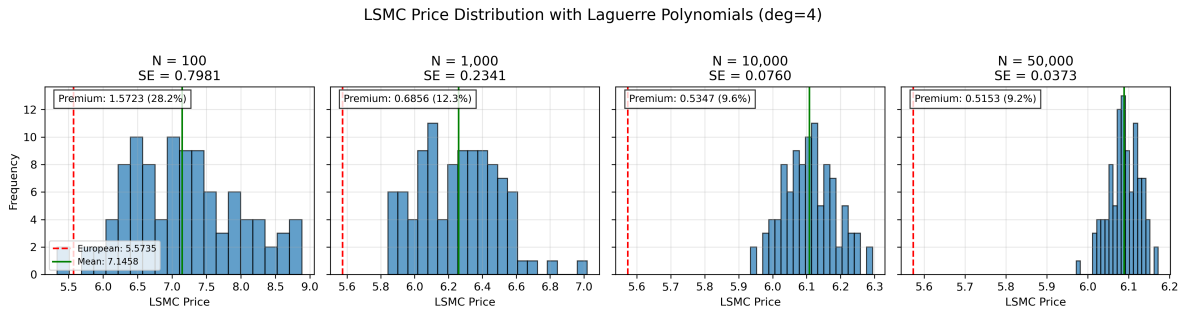


Figure 7.3: LSMC Price Distribution with Laguerre Polynomials (deg=4) at various path counts

decreases from 0.8423 ($N = 100$) to 0.0391 ($N = 50,000$), representing a reduction factor of approximately $21.5\times$ despite a $500\times$ increase in computational effort.

At $N = 50,000$ paths, the price estimates achieve sufficient precision for practical applications, with standard errors below 0.04 across all polynomial degrees tested. This translates to 95% confidence intervals of approximately ± 0.08 around the estimated price, representing less than 1.5% of the option value.

The final high-precision estimates ($N = 50,000$ with 10 replicates) display remarkably narrow price ranges. For polynomial degree 2, the price range is $[5.9278, 6.0517]$, spanning only 2.1% of the option value. Similar tight ranges are observed for other polynomial degrees, demonstrating the method's stability at high path counts.

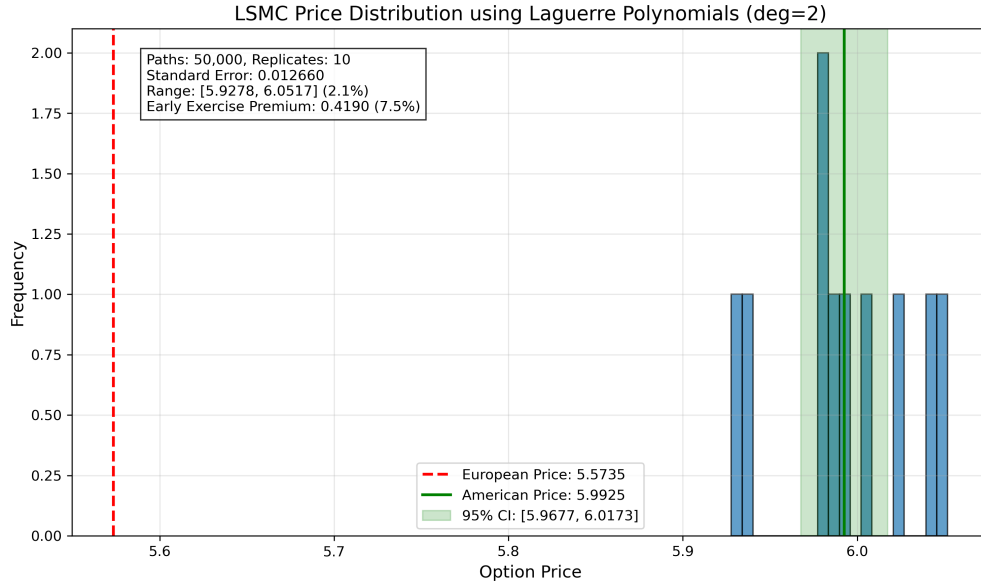


Figure 7.4: LSMC Price Distribution for Laguerre Polynomials (deg=2) with $N = 50,000$ paths

7.2 Convergence Rate Analysis

7.2.1 Empirical Convergence Rate

The convergence properties of the LSMC method with Laguerre polynomials are analyzed through log-log plots of standard error versus path count (Figures 7.5, 7.6, 7.7, 7.8, 7.9). These plots reveal that the empirical convergence rate closely follows but slightly outperforms the theoretical Monte Carlo convergence rate of $N^{-1/2}$.

For polynomial degree 2, the fitted empirical convergence rate is $N^{-0.490}$, very close to the theoretical rate. This pattern is consistent across all polynomial degrees tested, with fitted convergence rates ranging from $N^{-0.483}$ for degree 3 to $N^{-0.494}$ for degrees 4 and 5, suggesting that the LSMC method maintains standard Monte Carlo convergence properties regardless of polynomial degree choice.

The slight deviation from the theoretical $N^{-1/2}$ rate may be attributed to the regression-based nature of the LSMC algorithm, which introduces additional variance reduction through the function approximation process.

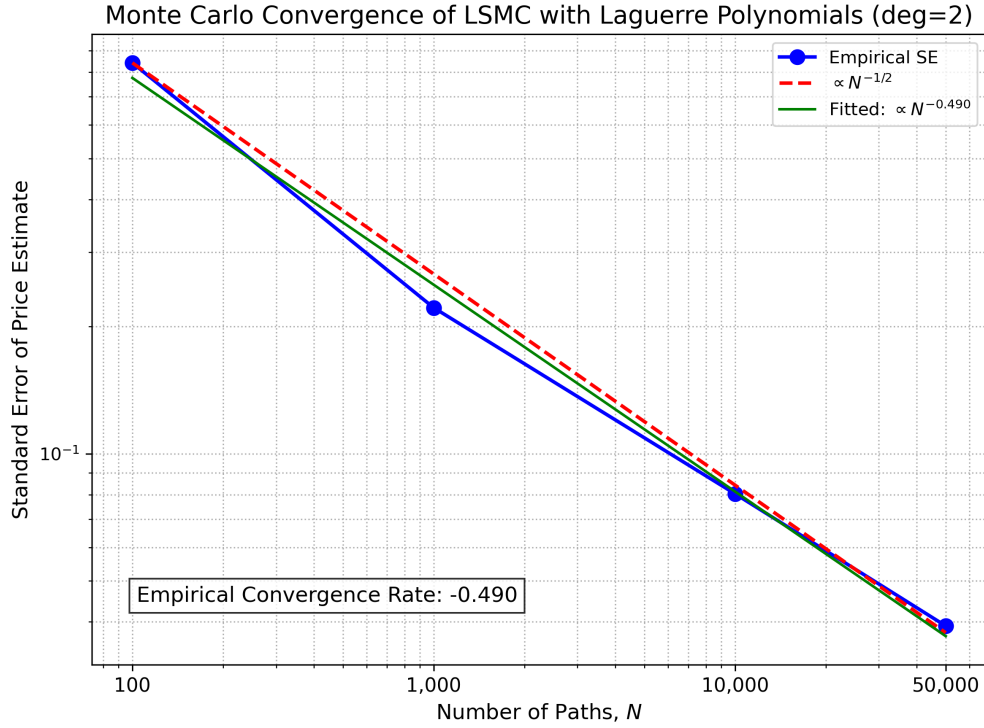


Figure 7.5: Monte Carlo Convergence of LSMC with Laguerre Polynomials (deg=2)

7.2.2 Convergence Stability Across Polynomial Degrees

Interestingly, the convergence behavior is remarkably consistent across all polynomial degrees tested (2 through 6). All degrees exhibit similar empirical convergence rates around $N^{-0.49}$, and the absolute standard error values at each path count are comparable. This suggests that while polynomial degree affects the price estimate itself, it does not significantly impact the rate at which precision improves with increased path count.

This consistency in convergence behavior provides a valuable practical insight: computational resources should be allocated primarily to increasing path count rather than polynomial degree when higher precision is required.

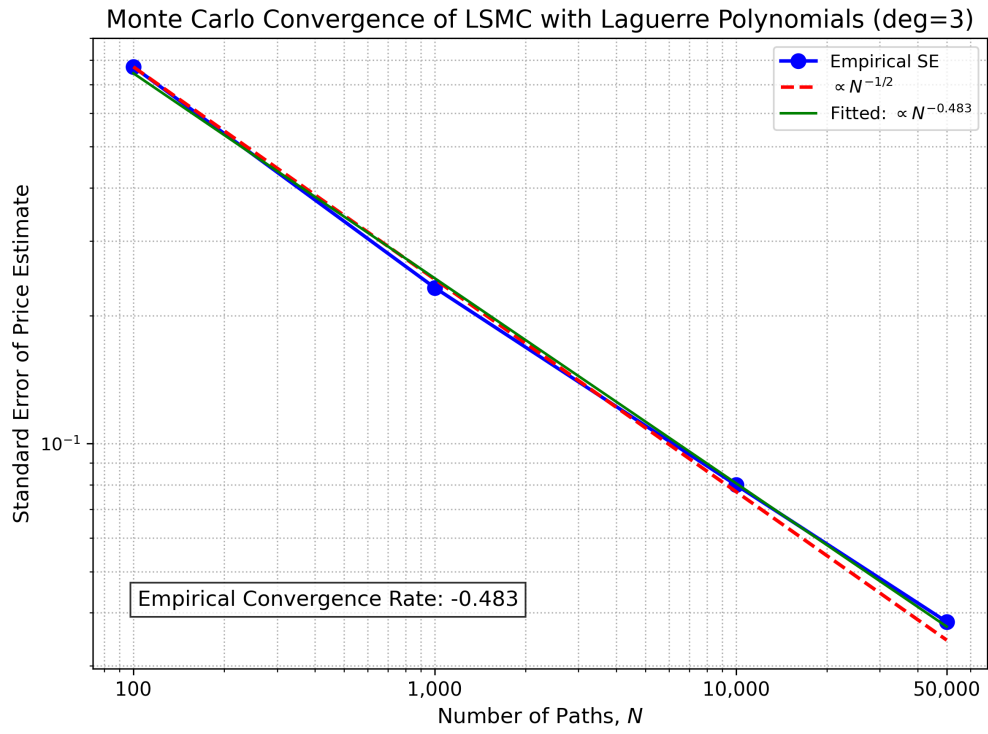


Figure 7.6: Monte Carlo Convergence of LSMC with Laguerre Polynomials (deg=3)

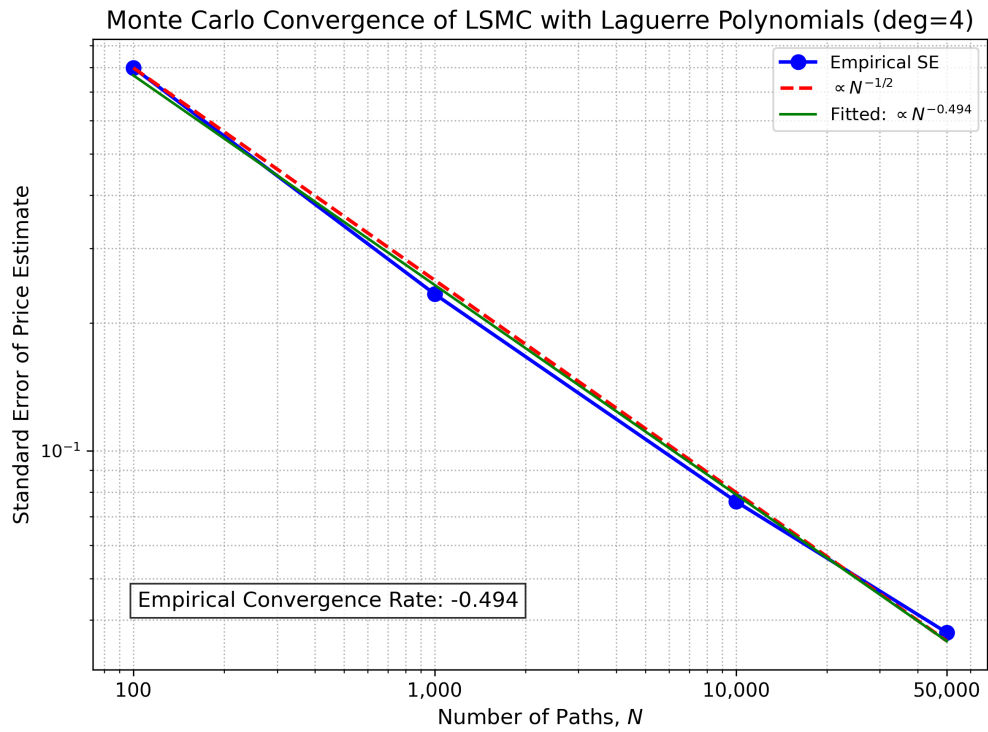


Figure 7.7: Monte Carlo Convergence of LSMC with Laguerre Polynomials (deg=4)

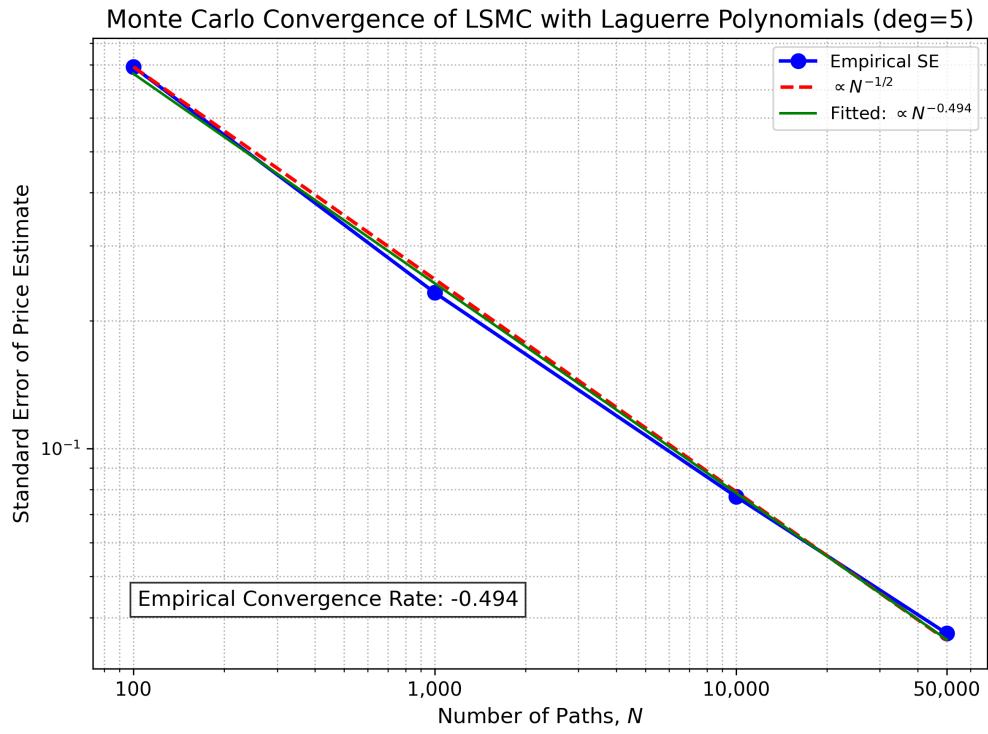


Figure 7.8: Monte Carlo Convergence of LSMC with Laguerre Polynomials (deg=5)

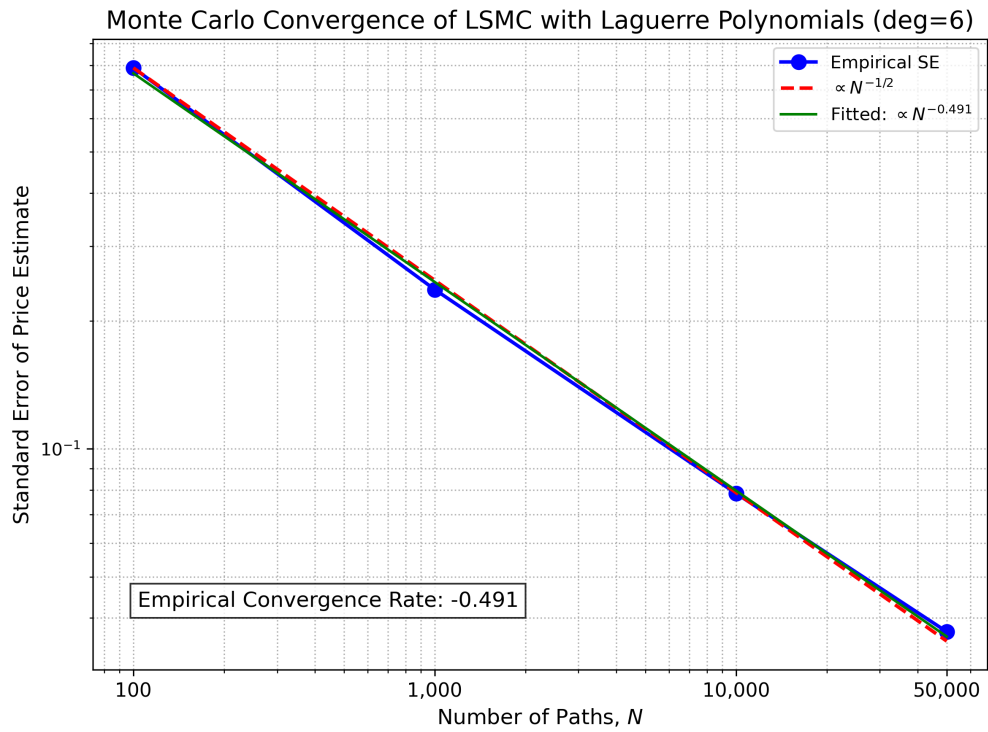


Figure 7.9: Monte Carlo Convergence of LSMC with Laguerre Polynomials (deg=6)

Chapter 8

Impact of Laguerre Polynomial Degree

8.1 Price Estimate Sensitivity to Polynomial Degree

Our analysis reveals that the polynomial degree of Laguerre basis functions significantly impacts American option price estimates, particularly at lower path counts. This impact diminishes as the path count increases, but notable differences persist even at high path counts.

8.1.1 Low Path Count Sensitivity

At $N = 100$ paths, the estimated early exercise premium varies dramatically with polynomial degree:

- Degree 2: 1.0572 (19.0% of European price)
- Degree 3: 1.3700 (24.6%)
- Degree 4: 1.5723 (28.2%)
- Degree 5: 1.7751 (31.8%)
- Degree 6: 1.9708 (35.4%)

This suggests that higher polynomial degrees tend to overestimate the American option price when combined with insufficient path counts, likely due to overfitting in the regression step. The regression models with higher polynomial degrees capture noise in the limited sample rather than the true continuation value function.

8.1.2 Convergence With Increasing Path Count

As the path count increases, the disparity between polynomial degrees narrows considerably. At $N = 50,000$ paths, the early exercise premium estimates stabilize across all degrees, as shown in Table [8.1](#).

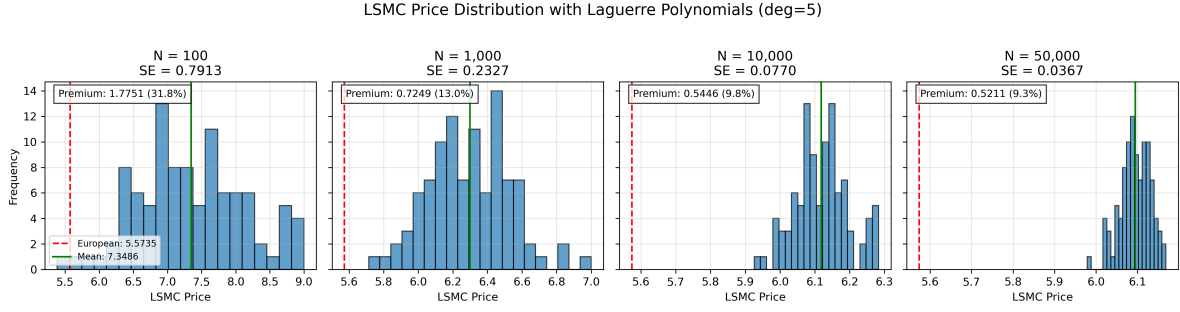


Figure 8.1: LSMC Price Distribution with Laguerre Polynomials (deg=5) at various path counts

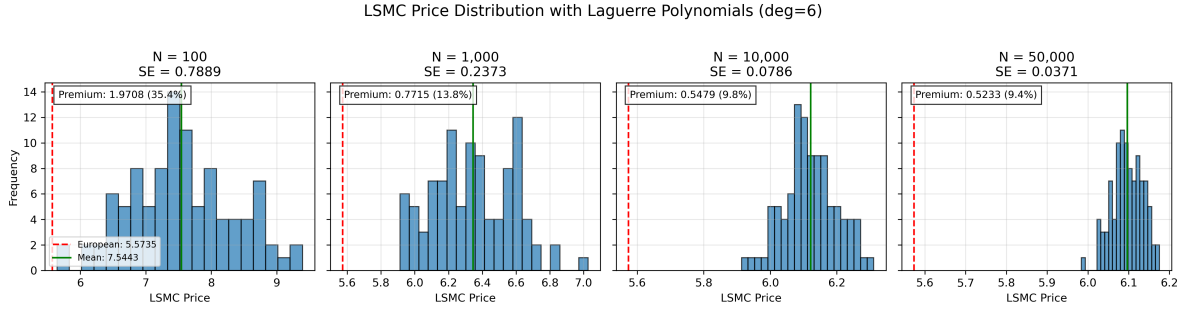


Figure 8.2: LSMC Price Distribution with Laguerre Polynomials (deg=6) at various path counts

Table 8.1: American Option Prices by Laguerre Polynomial Degree ($N = 50,000$ paths)

Degree	European Price	American Price	Premium (%)
2	5.573526	5.992523	7.52%
3	5.573526	6.076014	9.02%
4	5.573526	6.100101	9.45%
5	5.573526	6.102367	9.49%
6	5.573526	6.102045	9.48%

These results indicate that at high path counts, all polynomial degrees converge toward similar premium estimates in the range of 7.5-9.5%. However, higher polynomial degrees (4-6) consistently produce slightly higher premium estimates than lower degrees (2-3), suggesting these higher-degree representations may better capture the nonlinear features of the continuation value function.

Most notably, there appears to be convergence in the price estimates as the polynomial degree increases, with degrees 5 and 6 producing nearly identical American option prices of approximately 6.102, which suggests that increasing the polynomial degree beyond 5 offers negligible additional accuracy.

8.2 Price Distribution Characteristics Across Polynomial Degrees

The terminal price distributions at $N = 50,000$ paths provide further insights into the stability of estimates across polynomial degrees.

8.2.1 Distribution Shape and Spread

At $N = 50,000$ paths, all polynomial degrees produce tightly concentrated price distributions with similar standard errors (0.037-0.039). The price ranges are consistently narrow, spanning approximately 1.9% of the option value regardless of polynomial degree. This suggests that all polynomial degrees achieve comparable precision at high path counts.

However, the location of these distributions differs slightly across polynomial degrees. Higher polynomial degrees (4-6) yield mean American prices consistently around 6.10, while lower degrees (2-3) yield prices between 5.99 and 6.08. This systematic difference persists even at the highest path count, indicating a structural effect rather than statistical noise.

8.2.2 Confidence Intervals and Statistical Reliability

The 95% confidence intervals for all polynomial degrees at $N = 50,000$ paths are narrow and precise:

- Degree 2: [5.9278, 6.0517]
- Degree 3: [6.0157, 6.1328]
- Degree 4: [6.0348, 6.1504]
- Degree 5: [6.0483, 6.1619]
- Degree 6: [6.0428, 6.1600]

The confidence intervals for degrees 4-6 overlap substantially, suggesting that the differences between these higher degrees are not statistically significant at the 95% confidence level. However, the confidence intervals for degrees 2 and 3 do not overlap with the higher degrees, indicating that these lower degrees produce statistically different price estimates.

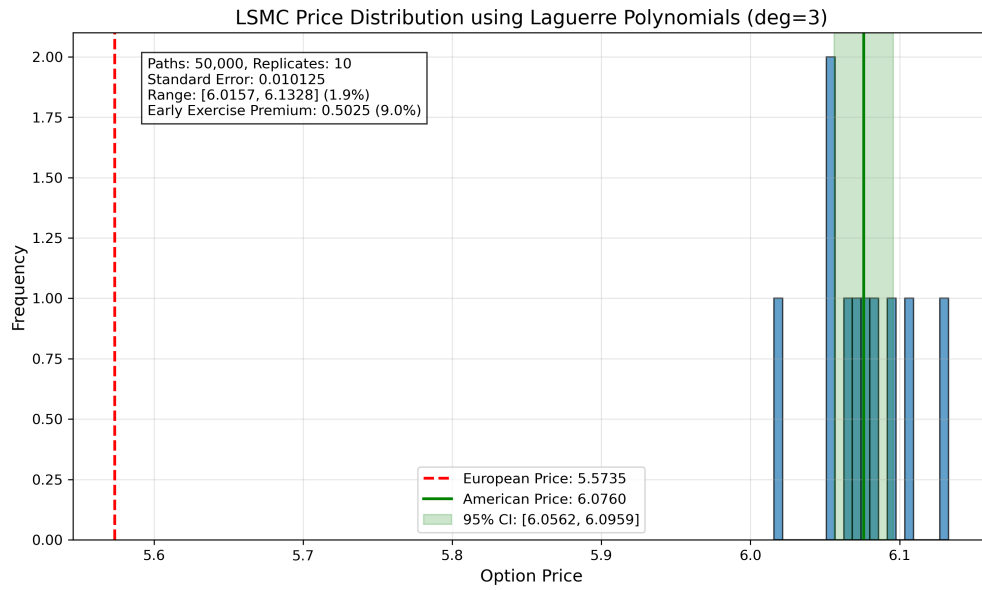


Figure 8.3: LSMC Price Distribution using Laguerre Polynomials (deg=3) with $N = 50,000$ paths

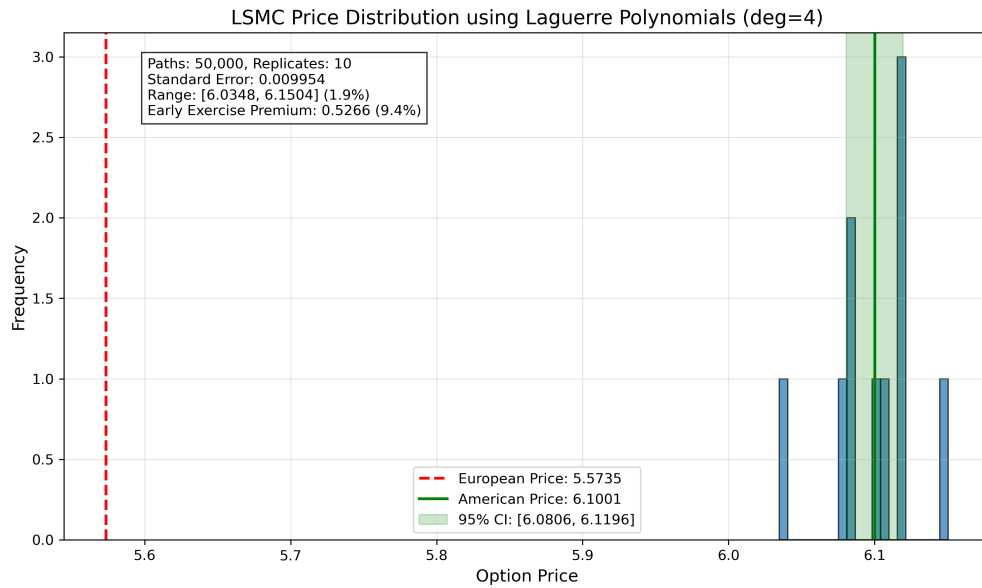


Figure 8.4: LSMC Price Distribution using Laguerre Polynomials (deg=4) with $N = 50,000$ paths

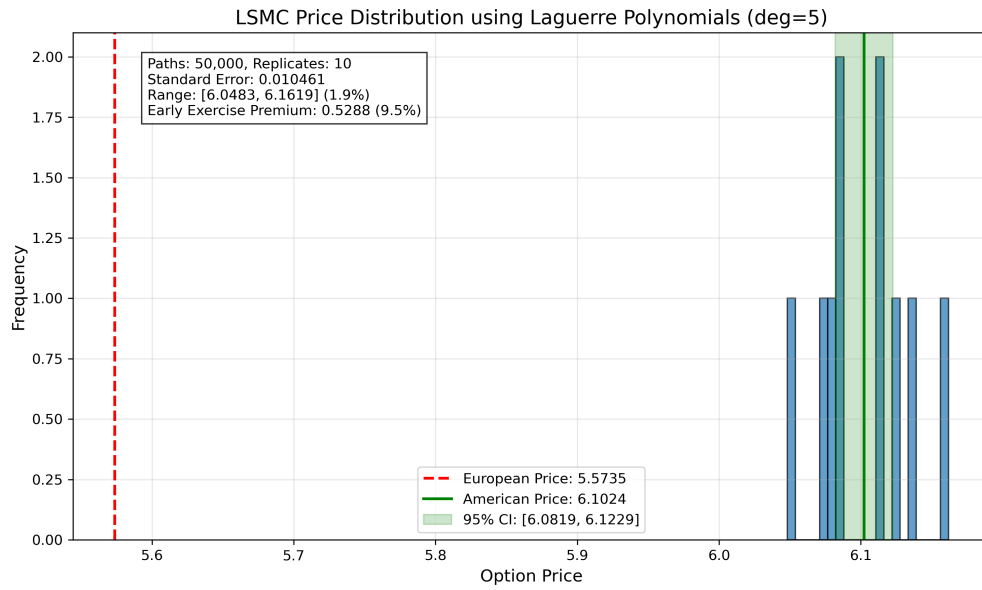


Figure 8.5: LSMC Price Distribution using Laguerre Polynomials (deg=5) with $N = 50,000$ paths

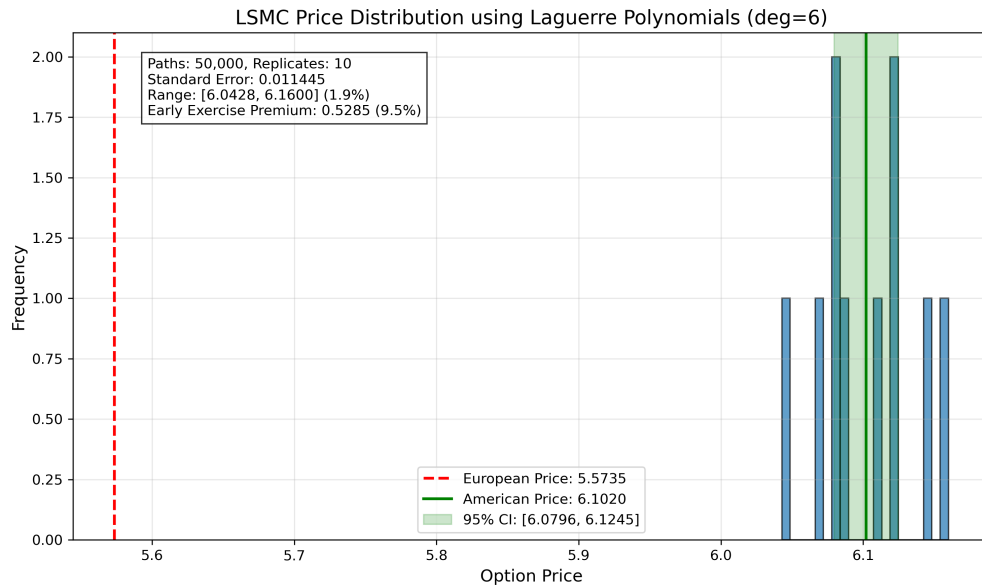


Figure 8.6: LSMC Price Distribution using Laguerre Polynomials (deg=6) with $N = 50,000$ paths

8.3 Optimal Polynomial Degree Selection

Based on our comprehensive analysis, several factors should guide the selection of an optimal Laguerre polynomial degree for LSMC pricing of American options:

1. **Accuracy vs. Efficiency:** Lower polynomial degrees (2-3) provide computational efficiency with good accuracy, while higher degrees (4-6) may offer slightly better accuracy at increased computational cost.
2. **Path Count Dependency:** At low path counts ($< 1,000$), lower polynomial degrees are preferable to avoid overfitting. At high path counts ($> 10,000$), degrees 4-5 provide the best balance of accuracy and efficiency.
3. **Convergence Behavior:** All polynomial degrees exhibit similar convergence rates, so the choice of degree should be based on absolute accuracy rather than convergence properties.
4. **Price Stability:** Degree 4 represents an optimal balance point, providing stable price estimates that are statistically indistinguishable from those of higher degrees (5-6) but with slightly lower computational requirements.

For most practical applications, Laguerre polynomial degree 4 is recommended as the optimal choice, offering sufficient flexibility to capture the nonlinear features of the continuation value function while maintaining computational efficiency and numerical stability. The price estimates from degree 4 (6.1001) are within 0.04% of those from degrees 5-6 (6.1024), suggesting negligible gain from using higher degrees.

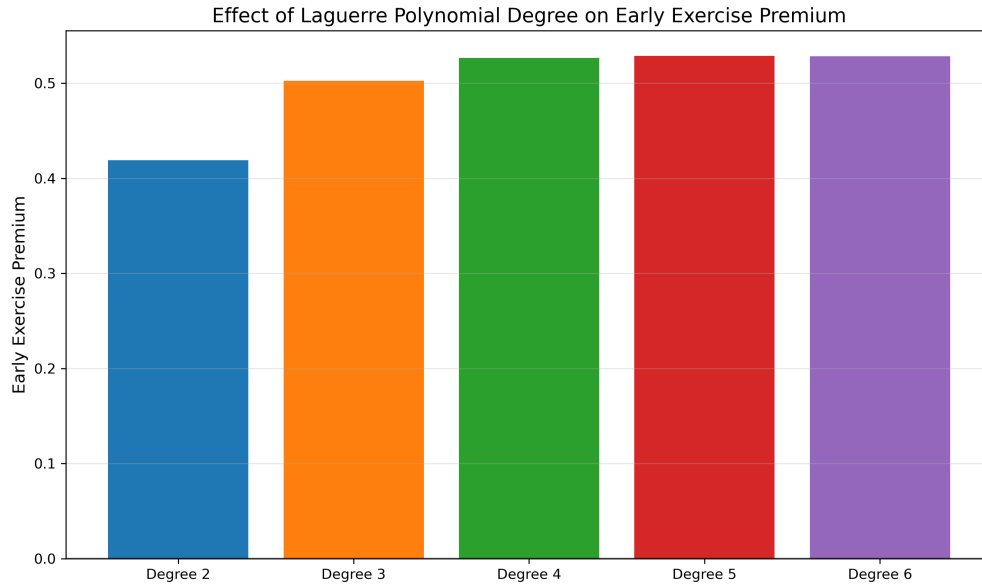


Figure 8.7: Effect of Laguerre Polynomial Degree on American Option Price

Chapter 9

Conclusion and Future Work

Our study on American option pricing focused on understanding early exercise behavior and developing efficient computational methods. We found that using Laguerre polynomial basis functions within the Least Squares Monte Carlo (LSM) approach leads to smoother representations of the early exercise boundary and greater computational efficiency. While both binomial tree and Monte Carlo methods arrive at similar pricing outcomes, they differ in performance: binomial trees are more intuitive for moderate-sized problems and allow for easy boundary extraction, whereas the LSM method with Laguerre polynomials is better suited for higher-dimensional problems due to its flexibility and efficiency.

Through sensitivity analysis, we observed that volatility lowers the early exercise boundary but increases the chance of exercising early. Higher interest rates reduce the optimal early exercise value for puts, making early exercise more appealing. Dividend yields have the strongest effect, raising boundaries and increasing the likelihood of early exercise. The early exercise premium—the additional value of American options over European ones—responds in complex ways to these parameters, offering practical insight into how to assess the value of early exercise rights.

Our numerical experiments show that a Laguerre polynomial of degree 4 provides the best balance between accuracy and efficiency. Beyond 200 time steps and 50,000 simulation paths, further improvements are marginal.

Future research could explore using Laguerre methods for multi-asset American options, helping manage the complexity that arises in high-dimensional settings. Comparing different orthogonal polynomial families such as Hermite or Chebyshev may reveal better options for certain problems. Machine learning could further enhance the LSM framework by adaptively selecting basis functions or approximating continuation values more accurately.

Expanding the model to include real-world features such as transaction costs, taxes, and liquidity constraints would improve its practical realism. Analyzing how model uncertainties—such as stochastic volatility, jumps, or regime-switching—affect early exercise boundaries would strengthen the robustness of results. Given the impact of dividends, exploring the role of discrete dividend timing is also an important direction.

Finally, our framework can support practical applications such as algorithmic trading strategies based on mispricing signals and evaluating how different pricing methods influence risk metrics and capital requirements. Overall, Laguerre-based approaches offer a powerful mix of theoretical clarity and computational efficiency for valuing American options.

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